Equipment Communication Specifications
HSMS Type Ver 1.6
for LG Display ToE

한글 버전



LG CNS

- Revision History -

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1.0	2017.06.22	Lee Sung Woo	Version 1.0 for LG Display ToE
1.1	2018.02.08	Kim Si Won	Version 1.1 for LG Display ToE
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0. From Ver AP4 1.4 to Ver ToE 1.0

No	Part or Section	Ver AP4 1.4 Page	Ver ToE 1.0 Page	Contents & Description
1	1.11 PD Down Request	p147 p181	p147 p180	Equipment PD Down Request S2,F95 → S2,F99 Changed (7.15 Chemical Change, Stream/Function Overlap)
2	19. Lot State Change Report	p152 p168	p152 p181	S1,F87 → S6,F81 Changed (CA Specification Standard)
3	6.9 Equipment Constant Recipe Request	p143 p184	p143 p183	S7,F81 → S7,F79 Changed (CA Specification Standard)
4	1.13 Multi Recipe Information Send	p149 p183	p149 p186	S7,F79 → S7,F81 Stream/Function Changed (CA Specification Standard)
5	S7,F78 : Recipe Body S7,F80 : Equipment Constant Recipe S15,F76 : Recipe Parameter Report S15,F77 : Recipe Parameter Change Send S15,F79 : Recipe Parameter Change Report S15,F96 : Equipment Constant Recipe Parameter Send S15,F97 : Equipment Constant Recipe Change Command S15,F99 : Equipment Constant Recipe Change Report	-	p128 p185 p195 p196 p199 p208 p209 p212	1. A[14] <RPARMNAME> → A[24] <RPARMNAME> Changed 2. A[14] <RPARMVAL> → A[24] <RPARMVAL> Changed

1. From Ver ToE 1.0 to Ver ToE 1.1

No	Part or Section	Ver 1.0 Page	Ver 1.1 Page	Contents & Description
1	Load Glass Data Send (S2,F87)	97-98	97-99	Message Format Modify
2	Unload Glass Data Report (S6,F63)	103-104	104-106	
3	Glass Move Request (S6,F67)	107	109	

2. From Ver ToE 1.1 to Ver ToE 1.2

No	Part or Section	Ver 1.1 Page	Ver 1.2 Page	Contents & Description
1	7. Scenario	30-35, 53,68, 69,71, 72,73, 149	30-35, 53,68, 69,71, 72,73, 149	S6,F73 Sequence Changed (ToE Specification Standard)

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3. From Ver ToE 1.2 to Ver ToE 1.3

No	Part or Section	Ver 1.2 Page	Ver 1.3 Page	Contents & Description
1	Load Glass Data Send (S2,F87)	97-99	98-100	Message Format Modify
2	Unload Glass Data Report (S6,F63)	104-106	105-107	

3. From Ver ToE 1.3 to Ver ToE 1.4

No	Part or Section	Ver 1.3 Page	Ver 1.4 Page	Contents & Description
1	Load Glass Data Send (S2,F87)	98-100	98-100	GSD Item Add
2	Unload Glass Data Report (S6,F63)	105-107	105-107	
3	Unload Glass Data Request (S2,F69)	93	93	2. B[2] <STAGEID> → 2. B[1] <STAGEID> Changed

4. From Ver ToE 1.4 to Ver ToE 1.5

No	Part or Section	Ver 1.4 Page	Ver 1.5 Page	Contents & Description
1	Load Glass Data Send (S2,F87)		100	GSD Item(Cut_Type) Change
2	Unload Glass Data Report (S6,F63)		107	

5. From Ver ToE 1.5 to Ver ToE 1.6

No	Part or Section	Ver 1.5 Page	Ver 1.6 Page	Contents & Description
1	Message Format Change S2 F80, S2 F87, S6 F63, S6 F65, S6 F73, S7 F61, S2 F82, S2 F84, S2 F93		96,98,99, 105,108, 113,123, 124,176, 178,182	<PNLIF> A[400]→A[2000] <SUBMDLIF> A[400]→A[2000] <SUBPNLIF> A[400]→A[2000]

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- Explanation -

1. 이 문서는 ECS에 적용된 Equipment Control System을 LG Display에 공급한 업체에 대한 장비 통신 사양서이다. 이 문서는 공통 장비 제어 통신을 다루기 위해 배포되었다.

2. 이 사양서는 LG Display의 비즈니스 절차에 따라 배포된다.

3. LG Display의 승인 없이, 이 명세서를 회사 밖으로 가지고 나가거나 배포/공유하는 경우, **해당 업체는 LG Display의 내부 방침에 따른 처벌과 보상을 해야 할 뿐만 아니라 '불공정 거래 방지법'과 '기업 비밀 보장법'에 의거 민/형사상 의 책임을 져야 한다.**

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Chapter Ⅰ. ECS-Equipment HSMS Standard Specification

For LG Display

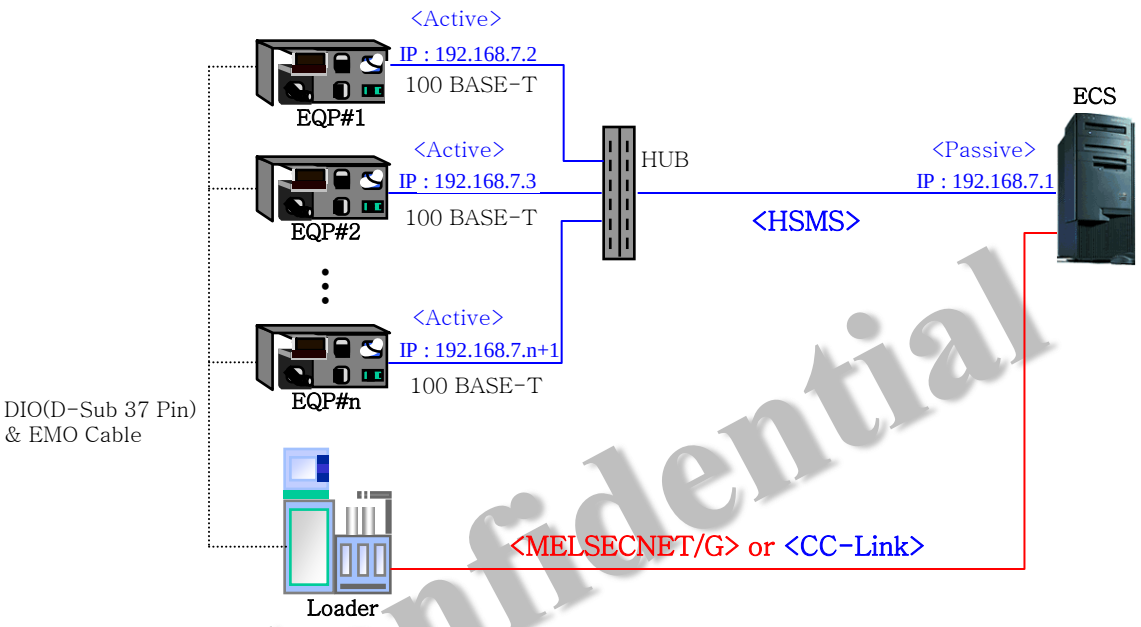
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1. System Format

1.1 System Configuration

본 문서는 장비와 ECS (Equipment Control System)간의 통신 사양에 적용된다. 이 사양은 하나의 Loader와 여러 종의 장비간 제어 시스템을 정의한다. 장비는 하나 또는 그 이상 존재할 수 있다.



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ECS 는 장비가 요청하는 Glass ID, LOT ID, CST ID, Recipe ID, Slot ID 등과 같은 data들을 주고 받는다. ECS와 장비간 통신은 HSMS Protocol에 의해 이루어진다.

주의) ECS와 장비가 로컬 네트워크로 연결되어 있는 경우, 다른 시스템과 절대 연결되어서는 안 된다.

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2. HSMS (High-speed SECS Message Services)

더 많은 정보를 얻고자 하는 사람은 "SEMI E37 HIGH-SPEED SECS MESSAGE SERVICES (HSMS) GENERIC SERVICES" 를 참고하기 바랍니다.

2.1 Physical Interface

2.1.1 Frame Type
IEEE 802.3u – Fast Ethernet (100Mbps)

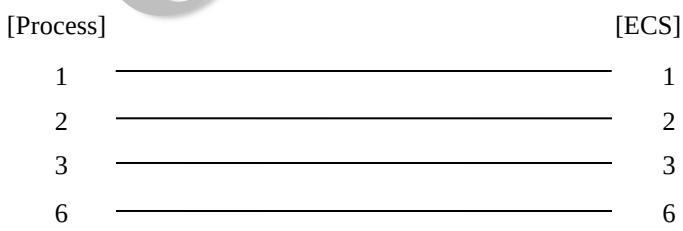
2.1.2 Connector RJ-45 (8-pin)

2.1.3 Ethernet Media
100BaseTX (Twisted Pair)

2.1.4 Signal Pins
Following table shows each pin of Connector functions .(Pin 1,2,3,6 used only.)

Pin	Signal	Description
1	Transmit +	Transmit Positive Signal
2	Transmit -	Transmit Negative Signal
3	Receive +	Receive Positive Signal
4	Not used	-
5	Not used	-
6	Receive -	Receive Negative Signal
7	Not used	-
8	Not used	-

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2.2 HSMS Message Format

2.2.1 Message Format
Maximum Message Length is 2 Mbytes.

Number of Bytes	Description
4 Bytes	Message Length. MSB First.
10 Bytes	Message Header
0 – n Bytes	Message Text

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2.2.2 Message Header

Following table shows the message header structure, and P type always uses "0"(SECS-II Encoding).

Bytes	Description
0~1	Session ID (Device ID).
2	Header Byte 2
3	Header Byte 3
4	P type
5	S type
6~9	System Bytes

Following table shows the selectable value in S type. Deselect and Reject Procedure is not used.

Value	Description
0	Data Message
1	Select.req
2	Select.rsp
5	Linktest.req
6	Linktest.rsp
9	Separate.req

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2.3 The method of communication

2.3.1 통신연결

TCP/IP (Transmission Control Protocol/Internet Protocol)

2.3.2 Parameter 설정

HSMS 구현은 반드시 아래의 parameters에 대한 시간정보 설정에 대비해야 한다. Protocol Parameter의 범위 및 해결방안은 최소한 아래 표와 같이 나타나야 한다. 각각의 설정치는 사용자에 의해 장비에서 수시로 변경 가능하도록 한다. 단 connect mode는 항상 장비가 ACTIVE 이고, ECS가 PASSIVE이다.

Symbol	Parameter Name	Typical Value	Value Range
DEVID	Device ID(Session ID)	0	0 ~ 32767
T3	Reply Timeout	90 sec	1 ~ 120 sec
T5	Separation Timeout	10 sec	1 ~ 240 sec
T6	Transaction Timeout	5 sec	1 ~ 240 sec
T7	NOT SELECTED Timeout	10 sec	1 ~ 240 sec
T8	Intercharacter Timeout	5 sec	1 ~ 120 sec
CT	Conversation Timeout	180 sec	0 ~ 600 sec
Connect Mode	Connect Mode (Active/Passive)	-	-
Local Entity	Local Entity IP Address & Port Number	-	-
Remote Entity	Remote Entity IP Address & Port Number	-	-

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2.2.3 Transaction

Transaction allowed in states.

Transaction	Allowed in State	Who initiates Transaction?
Select	HSMS Not Selected	Active Entity
Linktest	HSMS Selected	Either Entity
Separate	HSMS Selected	Either Entity

2.2.4 Procedures Description

2.2.4.1 “Select”(선택 절차)

선택 절차는 컨트롤 상태에서 선택 요청(Select.req) 메시지를 사용하는 TCP/IP 연결상의 HSMS통신을 설정할 때 사용된다. 선택 절차는 반드시 “NOT SELECTED” 상태에서만 가능하다. 성공적으로 0 선택 상태로 전환되지 못한 선택 절차는 반드시 TCP/IP 통신을 닫고, “Not CONNECTED” 상태로 전환되어야 한다.

2.2.4.2 “Linktest”(연결테스트 절차)

링크 테스트는 TCP/IP와 HSMS 통신의 동작상의 완결성을 결정짓는데 사용된다. “CONNECTED” 에서 이것의 사용은 언제라도 허용된다. 링크 테스트는 반드시 장비 연결 테스트에 적용된다.

- ‘SELECTED’상태에서 장비 (Active)는 ECS (Passive)로 반드시 주기적으로 ‘Linktest.req’를 보내야 한다.
- ‘SELECTED’상태, 그리고 (Active)로부터 120초 동안 아무런 메시지를 받지 못했을 때, ECS (Passive)는 장비 (Active)로 반드시 ‘Linktest.req’를 보내야 한다.

2.2.4.3 “Separate”(분리 절차)

분리 절차는 TCP/IP 통신이 끊어지기 전에 갑자기 HSMS 통신이 끊어졌을 때 사용된다. HSMS는 연결 상태가 ‘SELECTED’ 상태에서 분리를 사용했을 때 요구한다. 실제 응답이 보내지지 않고, 로컬 상태인지의 여부와는 상관없이 통신의 중단을 요구한다.

- 상태가 ‘SELECTED’에서 ‘NOT SELECTED’로 변경이 될 때, EQP는 반드시 ‘Separate.req’를 ECS로 보낸다.

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3. SECS - II

더 많은 정보를 원할 시에는 "SEMI E5-95 SEMI EQUIPMENT COMMUNICATIONS STANDARD 2 MESSAGE CONTENT (SECS-II)" 혹은 이후 버전을 참고하시기 바랍니다.

3.1 Abort Transaction (S*FO)

Function 0은 모든 Stream 에 존재하며 각각의 Stream에서 같은 특별한 의미를 가진다. Function 0 Message는 Transaction Timeout이 발생하는 것을 기다리지 않기 위해 Transaction을 종료한다. Function 0 은 Transaction Error 나 다른 몇 가지의 원인으로 인해 해석기가 기대되는 응답을 보낼 수 없을 때 그것을 대신하여 보내진다. 해석기가 Transaction을 종료하기 위해 Function 0을 보내는 것은 요구가 아니다.

3.2 Conversation Timeout

CT (Conversation Timeout) 는 ECS로부터 적정한 시간 내에 기대하는 응답을 받지 못했을 때 발생하는 일종의 timeout 메시지이다.

3.3 Interleaved Requests Interaction

비 동시적인 메시지의 상호작용은 작업 중 동시에 하나이상의 요청을 포함할 수도 있다. 예를 들어, ECS가 장비로부터 여러 가지의 동작 (응답 메시지)를 요구한다. 장비는 그 행동을 수행하고, 독립적으로 결과를 보내준다. ECS가 두 개(혹은 그 이상) 의 Primary 메시지를 순서대로 보낼 때, 장비는 반드시 모든 Primary 메시지에 대해 Secondary 메시지를 보내야 한다. ECS는 응답 메시지를 결정되지 않은 순서대로 받는다. 평행적인 요구에 대한 이 메커니즘은 분산처리 수행과 스레드 안전 프로그래밍의 복잡한 요구가 아닌 비 동시적인 메시지의 자연스러운 사용이다.

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3.4 A List of Stream/Function

3.4.1 A Standard Stream/Function

S	F (Primary)	F (Secondary)	Primary Direction	Name of Function
*	0	-	ECS ◆▶ EQP	Abort Transaction
1	1	2	ECS ◆▶ EQP	Are You There Request
1	61	62	ECS ⇦ EQP	All Equipment Status Report
1	63	64	ECS ⇨ EQP	All Equipment Status Request
1	67	68	ECS ⇦ EQP	All Glass Position Data Report
1	75	76	ECS ⇦ EQP	Ionizer Status Report
2	17	18	ECS ⇦ EQP	Date and Time Request
2	25	26	ECS ◆▶ EQP	Loopback Diagnostic Request
2	31	32	ECS ⇨ EQP	Date and Time Data Send
2	61	62	ECS ⇦ EQP	Control State Change Request
2	63	64	ECS ⇦ EQP	Equipment Operation Mode Report
2	65	66	ECS ⇦ EQP	Local LOT Status Change Request
2	67	68	ECS ⇦ EQP	Cycle Stop Status Report
2	69	70	ECS ⇨ EQP	Unload Glass Data Request
2	75	76	ECS ⇦ EQP	Chemical Change Status Report
2	79	80	ECS ⇦ EQP	Glass Data Request
2	87	88	ECS ⇨ EQP	Load Glass Data Send
2	95	96	ECS ⇦ EQP	Chemical Change Enable Request
2	97	98	ECS ⇨ EQP	Chemical Change Enable Send
5	1	2	ECS ⇦ EQP	Alarm Report
6	61	62	ECS ⇦ EQP	Load Glass Data Request
6	63	64	ECS ⇦ EQP	Unload Glass Data Report
6	65	66	ECS ⇨ EQP	LOT End Data Send
6	67	68	ECS ⇦ EQP	Glass Move Request
6	69	70	ECS ⇦ EQP	Cassette Map Request
6	71	72	ECS ⇦ EQP	Sampling Slot Request
6	73	74	ECS ⇦ EQP	Glass Actual Processing Data Report
6	83	84	ECS ⇦ EQP	LOT Actual Processing Data Report
6	85	86	ECS ⇦ EQP	Dummy Lot Request
6	89	90	ECS ⇦ EQP	Rework Request
6	91	92	ECS ⇦ EQP	Equipment Status Change Report
6	93	94	ECS ⇦ EQP	Scrap Data Report

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6	95	96	ECS ⇦ EQP	OCR(VCR) Status Report
7	19	20	ECS ⇨ EQP	Current EPPD Request
7	61	62	ECS ⇨ EQP	Recipe ID Confirm Request
7	63	64	ECS ⇦ EQP	Sampling Data Report
7	67	68	ECS ⇨ EQP	Current Recipe Group List Request
7	69	70	ECS ⇨ EQP	Current Recipe ID List Request by Group
7	77	78	ECS ⇨ EQP	Recipe Body Request
9	1	-	ECS ⇦ EQP	Unrecognized Device ID
9	3	-	ECS ⇦ EQP	Unrecognized Stream Type
9	5	-	ECS ⇦ EQP	Unrecognized Function Type
9	7	-	ECS ⇦ EQP	Illegal Data
9	9	-	ECS ⇦ EQP	Transaction Timer Timeout
9	11	-	ECS ⇦ EQP	Data Too Long
9	13	-	ECS ⇦ EQP	Conversation Timeout
10	3	4	ECS ⇨ EQP	Terminal Display, Single

Note) *EQP : Equipment,*
ECS : Equipment Control System
S/F : Stream/Function

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3.4.2 A Exception Stream/Function

S	F (Primary)	F (Secondary)	Primary Direction	Name of Function
1	69	70	ECS ⇨ EQP	Change Glass Count Send
1	71	72	ECS ⇦ EQP	Remained Glass Count Request
1	73	74	ECS ⇨ EQP	Monitoring Lot Status Data Send
1	77	78	ECS ⇦ EQP	Equipment's Part Data Report
1	79	80	ECS ⇨ EQP	Buffer Lot Status Data Send
1	81	82	ECS ⇨ EQP	Port State Data Send
1	83	84	ECS ⇦ EQP	Port State Data Report
1	85	86	ECS ⇦ EQP	Port Status Change Data Report
1	87	88	ECS ⇦ EQP	Lot State Change Report
1	89	90	ECS ⇦ EQP	Mask Data Report
1	91	92	ECS ⇦ EQP	All Mask Position Data Report
2	71	72	ECS ⇦ EQP	Carrier Transfer Mode Change Report
2	77	78	ECS ⇦ EQP	Glass Call Request
2	81	82	ECS ⇦ EQP	Lost Glass Data Request
2	83	84	ECS ⇦ EQP	Injecting Glass Data Request
2	85	86	ECS ⇦ EQP	Unload Process Report
2	91	92	ECS ⇨ EQP	Batch Job (Glass Injection) End Data Send
2	93	94	ECS ⇦ EQP	Various Data Report
2	99	100	ECS ⇨ EQP	Equipment PD Down Request
6	97	98	ECS ⇦ EQP	Chamber Status Change Report
7	79	80	ECS ⇨ EQP	Equipment Constant Recipe Request
7	81	82	ECS ⇨ EQP	Multi Recipe Information Send
15	65	66	ECS ⇨ EQP	Current Recipe ID List Request
15	71	72	ECS ⇨ EQP	Recipe List Report Request
15	73	74	ECS ⇨ EQP	Lower Recipe List Report Request
15	75	76	ECS ⇨ EQP	Recipe Parameter Report Request
15	77	78	ECS ⇨ EQP	Recipe Parameter Change Send
15	79	80	ECS ⇦ EQP	Recipe Parameter Change Report
15	81	82	ECS ⇨ EQP	Operation Reference Data Send
15	83	84	ECS ⇦ EQP	Equipment Operation Reference Data Apply Report
15	95	96	ECS ⇨ EQP	Equipment Constant Recipe Parameter Request
15	97	98	ECS ⇨ EQP	Equipment Constant Recipe Change Command
15	99	100	ECS ⇦ EQP	Equipment Constant Recipe Change Report

Note) *EQP : Equipment,*
ECS : Equipment Control System
S/F : Stream/Function

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4. Software Specifications

4.1 Codes and ID

4.1.1 CST ID (Cassette ID)

CST ID는 개개의 Cassette를 나타내는 고유한 이름이다.

4.1.2 LOT ID

1 - LOT 은 Cassette에 로딩된 Glass (Panel)에 대한 논리적인 ID로서, 특별한 경우를 제외하고 1-Cassette 와 동등하다.

4.1.3 Recipe ID (PPID)

Recipe ID는 장비가 공정을 진행하기 위해 작성하는 Process Program에 대한 ID로서, 대소문자 구별 없이 사용한다.

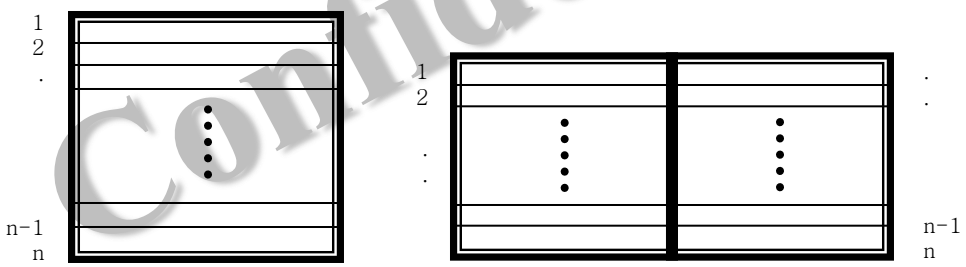
4.1.4 GLS ID (PNL ID)

이것은 각각의 Glass (Panel)에 대한 유일한 이름으로서, ECS로부터 다운로드 받거나 OCR (VCR)이 장착된 장비는 Glass (Panel)로부터 OCR (혹은 VCR)을 통해 읽어 들인다.

4.1.5 장비는 CST ID, LOT ID, GLS ID, PNL ID,RECIPEID 같은 문자열을 확인한다. 만약 ID가 공백이나 제어 불가능한 문자나 정확하지 않는 ID길이를 포함할 시 Error처리하고, 재입력을 요청한다.

4.1.6 SLOT ID

SLOT ID 는 Cassette에 Glass (Panel)가 들어가는 고유한 위치를 나타낸다. SLOT ID의 발번 규칙은 아래 그림과 같이 정면에서 보았을 때 상단에서 하단으로 1번에서부터 '1'씩 증가한다.



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4.1.7 PORT ID

PORT ID 는 각각의 Port를 구별하는 고유한 번호이다.

4.1.8 Unit ID (for EQP ↔ ECS)

장비의 개별 Unit에 매겨지는 고유한 번호이다. Unit ID는 LGD(or ECS) 에 의해 할당되고, 추후 상세항목에 대한 협의가 필요하다.

아래 표는 단순히 참고사항입니다.

<UNIT ID Rule>	
Rule	Controller ID + Unit ID (ex. 201, 202)
Explain	'Controller ID' is unique number of equipment's controller. Maybe, it can be orderly defined from '2' because Loader is defined as '1'. Controller ID is numeral and is 1 or more word. (ex. '2', '3', '11') 'Unit ID' is unique number of equipment's each partial units. It's used for identification of equipment part. Anyway it is assigned by LG Display (or ECS). Unit ID is numeral and is 2 word (ex. '01', '02').

4.1.9 Sub Unit ID

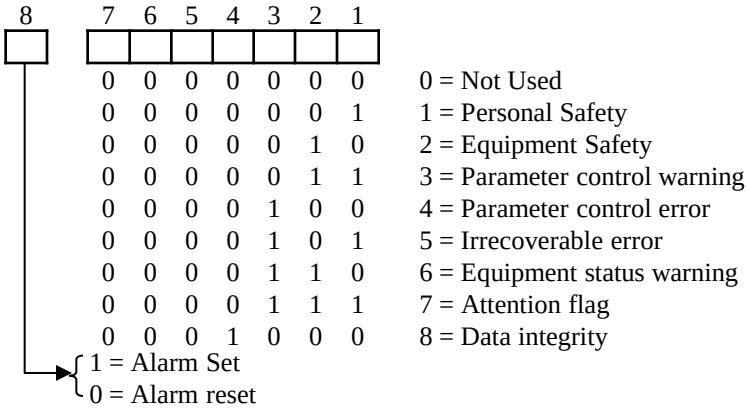
장비의 각 부분과 세부 Unit에 사용되는 고유 번호이다. 이는 Sub Unit ID 와 같을 수도 있고, 다를 수도 있다. 이에 대한 추후 상세 협의가 필요하다.

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4.1.10 Alarm Code (<ALCD>)

ALCD is divided Alarm Set/Cleared bit and Alarm Category bit.

ALCD Bit



4.1.11 Different ID Code (<DICODE>)

0 : ECS에서 받은 Glass ID와 VCR 혹은 OCR을 통해 장비로부터 읽어 들인 Glass ID 가 정확하게 일치할 때

1 : ECS에서 받은 Glass ID와 VCR 혹은 OCR을 통해 장비로부터 읽어 들인 Glass ID 가 일치하지 않을 때

2 : 각각의 Glass ID가 장비에 있는 VCR or OCR에 의해 확인되지 않았을 때

4.1.12 Scrap Type Code (<SCRPTYPE>)

Glass 또는 Panel에 Scrap 이 발생함을 나타낸다.

- G : Glass에 Scrap 발생.
- P : Panel에 Scrap 발생.

4.1.13 Port Transfer State Code(<PORTSTATE>)

- LR : Load 요청
- LC : Load 완료
- UR : Unload 요청
- UC : Unload 완료

4.1.14 Judgment Information(<PNLIF>,<SUBMDLIF>,<GLSJUDGE>,<LOTJUDGE>)

장비에서의 각각의 판정의 사용과 설정치는 LG Display와 협의해야 한다.

<PNLIF> : Panel 판정 정보 . Panel에 대한 판정은 영문 한 글자로 표현되며, 이는 Good / NG / Repair / Rework 을 의미한다.

<SUBMDLIF> : Sub Panel 판정 정보 . Sub Panel에 대한 판정은 영문 한 글자로 표현되며, 이는 Good / NG / Repair / Rework을 의미한다.

<GLSJUDGE> : Glass Judge. 이는 한 글자 또는 두 글자의 영문으로 표현되며, 이는 Good / NG / Rework을 의미한다.

<LOTJUDGE> : LOT Judge. 이는 한 글자 또는 두 글자의 영문으로 표현되며, 이는 Good / NG / Rework / 1 Step Skip / 2 Step Skip / Hold를 의미한다.

* 장비는 반드시 최소한 16개 이상의 (<PNLIF>, <SUBPNLIF>, <GLSJUDGE>,<LOTJUDGE>) judgment에 대해 응답할 수 있어야 한다.

4.1.15 Glass Key Code (<KEYID>)

Glass 데이터에 대한 고유한 값이다. 이는 동일 LOT의 모든 Glass는 반드시 같은 값을 가진다.

- 형태 : <LOTID(14Byte)_YYYYMMDDhhmmss(14Byte)>
- 예 : <7EA001000000_20050301115623>

4.1.16 Glass Order (<GLSODR>)

0 : 첫 번째 Glass

1 : 첫 번째 Glass와 마지막 Glass를 제외한 모든 Glass

2 : 마지막 glass (이 경우 , 이 값은 Scrap 또는 LOT Abort 에 대해서는 전달되지 않는다)

3 : LOT에 Glass가 한 장 밖에 없을 경우, 유일한 Glass를 나타낸다.

4.1.17 Cassette Port Status (<CSTREADY>)

0 : Port에 Cassette가 존재하지 않을 때.

1 : Port에 Cassette가 존재할 때.

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4.1.18 Stage ID (<STAGEID>)
장비에서 Glass가 투입 / 취출되는 영역에 대한 고유한 번호를 나타낸다.

4.1.19 Naming rule of Actual Processing Data (APD)

4.1.19.1 Numeric Data

4.1.19.1.1 Data가 숫자일지라도, 프로세스는 ASCII로 보고해야 한다.

4.1.19.1.2 부호가 처음에 위치하고, 좌측에서 우측으로 채워나가고, 나머지는 공백으로 채운다.
만약 첫 번째 위치하는 부호가 공백일 때는 무효로 처리한다.
(예외 : "[]"는 문자 리스트의 나눔, "{}"는 생략가능, " "은 공백)
-고정소수점 : [{+}-][string]{[소수점][string]}[space…]
-가변: [{+}-][string]{[소수점][string]}[E][{+}-][space…]
예) "849_____"
"-3.14E-10___"

4.1.19.2 Character String Data

부호는 좌측에서 우측으로 채워나가고, 나머지는 공백으로 한다. 만약 첫번째 부호가 공백이면 무효이다.
예) "ABCDEF_____"

4.1.20 Part Number (<PARTNUM>)
LOT 의 Model 종류를 말한다. 해당 Data 사용시 LG Display와 협의한다.

4.1.21 Prod Type (<PRODTYPE>)
공정을 시작하려는 LOT의 속성을 말한다. 장비는 PRODTYPE 을 참조하여 공정을 변경하거나 Glass 의 장비 내 이동을 설정할 수 있다. 자세한 사항에 대하여는 LG Display와 협의한다

4.1.22 Attribute (<ATTRIBUTE>)
공정을 시작하려는 LOT의 특이 사항 Data이다. 해당 LOT의 Model Type 또는 공정 관련한 특이 Data를 전송한다. 장비는 ATTRIBUTE 을 참조하여 공정을 변경하거나 Glass 의 장비 내 이동을 설정할 수 있다

4.1.23 Recipe Parameter Name (< RPARMNAME>)
Recipe Parameter Name을 의미한다. Parameter Name은 RPARMVAL과 짝을 이루며, RPARMNAME은 PPAMRVAL의 Name이다. 항목의 선정은 LG Display와 협의하며, 꼭 영문 대문자와 '_'만을 사용한다

4.1.24 Recipe Parameter Value (<RPARMVAL>)
RPARMNAME에 대한 Recipe Parameter Value 이다. List내에 배치되는 순서와 관계 없이 짝을 이루는 RPARMNAME에 해당하는 Value여야 한다.

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4.2 The conservation of ECS communication history

(Log - Interlock Communication & SECS Message)

장비는 적어도 최근 3일 간의 ECS 통신 데이터와 Interlock 통신 데이터를 저장할 능력을 가져야 한다.
유효기간이 만료되었을 시 Log는 자동으로 삭제된다.

4.3 Change of HSMS communication parameter.

HSMS 통신 parameter 는 장비 내에서 사용자에 의해 언제라도 변경이 가능하다.

4.4 Manual Correction of Glass Information

예기치 않게 ECS로부터 받은 Glass 정보가 사라졌을 때, 장비는 반드시 ECS로 수동으로 Glass 정보 (e.g. GLS ID, Original Port ID, Original Slot ID, 등 …) 를 입력하여 보낼 수 있어야 한다.

4.5 Injection Glasses of Multi LOTs

진행중인 장비에 복수 개의 LOT을 투입할 수 있어야 한다.하지만 Glass는 순차적으로 투입해야 한다.
장비가 한 LOT의 Glass를 처리하고 있을 때, 다른 LOT의 Glass를 장비에 투입할 수 있어야 한다. 모든 형태의 장비 (Chamber or 다른 형태) 가 가능해야 한다.

4.6 Caution about Glass Order(<GLSODR>)

<GLSODR> = 2일 떄 , 이 값(= 2)은 Glass Scrap 혹은 LOT Abort 때문에 전달되지 않을 수도 있다.
만약 장비에서 어떠한 GLSODR 정보를 사용하여 무언가 할 가능성이 있다면, 장비업체는 LG CNS와 협의해야 한다.

4.7 Using of Glass Data Request (S2F79, S2F80)

이 stream function 은 수동사용 상황, 장비에서 Glass 정보를 상실했을 경우에 사용된다. 하지만 이때 장비는 'Glass ID or Glass Code' (또는 에러 상황에서 작업자의 Glass ID 또는 Glass Code 입력)를 가지고 있다.
만약 장비가 이 stream function을 사용하면, 수동 조작 하에서 S2F79를 요청함에 있어 사용자 접속환경을 제공한다.(예. 작업자가 GLSID또는 GLSCODE를 UI화면에서 입력하고, 작업자의 확인 버튼.. 등)

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5. Mode Definition

기본적으로 장비는 각각의 운영 모드로 변경하는 기능을 가지고 있다. 필요한 경우가 발생하면, ECS와 장비 업체간에 협의를 통해 ECS는 이를 할 수 있다.

Communication State	Control state		Operation Mode	Description
Communicating	Online	Remote	장비의 특징적인 운영 모드 *NOTE) It needs conference with ECS later.*	장비로의 Glass 투입은 ECS가 결정 한다
		Local		장비는 Glass 가 장비로 투입될지를 결정하고 ECS로부터 Glass 를 요청한다
	Offline		N/A	이 상태에서 장비는 S2F25를 제외하고 ECS로부터 받는 모든 최초의 메시지에 대해서 S*F0로 응답한다. 그리고 ECS는 장비의 S1F1/ S2F25/ S2F61를 2차 메시지로 응답한다. 또한 장비는 S1F1/ S2F25/ S2F61를 제외한 어떤 1차 메시지를 보내서는 안 된다. ECS는 S2F25를 제외한 어떤 메시지도 보낼 수 없다. 장비가 Offline 상태일 때, 장비는 반드시 ECS로 부터 받은 이전 진행상태의 데이터를 가지고 있어야 한다.
Not communicating	N/A			

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5.1 Communicating

물리적으로 통신이 이루어진 상태

5.2 Not communicating

물리적으로 통신이 이루어지지 않은 상태

5.3 Online – Remote

HSMS 통신이 성립됨. 이 상태에서는 ECS Glass 처리에 대한 권한을 가진다.
즉, 장비가 Glass 투입을 요청하면, ECS는 적절한 시간에 Glass를 장비에 투입한다.
Online – Remote 요청은 언제나 가능하다.

5.4 Online – Local

HSMS 통신이 성립됨. 이 경우, 장비는 Glass 처리에 대한 권한을 가진다.
즉, 장비가 Glass의 진행에 대한 결정을 하고 , ECS로부터 Glass를 요청 받는다 .
Online – Local 요청은 언제나 가능하다.

5.5 Offline

통신이 물리적으로 연결된 상태. 하지만 ECS와 장비 양쪽모두 목적에 맞는 메시지를 보내지 못한다.
이 경우, ECS와 장비는 위 테이블에서 정의한 메시지를 제외한 모든 메시지에 대해 처리 실패를 수행한다.
Offline 요청은 언제나 가능하다.

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6. Required Items of the Counter Specification Document

대응 사양서는 최소한 다음과 같은 사항을 포함하여야 한다.

6.1 Control unit function을 포함한 장비 구성도

☞ 장비 내부 제어 구성도와 Subunit구성 및 구성도

6.2 전기적 interface 특성과 통신규약 (communication protocol)

6.3 지원되는 Stream Function 목록과 Message Block의 세부사항

6.4 Stream Function내의 각 Data(혹은 code)에 대한 기술

☞ 언제, 어떻게, 무엇을, 수서에 어떤 영향을 미치는가 등등..

6.5 ECS와 통신시, 상태, 모드, 시점 등에 대한 세부적인 기술

☞ Alarm 발생 / 해제 시점 기술

☞ 장비 상태에 대한 정의와 변경 시점

☞ Glass loading / unloading에 대한 정의와 요청 시점

6.6 Transaction Sequence에 대한 세부적인 기술

다음 순서에 대한 설명을 위해 적절한 차트가 필요하다. Message Block 안의 각 단계의 코드 상태를 포함한다.

☞ 최초의 시작 순서

☞ Glass 작업 순서

☞ LOT End 진행 (Last Glass 진행) 순서

☞ 각 운영 모드의 순서

note) 각 Sequence 상세 묘사에는 Loader와 장비간의 하드웨어 인터페이스인 Interlock diagram을 포함한다.

6.7 장비와 각 unit별(ALID, ALCD, ALTX) Alarm ID 표

6.8 A.P.D / Recipe Body list

☞ 보고 data (Data에 대한 설명, 설정치, 이름…)

☞ Data 단위 (℃, Å, ㎖ etc)

☞ 각 Data의 허용치(범위), 보고된 각각의 Data의 명확한 기준(약속) (Ex : -10↔100℃)

☞ 각 Data의 보고 형식 (Ex : 10E-3 or 10.000 etc)

☞ Glass unit과 LOT unit의 A.P.D format이 같다 하더라도 이 둘은 구분할 수 있어야 한다.

6.9 RECIPE ID 세부사항(명명규칙, 최대 / 최소 등록 개수 etc)

6.10 장비 통신 사용자 환경

6.11 각 운영 모드의 설명

☞ 동의에 의해 수용된 운영 모드의 세부사항

☞ 만약 운영상에 제약이 있다면, 이는 기술되어야 한다.

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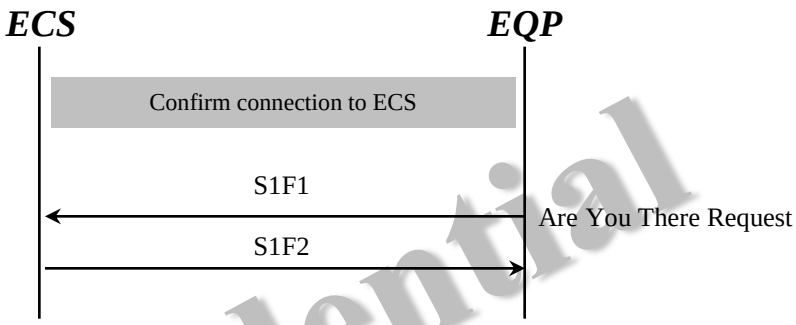
7. Scenario

이번 단락에서는 각 경우의 일반적인 순서를 보여준다. 이는 LG의 필요에 의해 더해지거나 변경 가능하다.

7.1 Control State and Equipment Operation Mode Change

7.1.1 Are You There Request

- 장비의 운영이 시작되고 초기화 될 때, 장비는 ECS와 연결한 후 ECS로 'Are You There Request' 메시지를 보내야 한다.

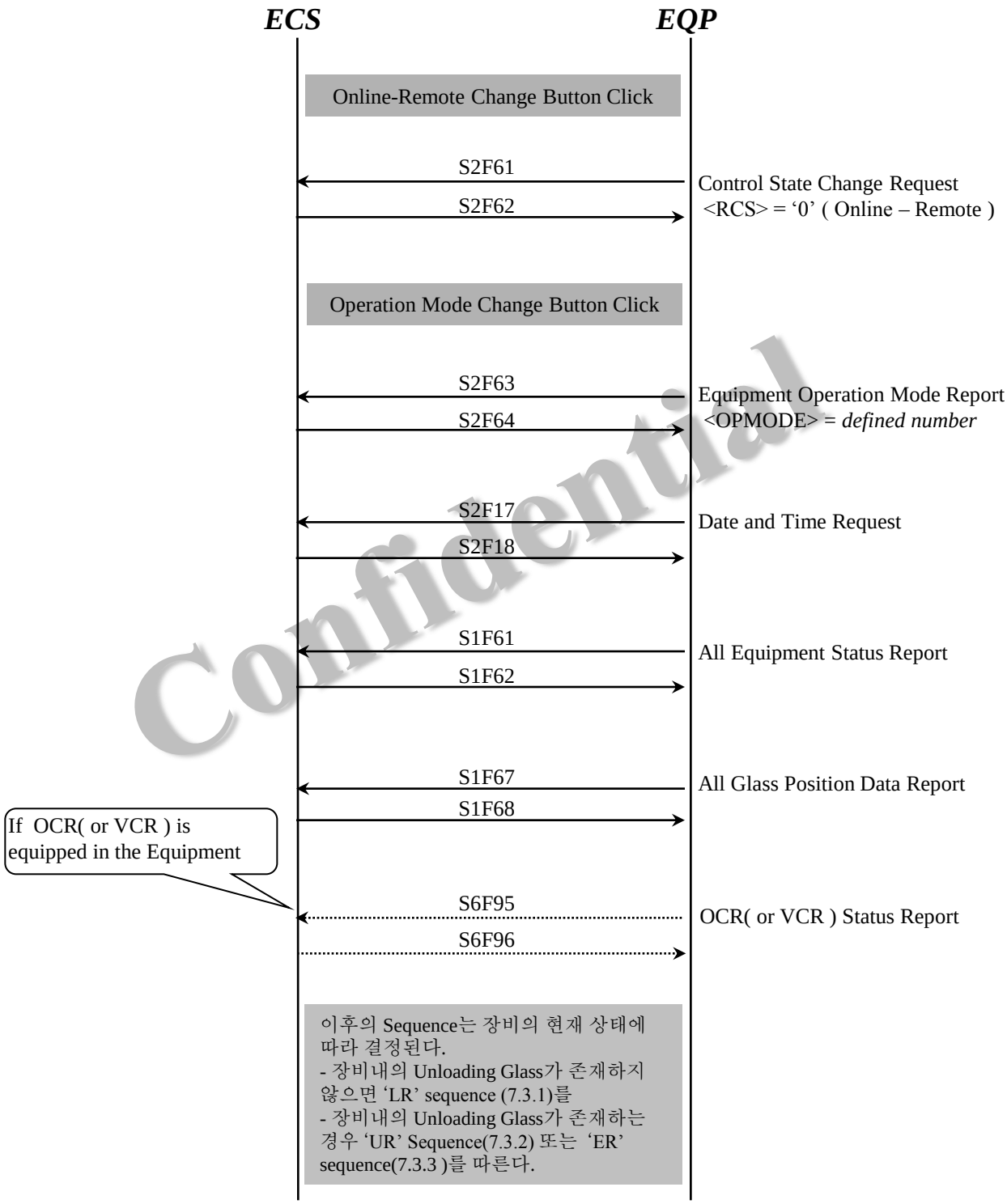


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7.1.2 Offline → Online Remote

- Offline 에서 Online-Remote 로 상태 변경은 아래의 Scenario를 따른다.
- 만약 응답이 없거나 통신 에러가 발생하면, Offline 상태를 계속 유지된다

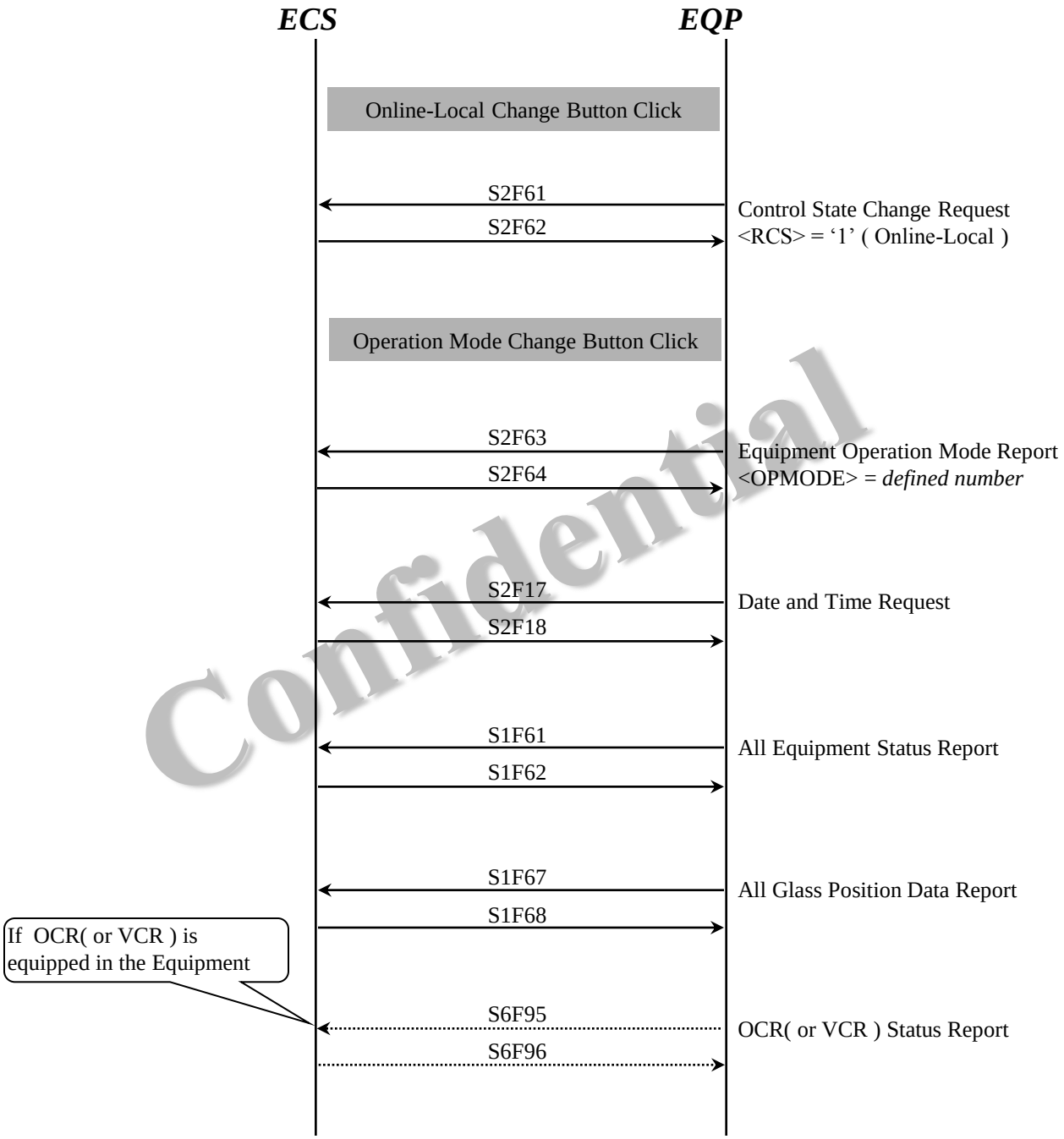


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7.1.3 Offline → Online Local

- Offline 에서 Online Local 로 상태 변경은 아래의 Scenario를 따른다.
- 만약 응답이 없거나 통신 에러가 발생하면, Offline 상태를 계속 유지된다.

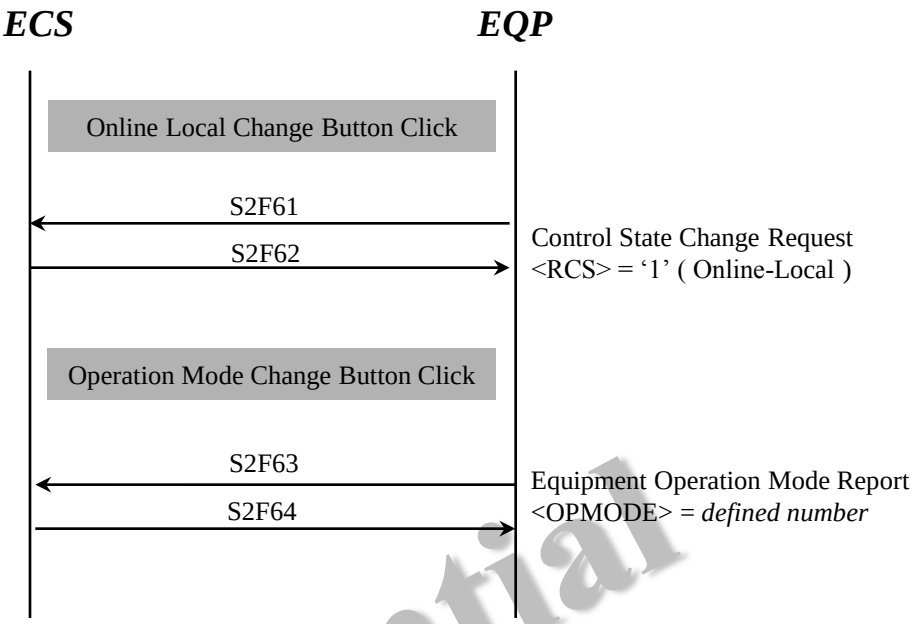


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7.1.4 Online Remote → Online Local

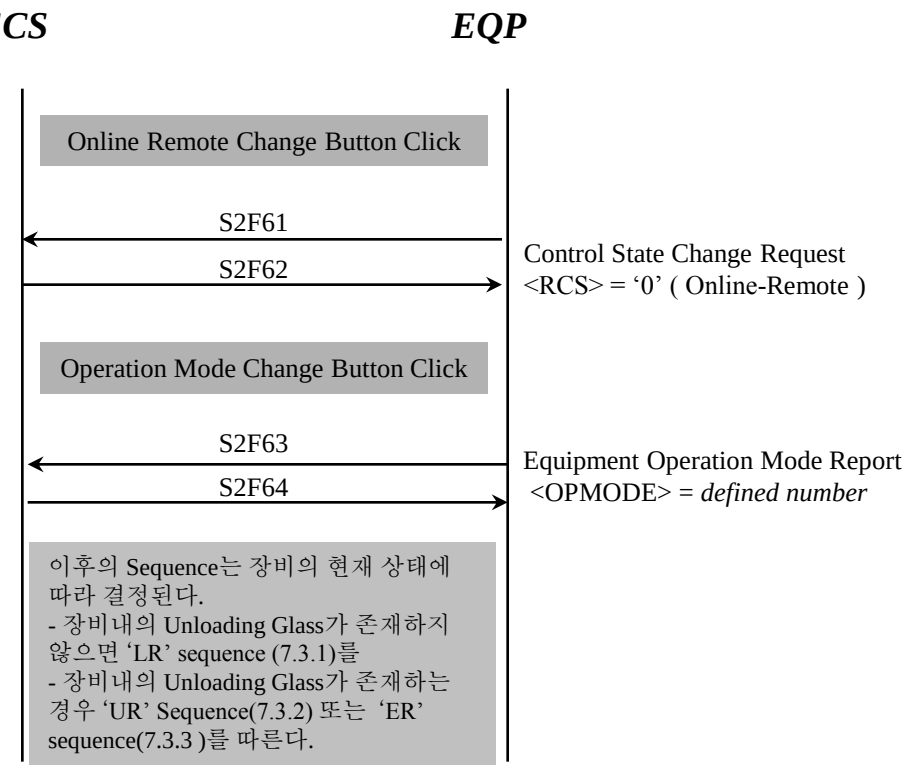
- Online Remote 에서 Online Local 로 상태 변경은 아래의 Scenario를 따른다.



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7.1.5 Online Local → Online Remote

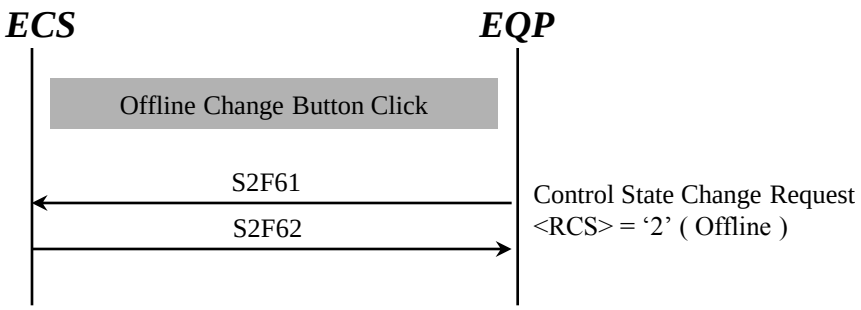
- Online Local 에서 Online Remote 로 상태 변경은 아래의 Scenario를 따른다.



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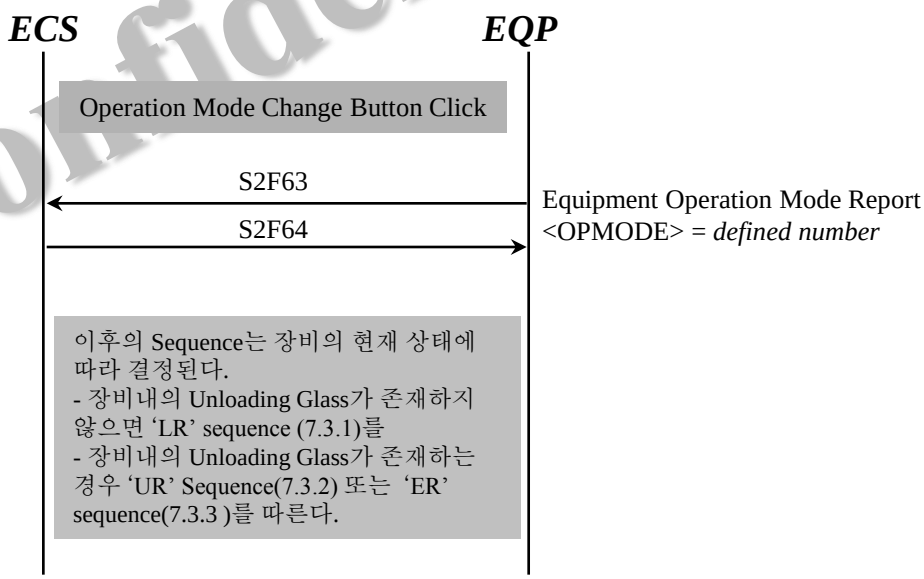
7.1.6 Online(Remote or Local) → Offline

- Online Remote에서 Offline으로 상태 변경은 아래의 Scenario를 따른다.
- Online에서 Offline으로 전환 시, 응답이 없거나 통신에러가 발생하더라도 off-line으로 전환된다.



7.1.7 Equipment Operation Mode Change (Under same Control State)

- 같은 통제 상태 하에서 운영모드의 변경은 다음 Scenario를 따른다. 이 경우, 장비는 통제 상태 보고를 보내지 않는다.
- 만약 응답이 없거나 통신 에러가 발생하면, 현재의 모드를 유지한다.

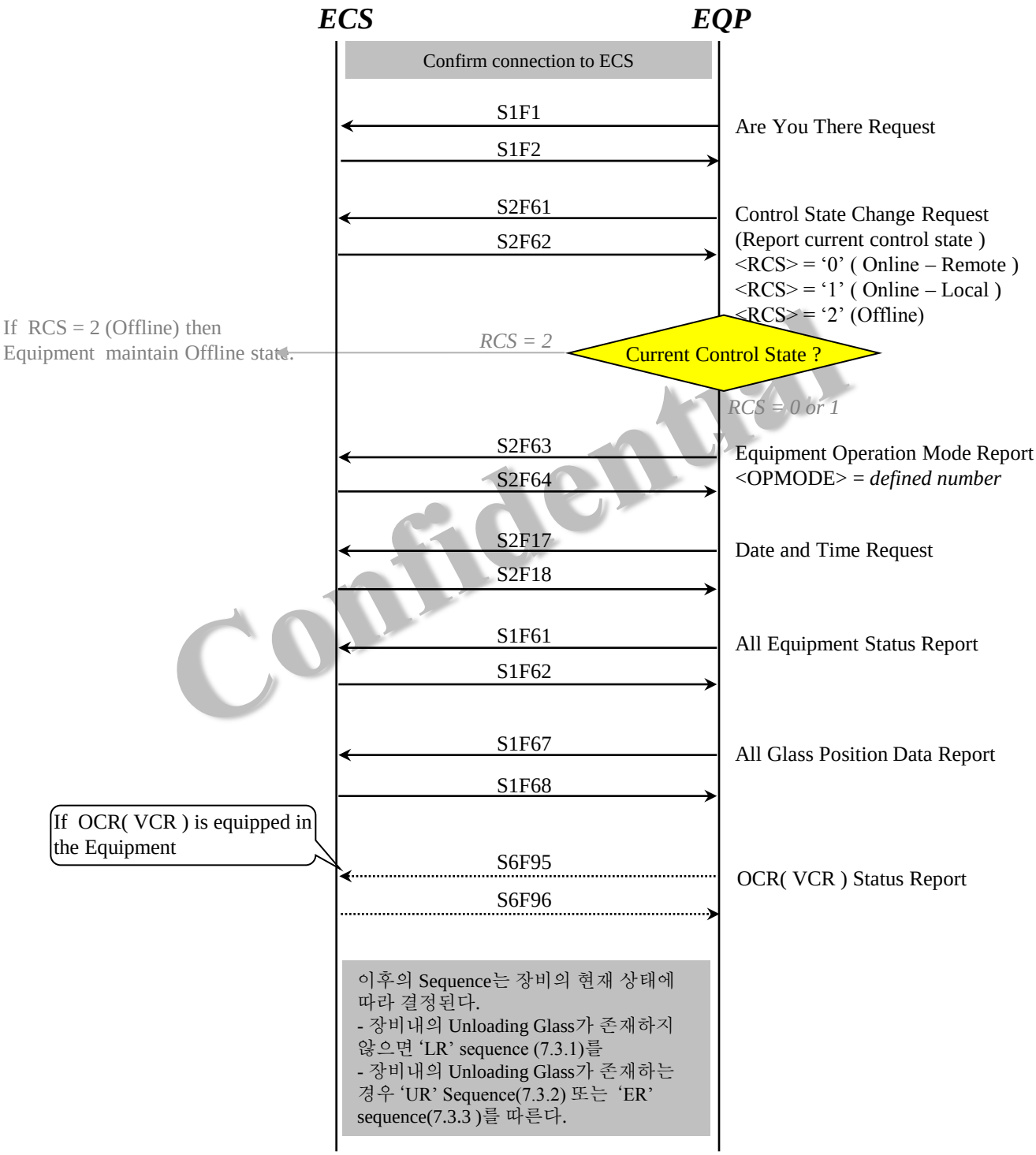


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7.1.8 ECS Restart

- ECS 재기동시, 장비와 ECS간의 통제 상태는 다음 Scenario에 의해 변경된다.

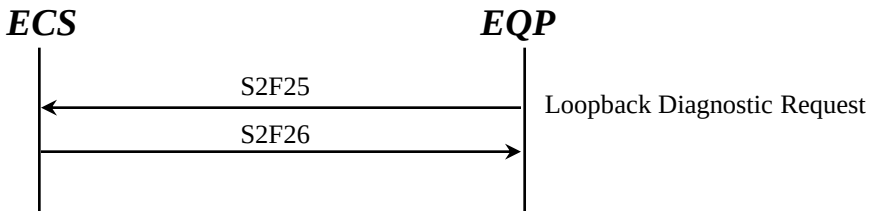


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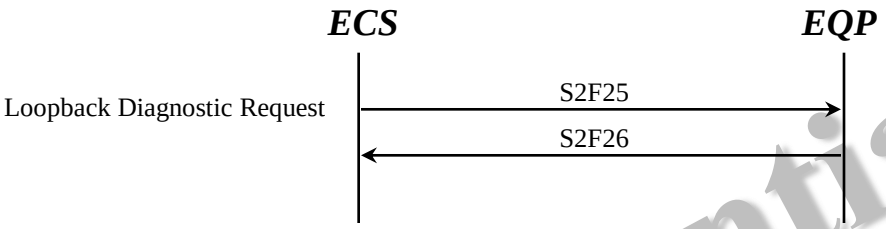
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7.2 Loop Back Diagnostics

7.2.1 Equipment initiates Loop Back Test (ECS ← EQP)



7.2.2 ECS initiates Loop Back Test (ECS → EQP)



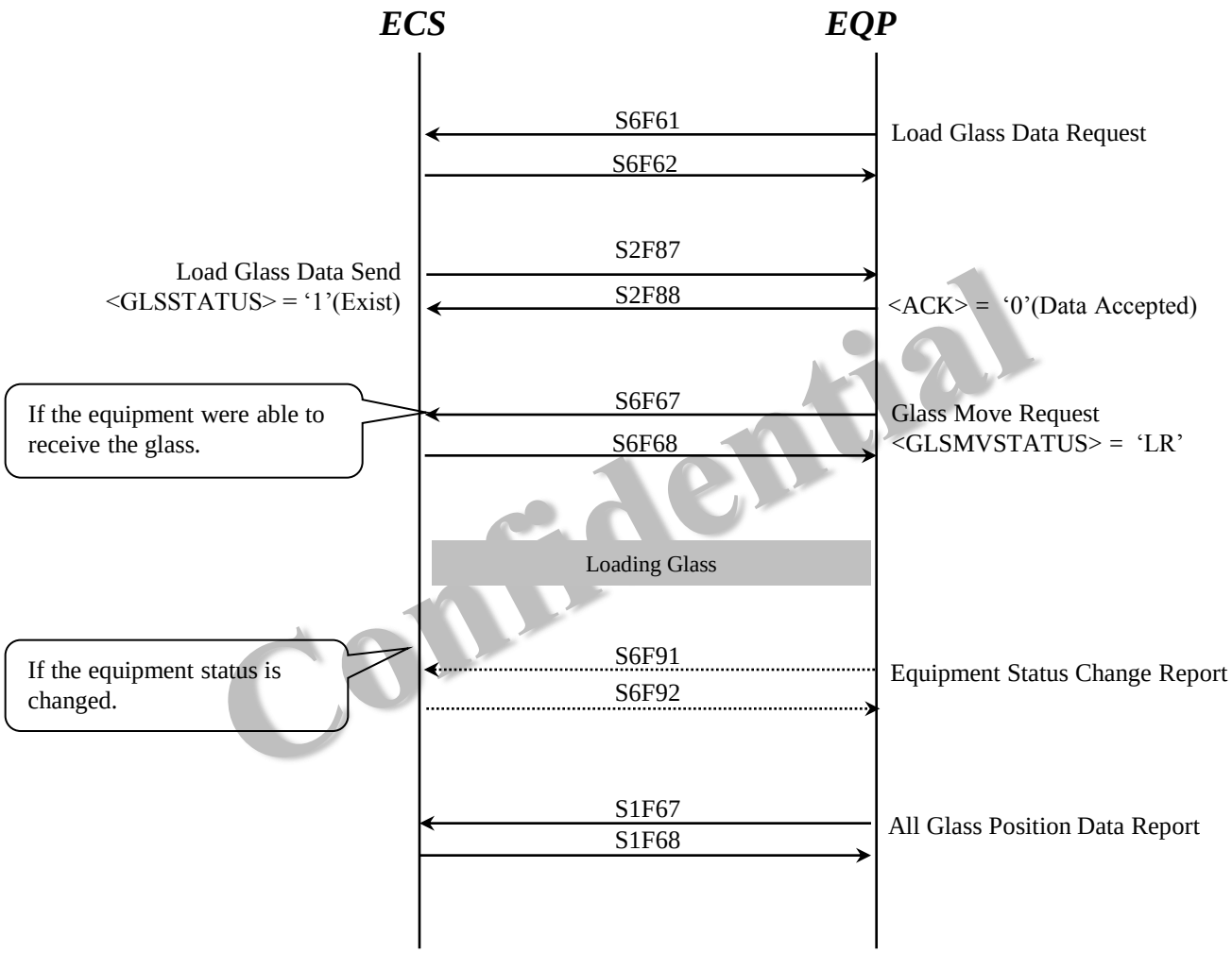
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7.3 Glass Transfer Sequence

7.3.1 Glass Move State ‘LR’ Request(Load Request)
‘Load Glass Data Request(S6F61)’는 장비에서 실제 Glass Load (‘LR’)가 발생하기 5 ~ 10초 이전에 보낼 수 있어야 한다.
장비의 특성에 따라 더 많은 시간이 요구 될 때는 LG측과 협의가 필요하다.

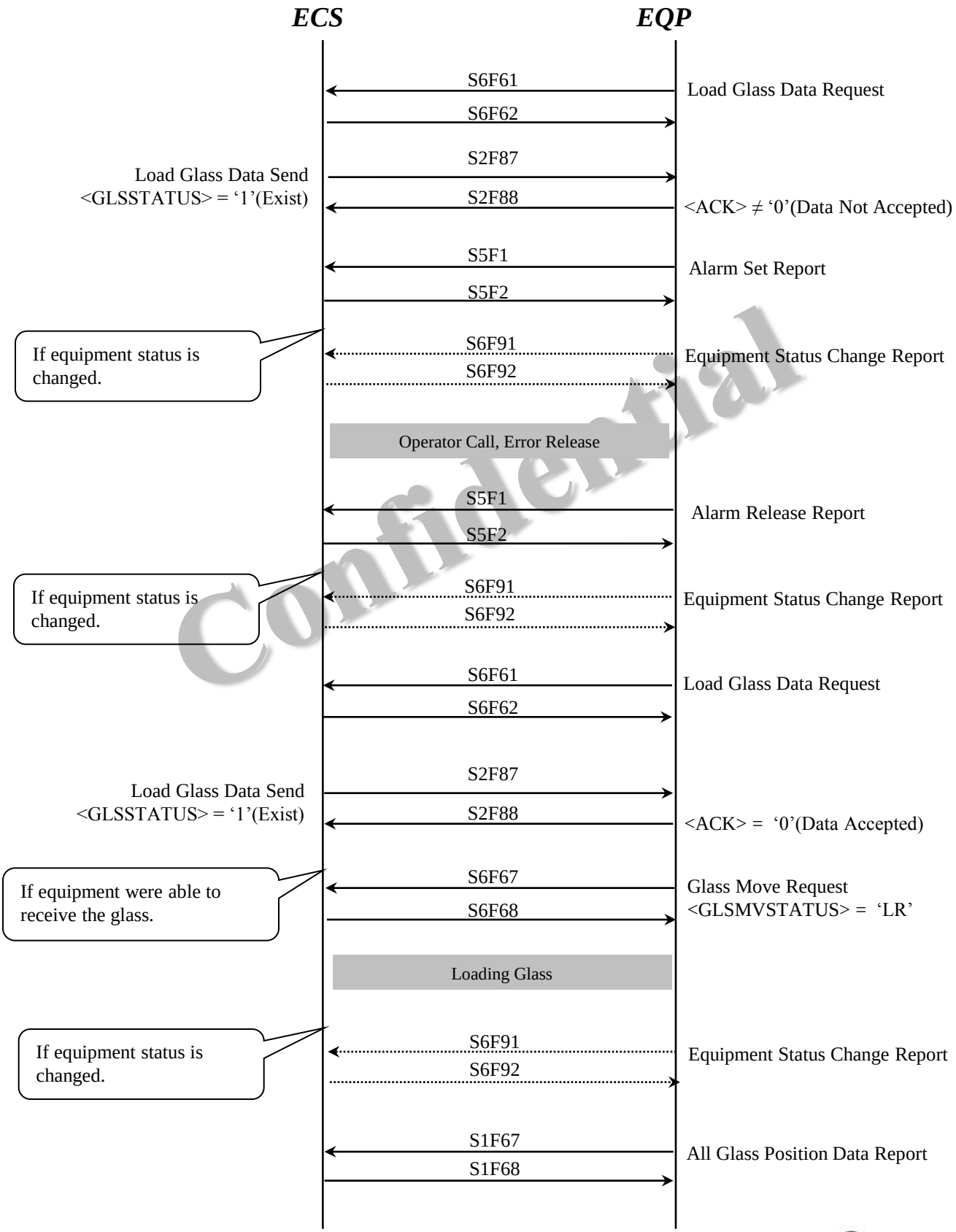
7.3.1.1 Load Glass Data Accepted



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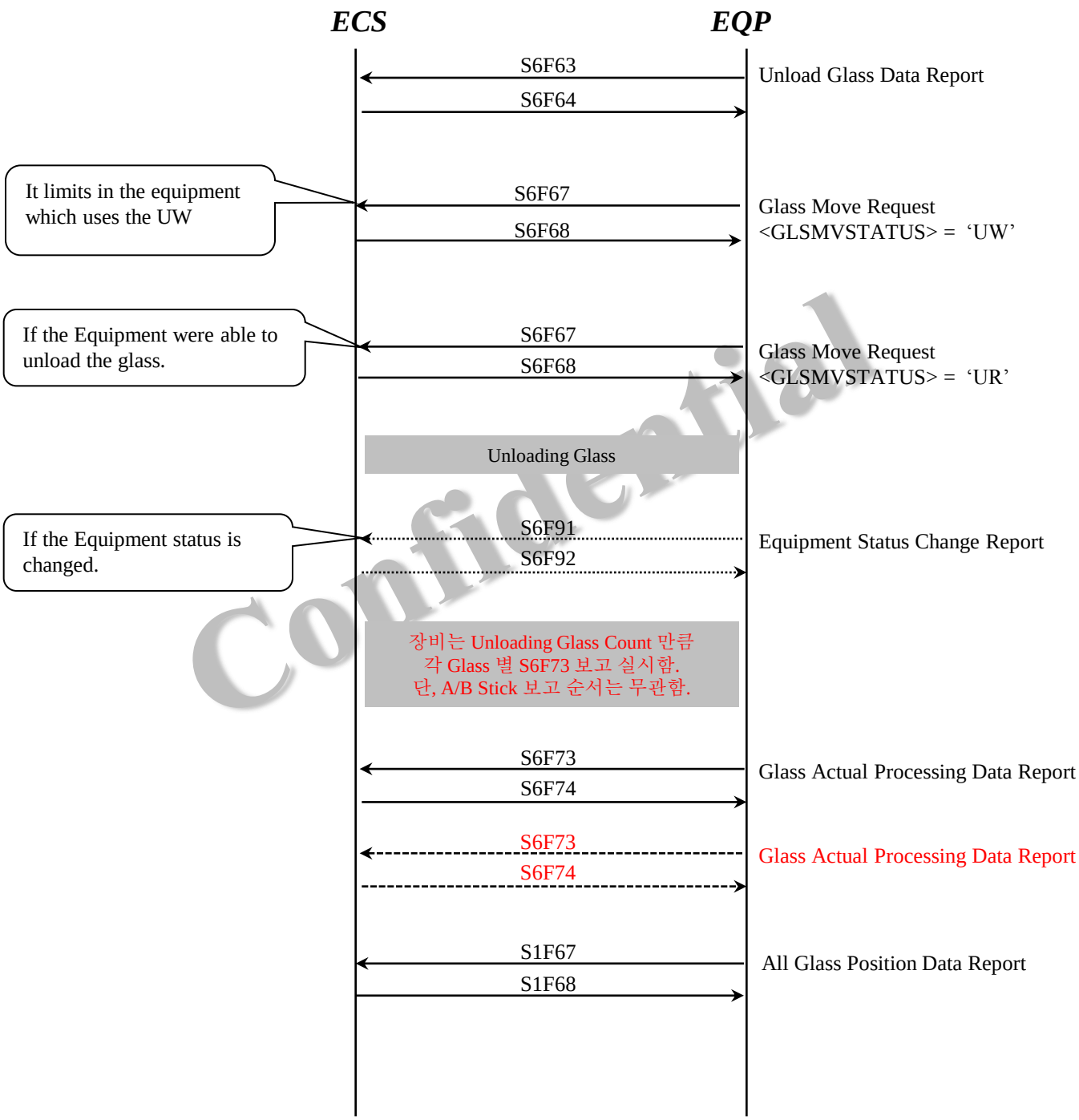
7.3.1.2 Load Glass Data Not Accepted



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7.3.2 Glass Move State 'UR' Request(Unload Request) - Only Unload Stage Type
'UW(S6F67)'는 장비에서 실제 Glass Unload ('UR')가 발생하기 5 ~ 10초 이전에 보낼 수 있어야 한다.
장비의 특성에 따라 더 많은 시간이 요구 될 때는 LG측과 협의가 필요하다.



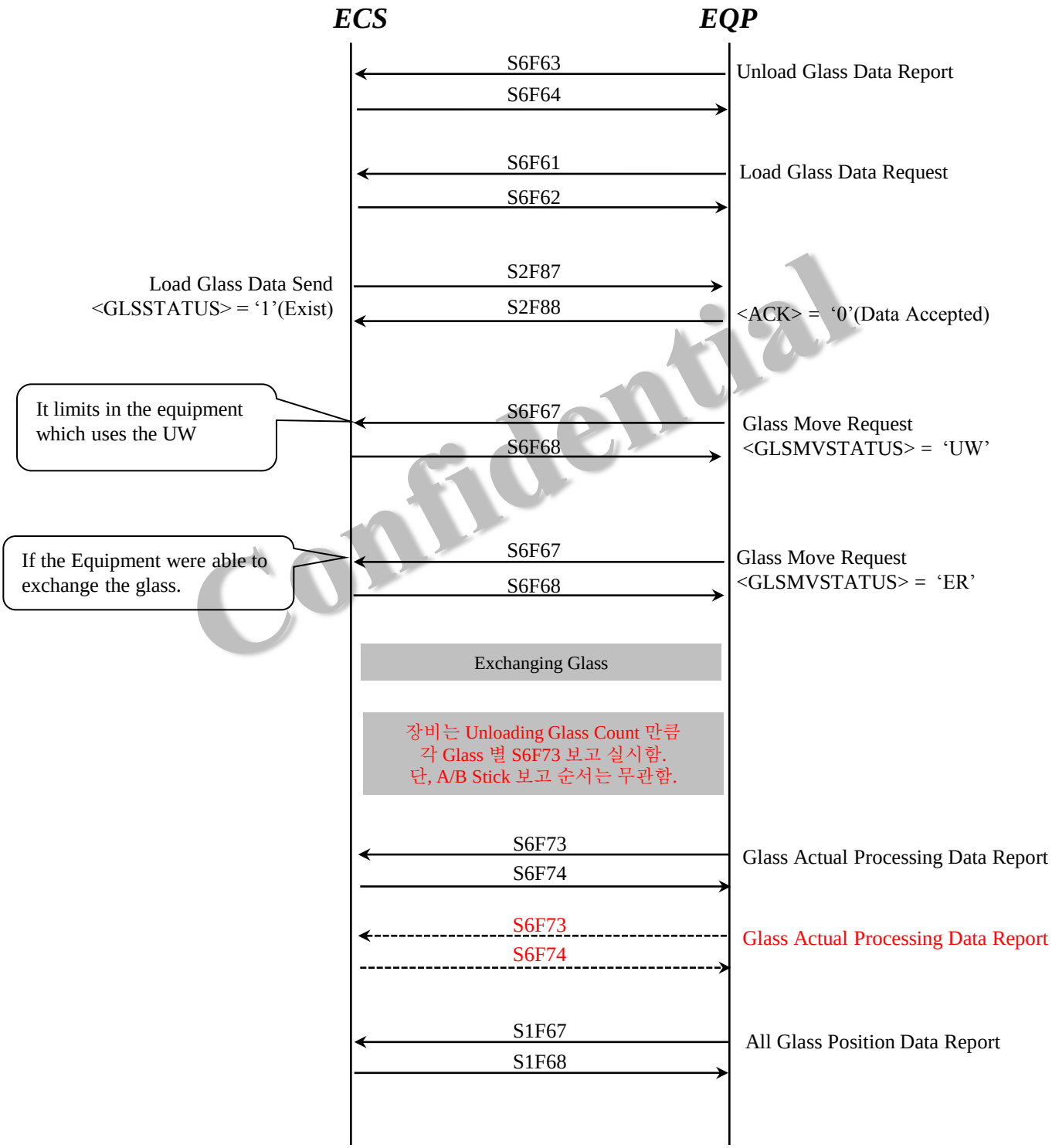
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7.3.3 Glass Move State 'ER' Request

'UW(S6F67)'는 장비에서 실제 Glass Exchange ('ER')가 발생하기 5 ~ 10 초 이전에 보낼 수 있어야 한다.
장비의 특성에 따라 더 많은 시간이 요구 될 때는 LG측과 협의가 필요하다.

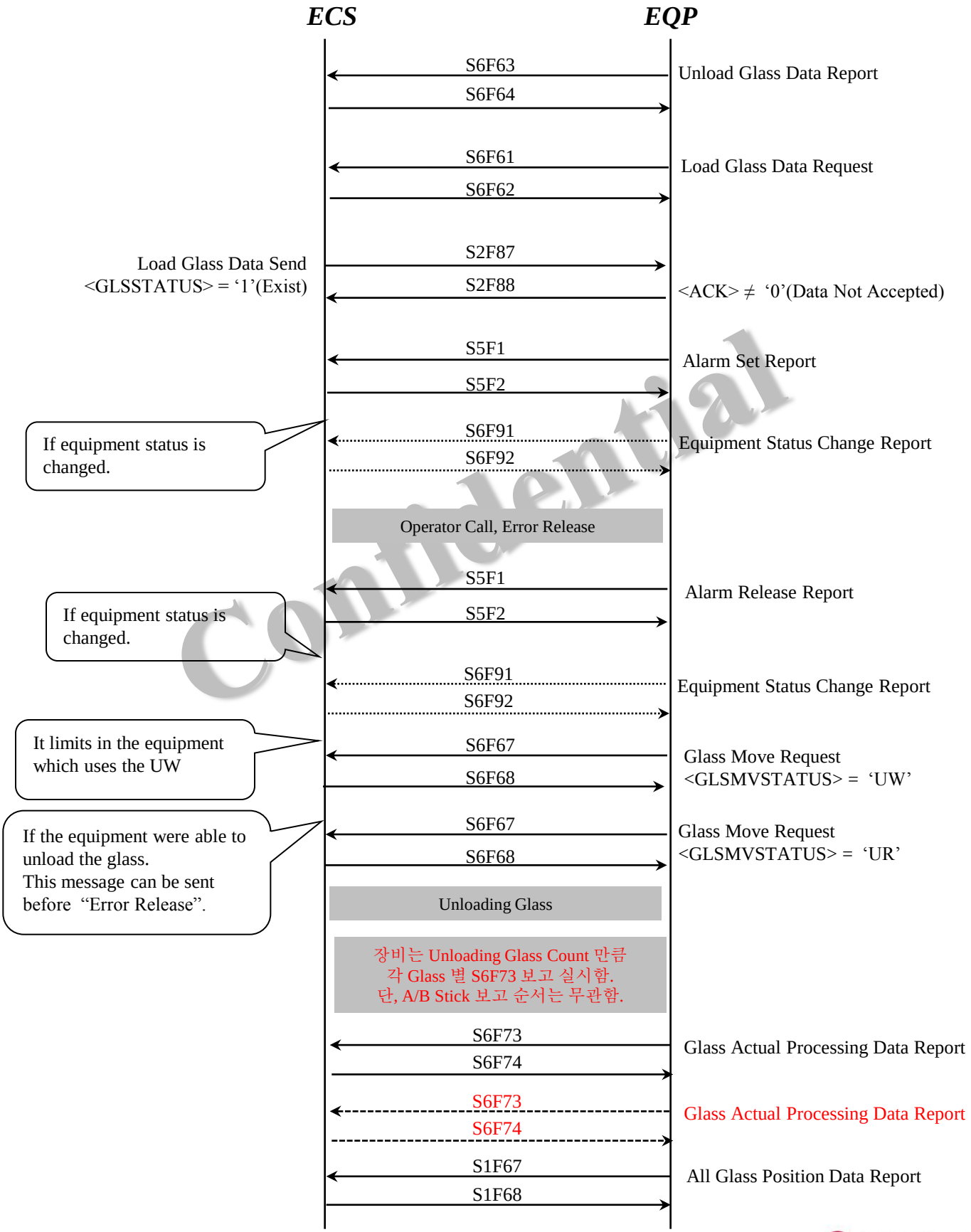
7.3.3.1 Load Glass Data Accepted



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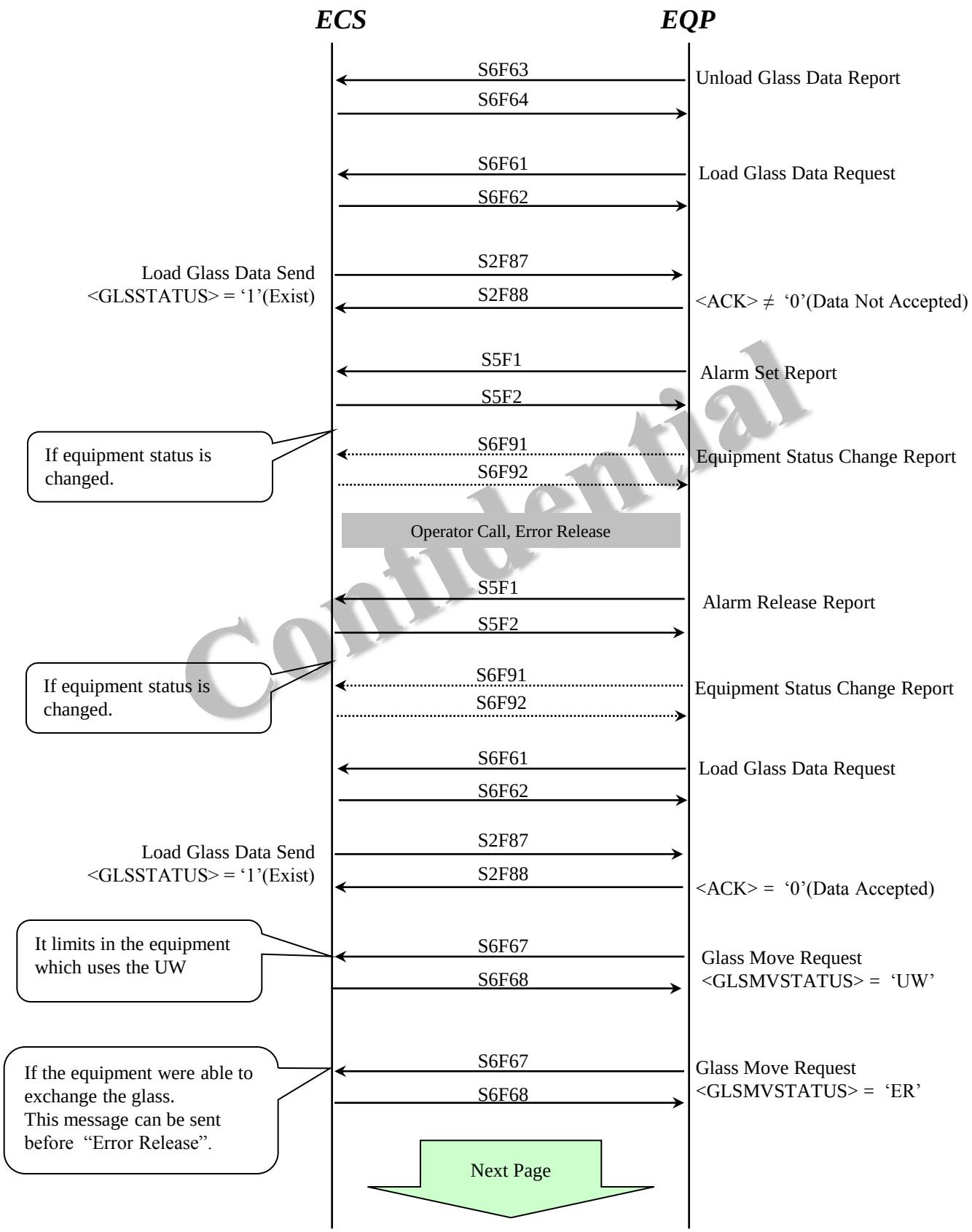
7.3.3.2 Load Glass Data Not Accepted – Glass Unloading Sequence after data not accept error release



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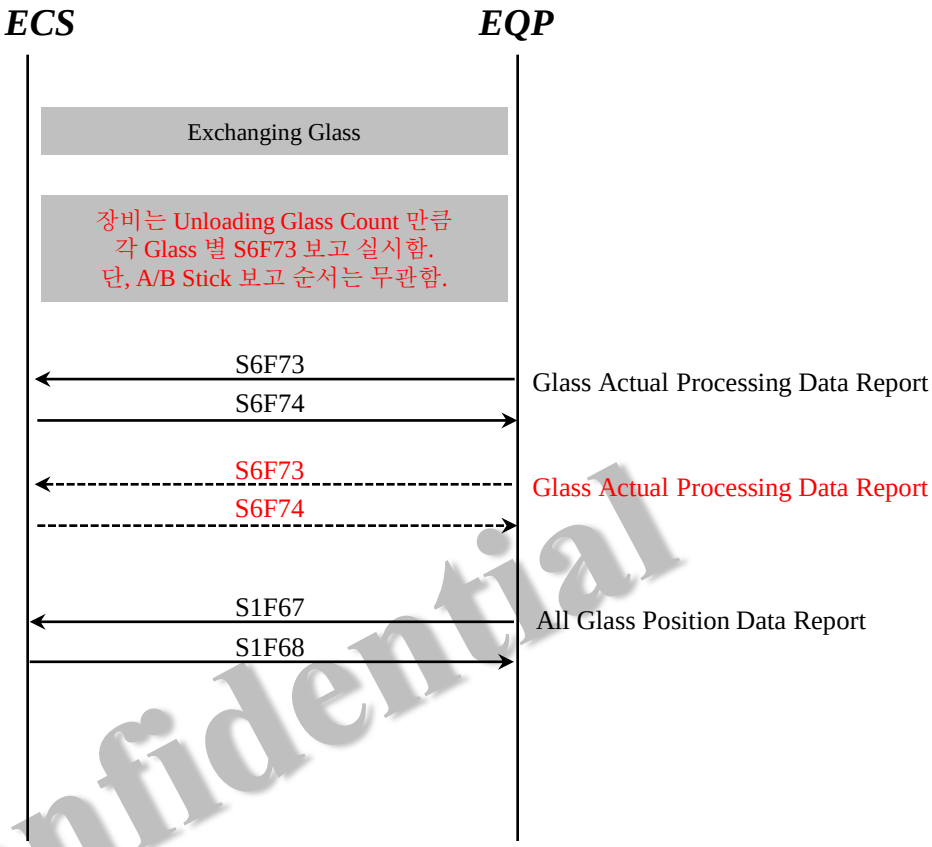
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7.3.3.3 Load Glass Data Not Accepted – Glass Exchange Sequence after data not accept error release



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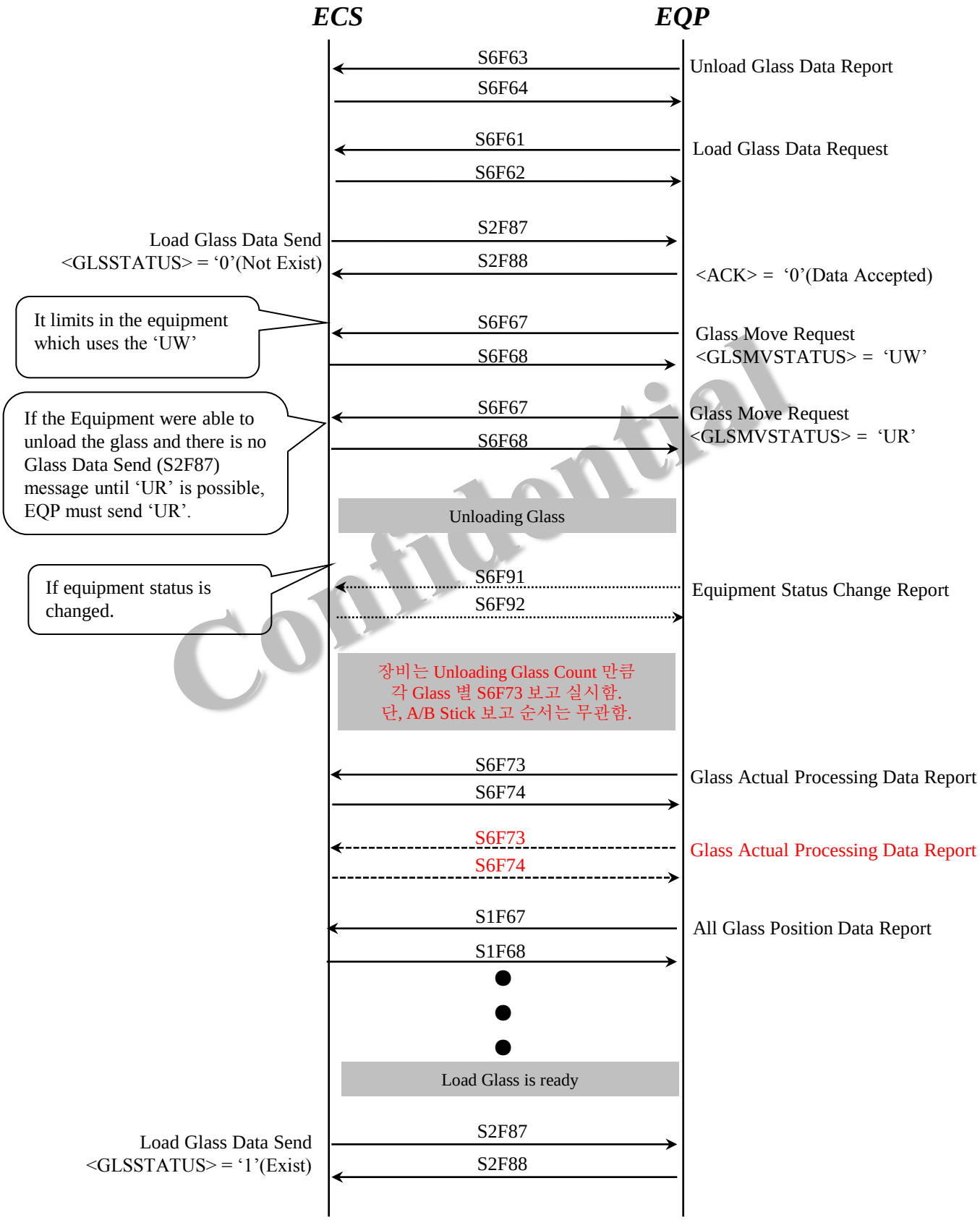
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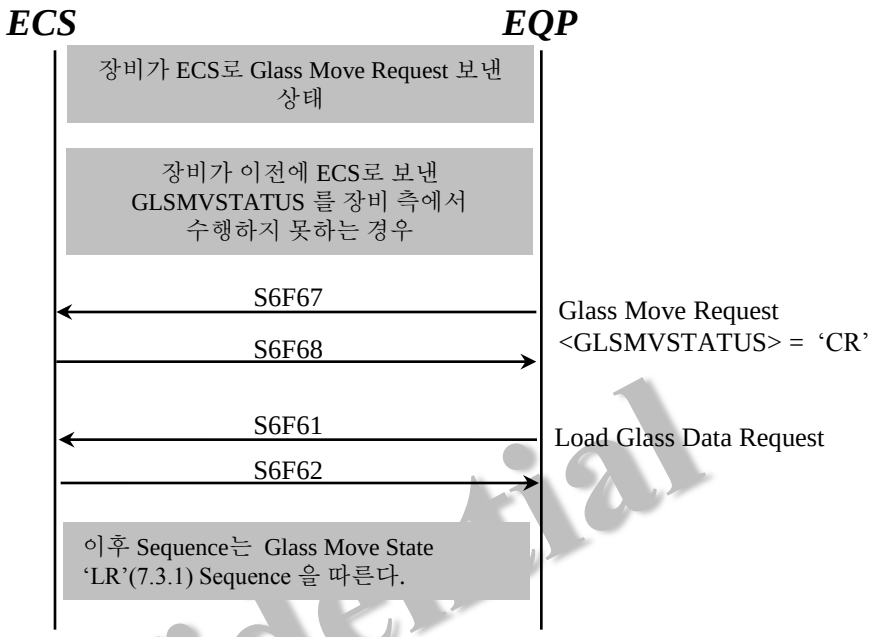
7.3.3.4 Load Glass Not Exist



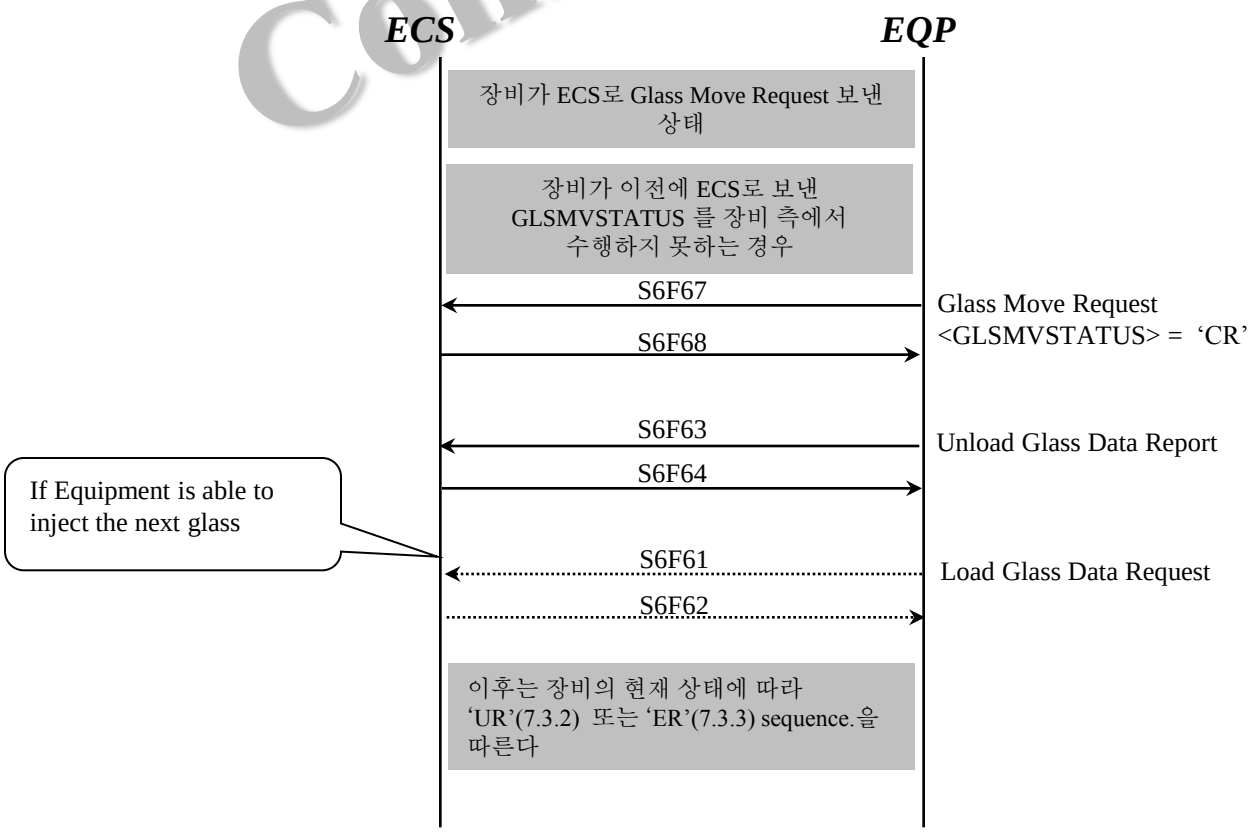
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7.3.4 Glass Move State ‘CR’ Request(Cancel Request)
- 세부 사항은 LG CNS, LG Display와의 협의가 필요하다.

7.3.4.1 Case of Unload Glass Not Exist



7.3.4.2 Case of Unload Glass Exist



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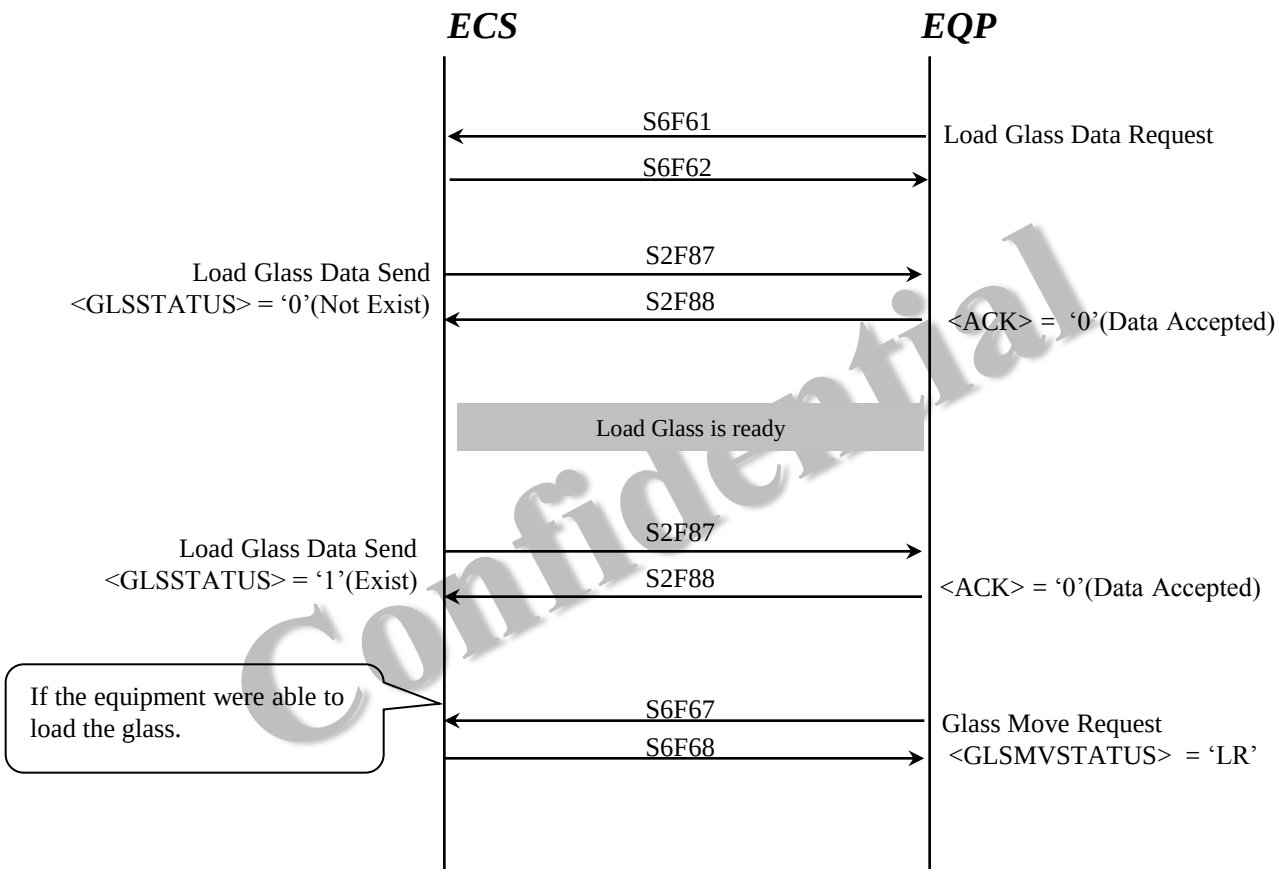
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7.3.5 Load Glass Data Change Case

- 장비로 투입될 Glass의 정보가 변경되었을 때, ECS는 장비로 이 순서를 보낸다.
- Glass data change message는 언제라도 보내질 수 있다.
- 장비는 ECS가 전송하는 S2F87 수신 후 Glass 존재 여부에 따라 S6F67 송신 여부를 판단한다.
- 세부 사항은 LG CNS, LG Display와의 협의가 필요하다.

7.3.5.1 Change form Not Exist to Exist

1) Unload Glass Not Exist

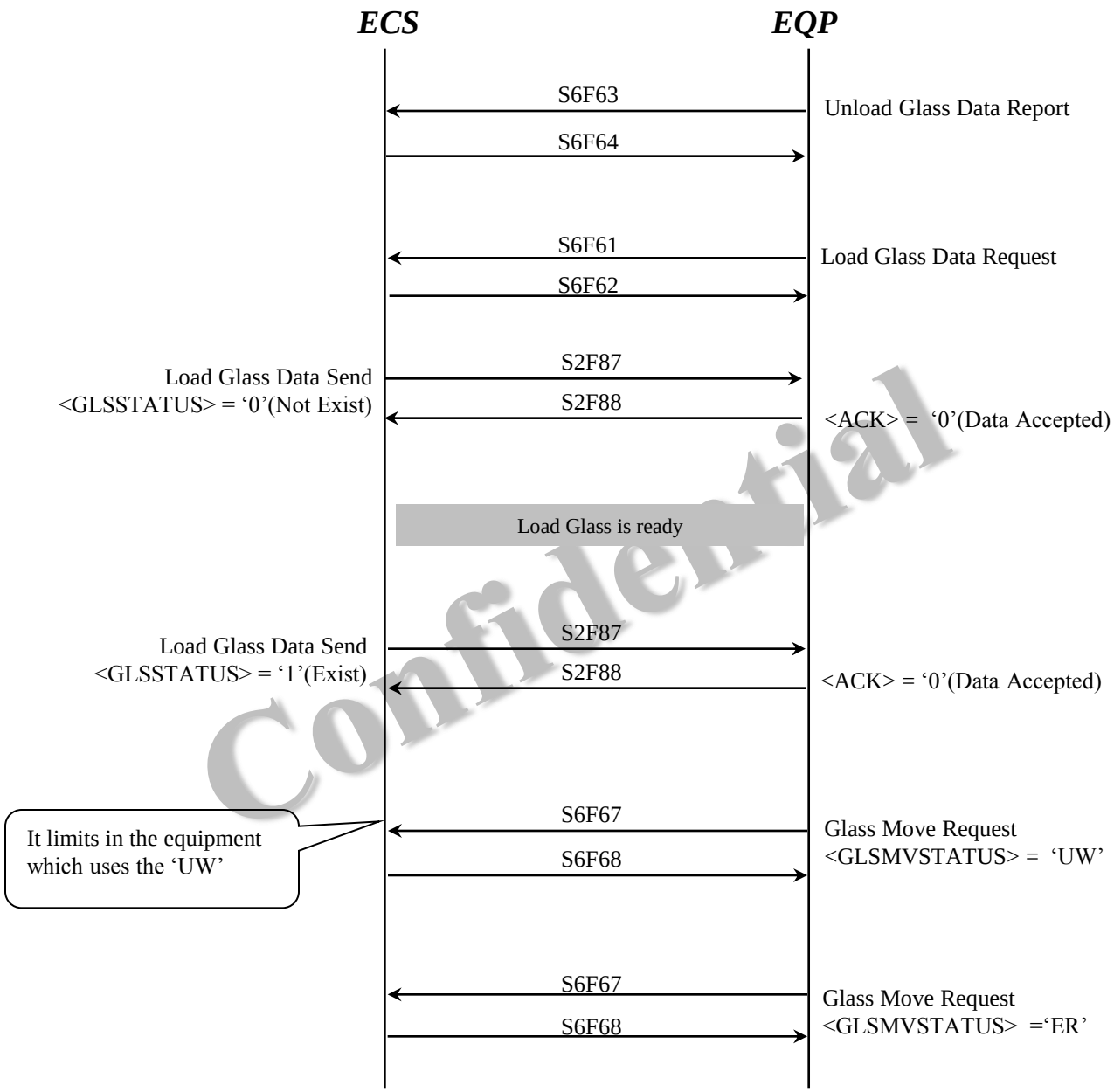


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2) Unload Glass Exist

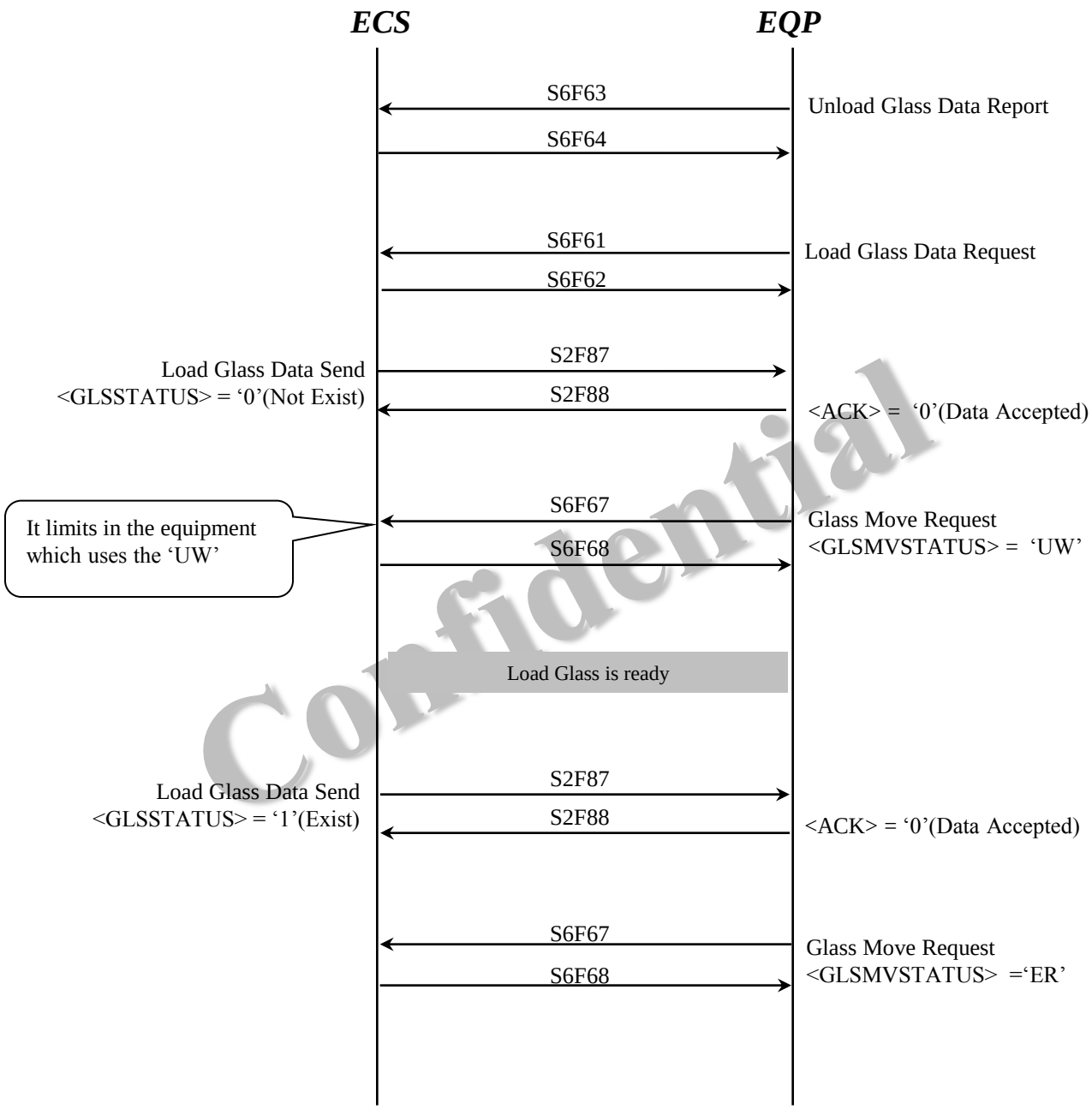
- Case ⓐ : Glass Data Change Send (S2F87) before 'UW'



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2) Unload Glass Exist
- Case ⓑ : Glass Data Change Send (S2F87) after 'UW'

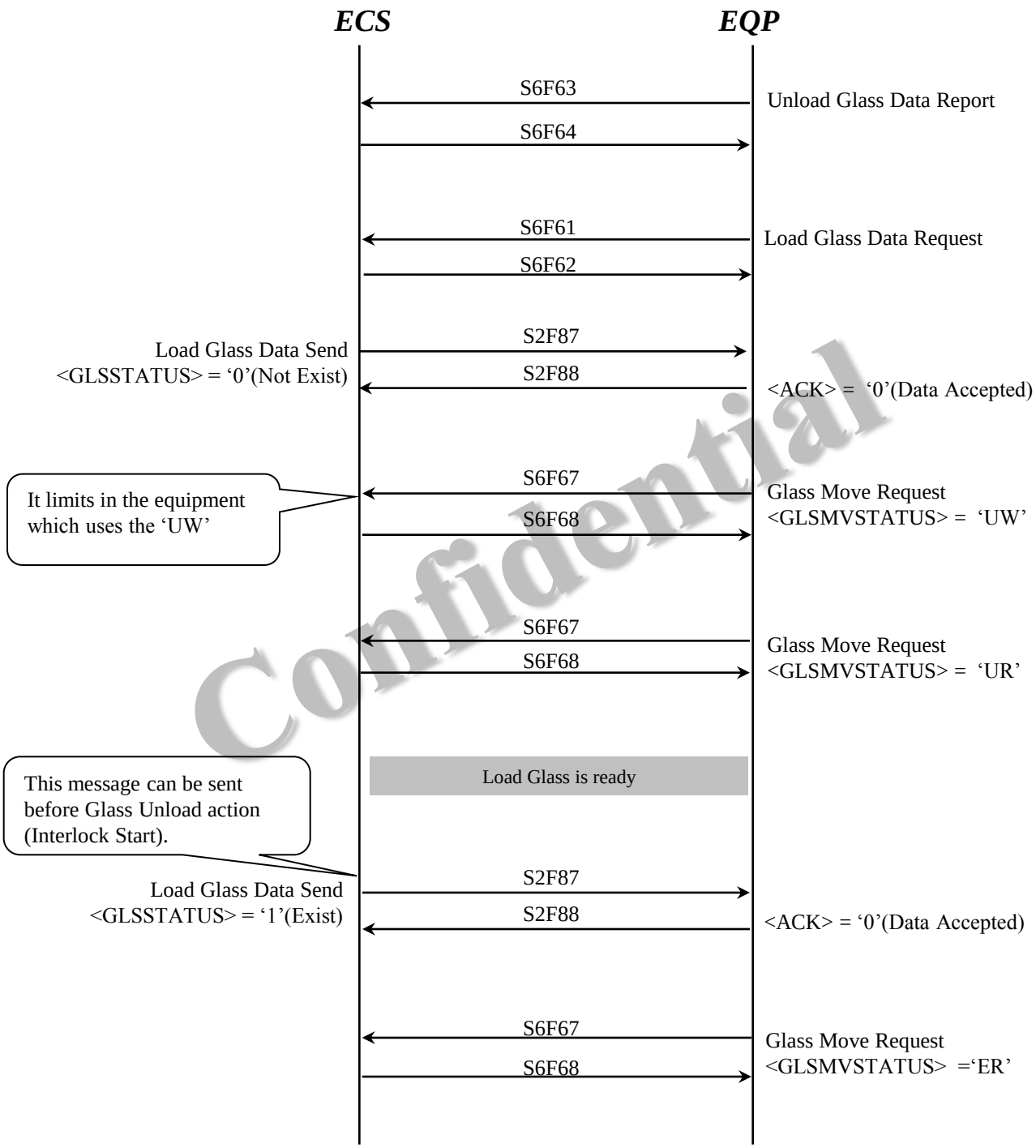


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2) Unload Glass Exist

- Case ⓒ : Glass Data Change Send (S2F87) after 'UR'

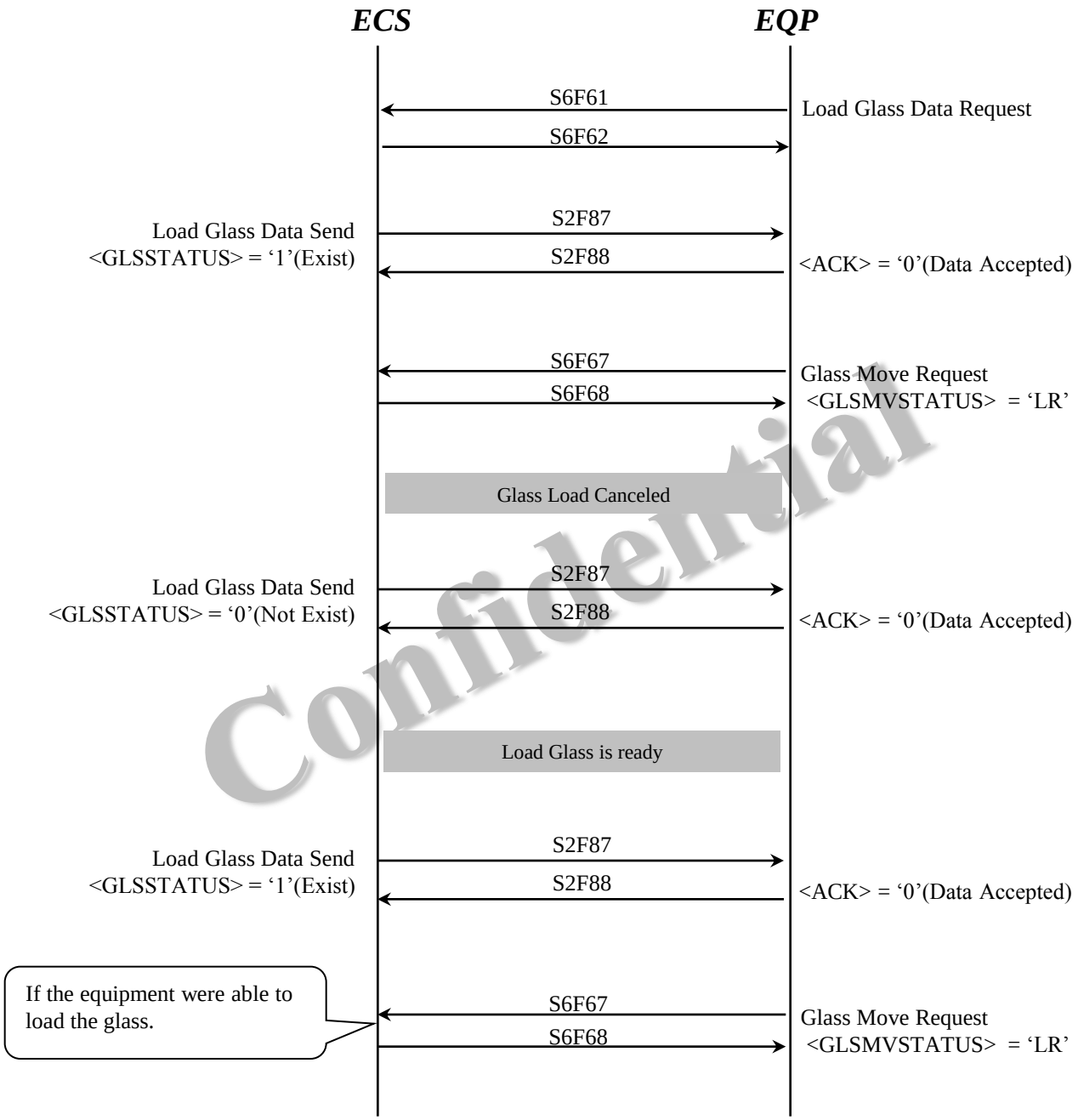


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7.3.5.2 Change form Exist to Not Exist

1) Unload Glass Not Exist

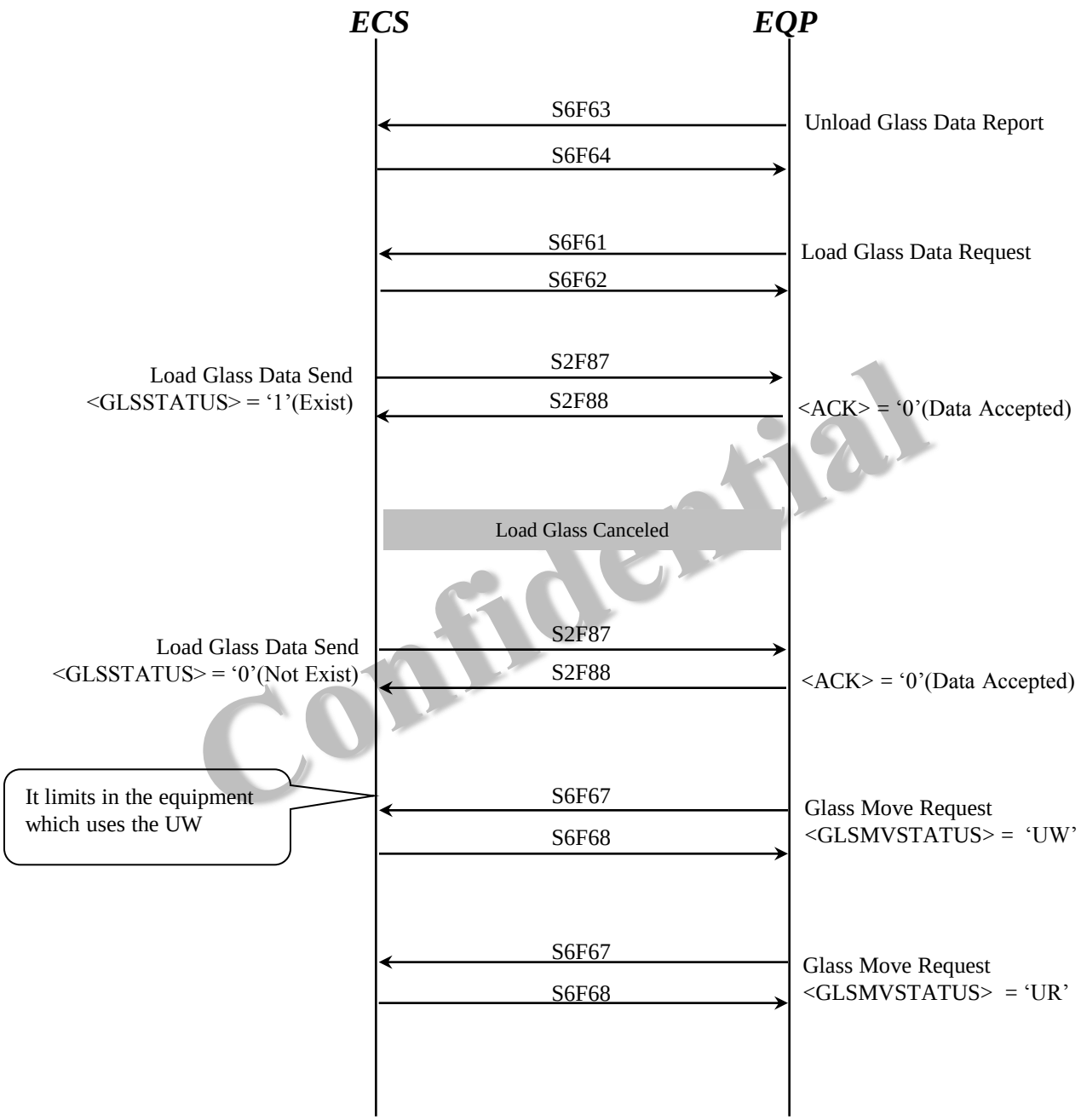


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2) Unload Glass Exist

- Case ⓐ : Glass Data Change Send (S2F87) before 'UW'

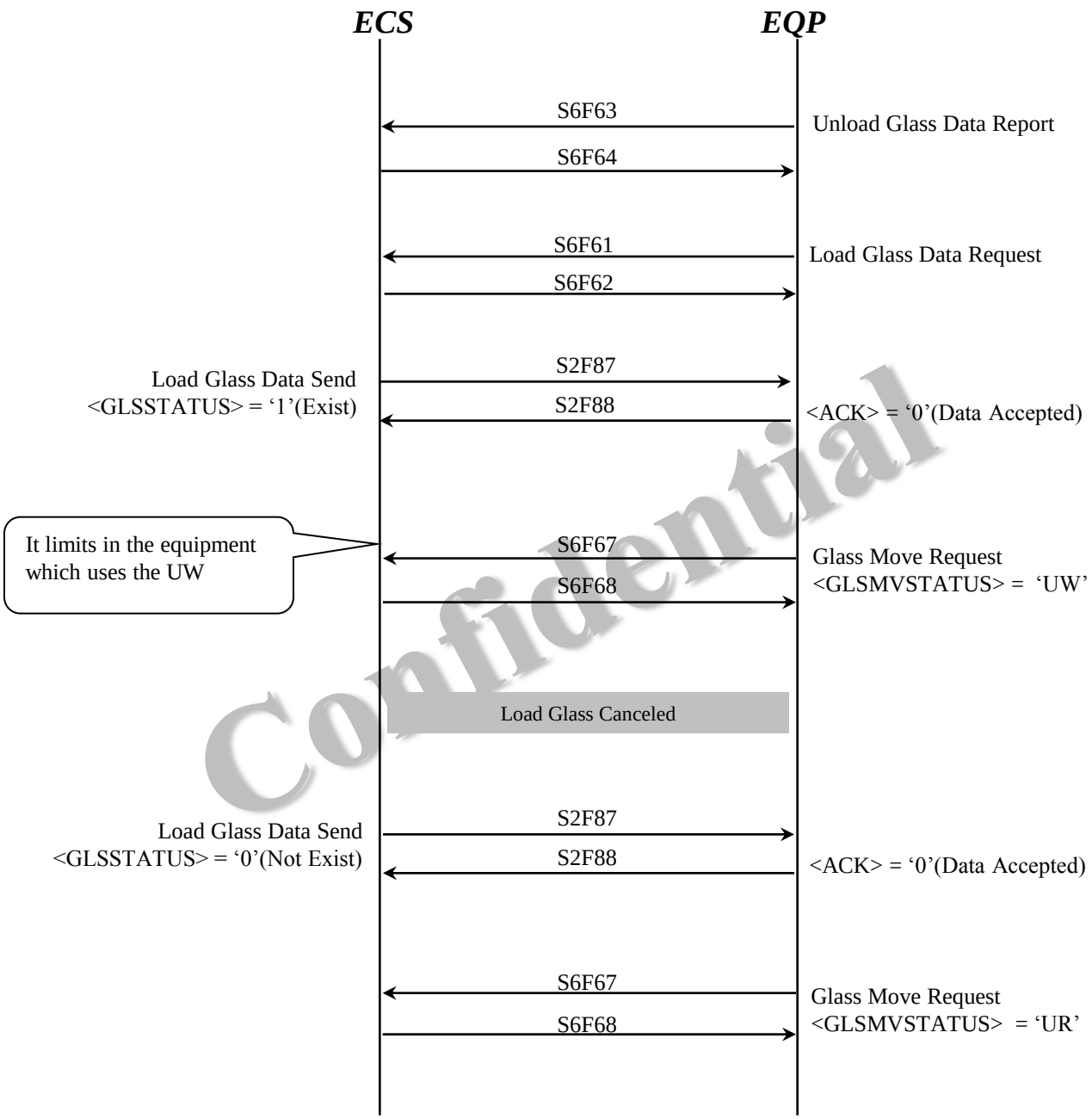


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2) Unload Glass Exist

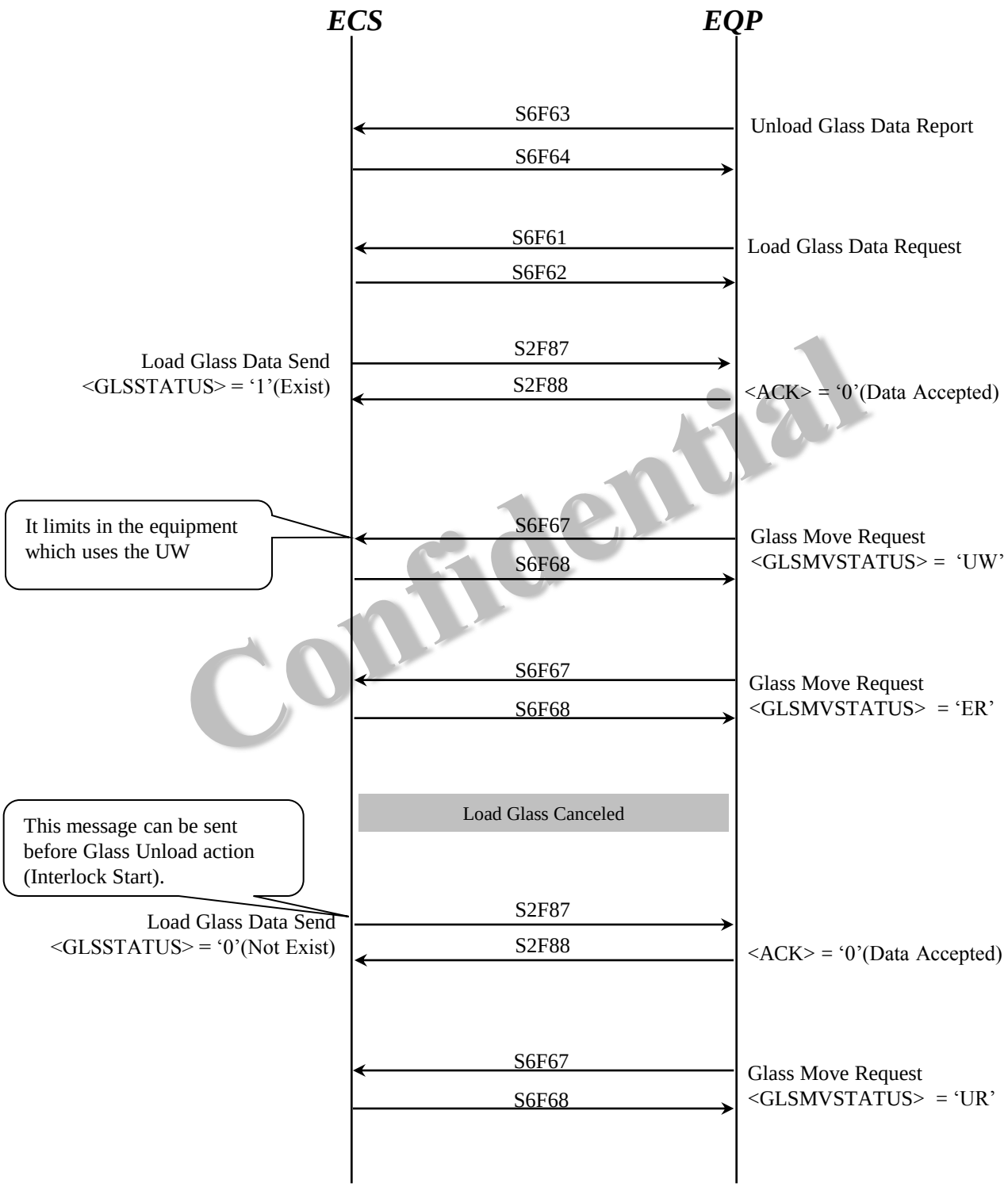
- Case ⓑ : Glass Data Change Send (S2F87) after 'UW'



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2) Unload Glass Exist
- Case ⓒ : Glass Data Change Send (S2F87) after 'ER'

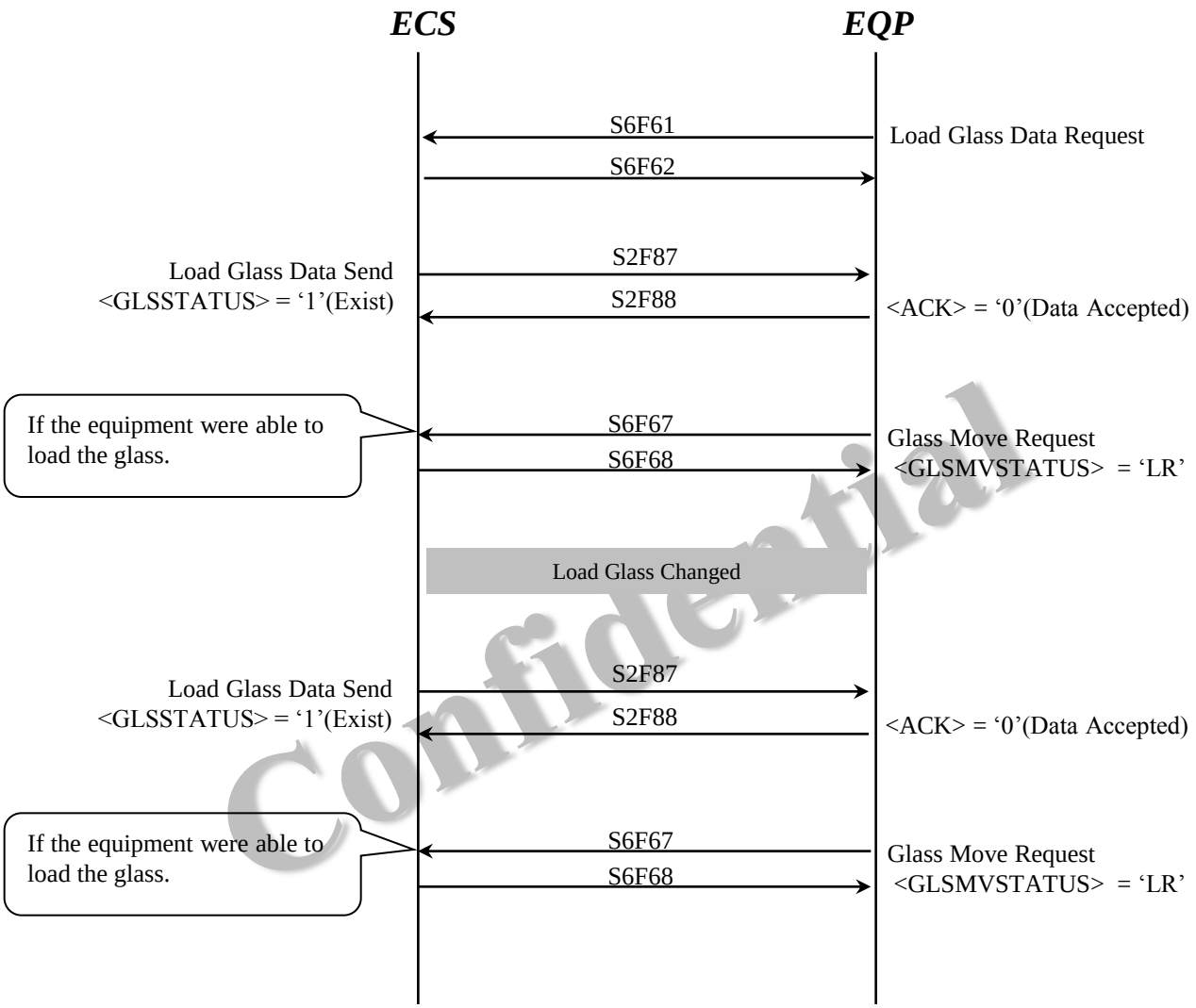


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7.3.5.3 Change form Exist to Exist

1) Unload Glass Not Exist

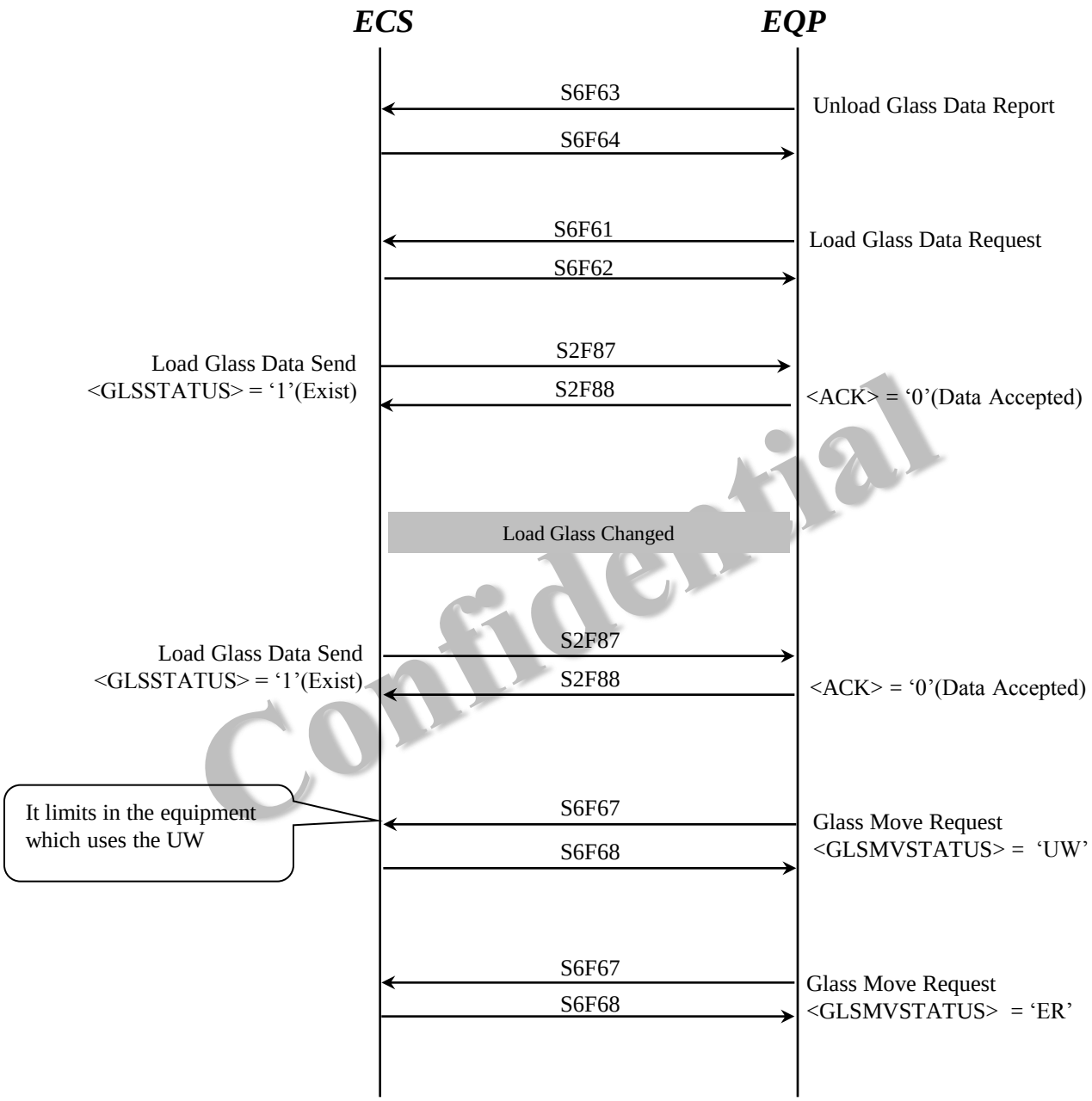


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2) Unload Glass Exist

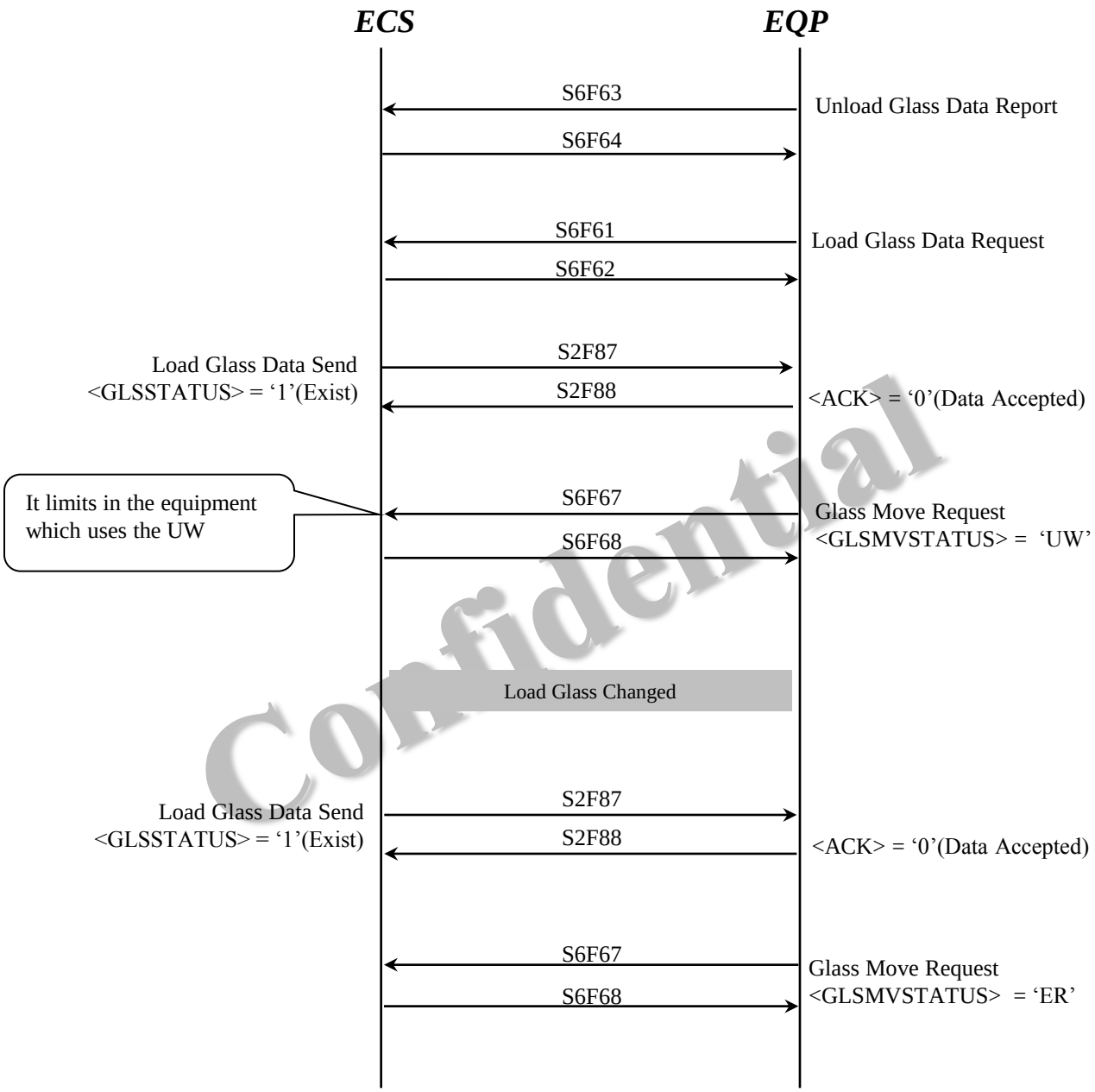
- Case ⓐ : Glass Data Change Send (S2F87) before 'UW'



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2) Unload Glass Exist
- Case ⓑ : Glass Data Change Send (S2F87) after 'UW'

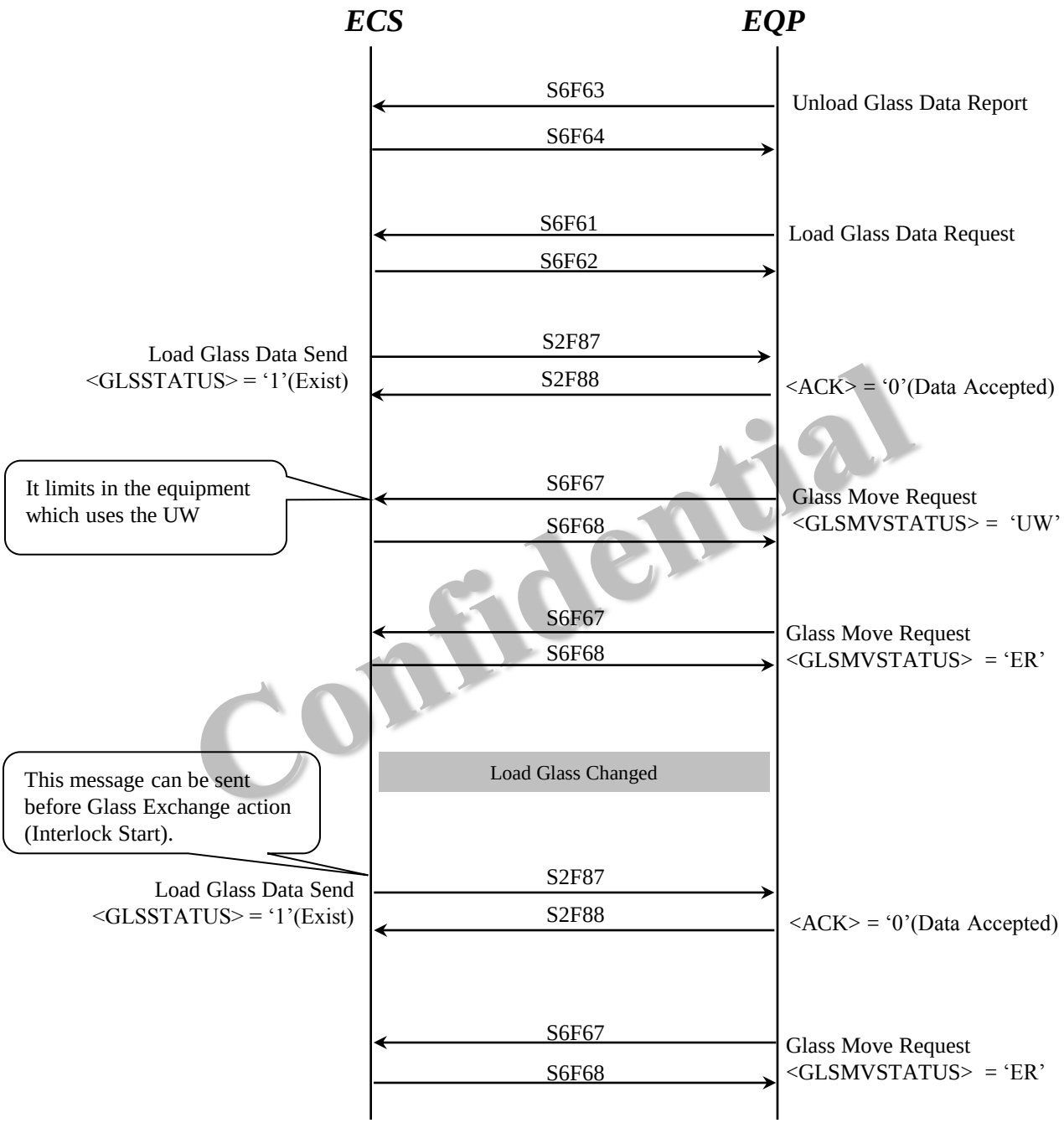


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2) Unload Glass Exist

- Case ⓒ : Glass Data Change Send (S2F87) after 'ER'

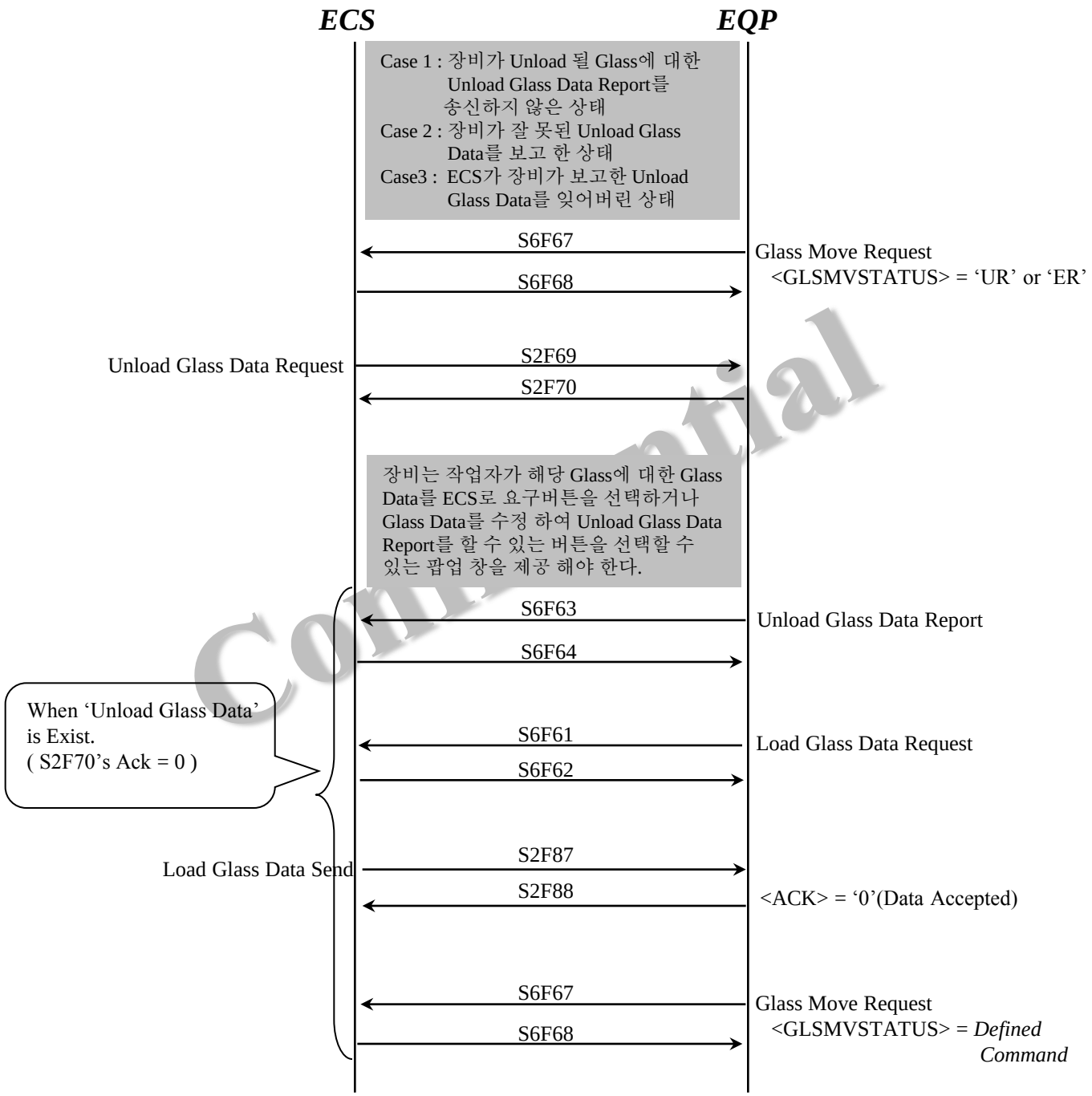


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7.3.6 Unload Glass Data Request

- 이 메시지는 ECS가 'unload glass data report'를 받지 못했을 때 사용된다.
- 세부적인 사용에 대해서는 LG CNS, LG Display와의 협의가 필요하다

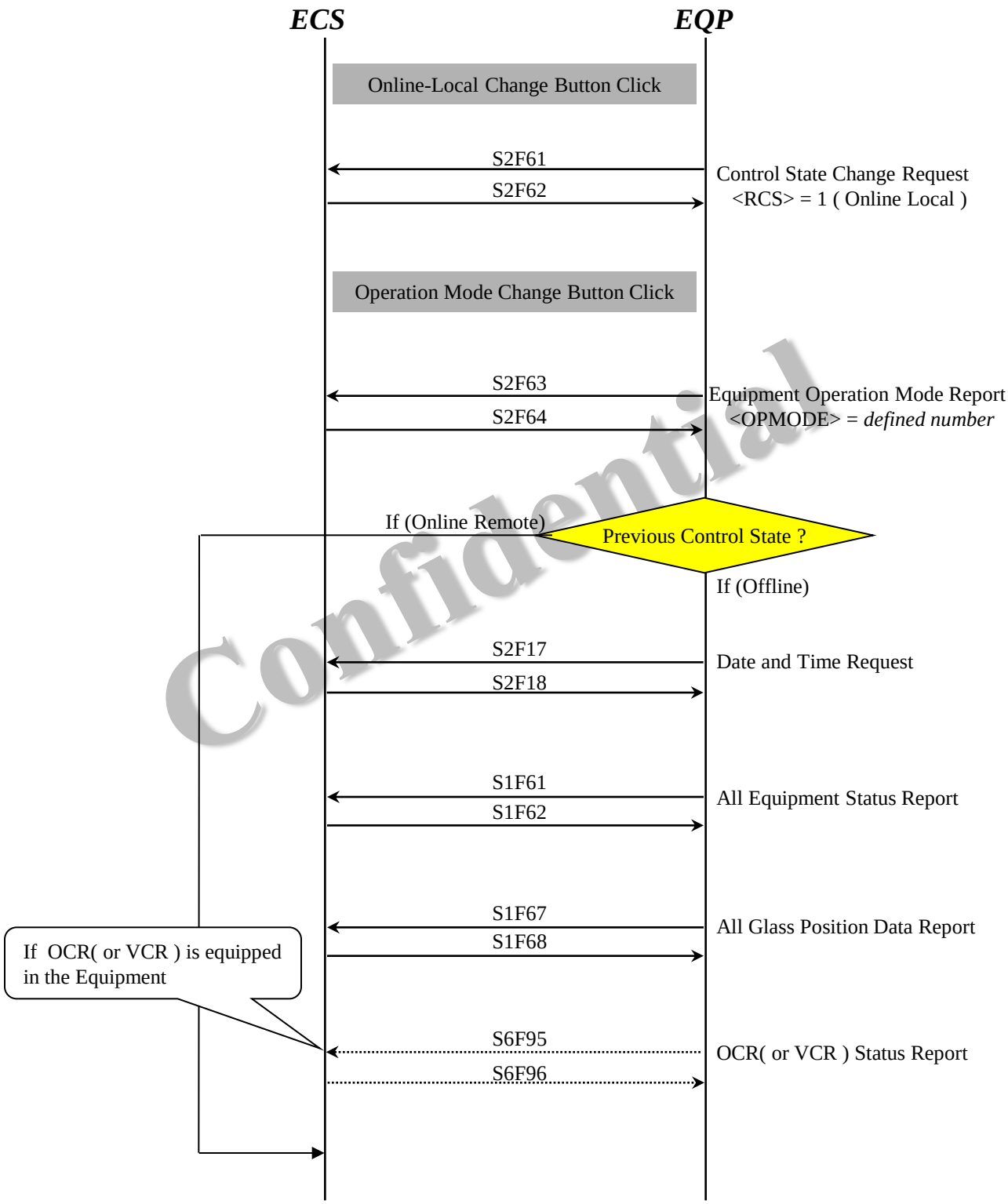


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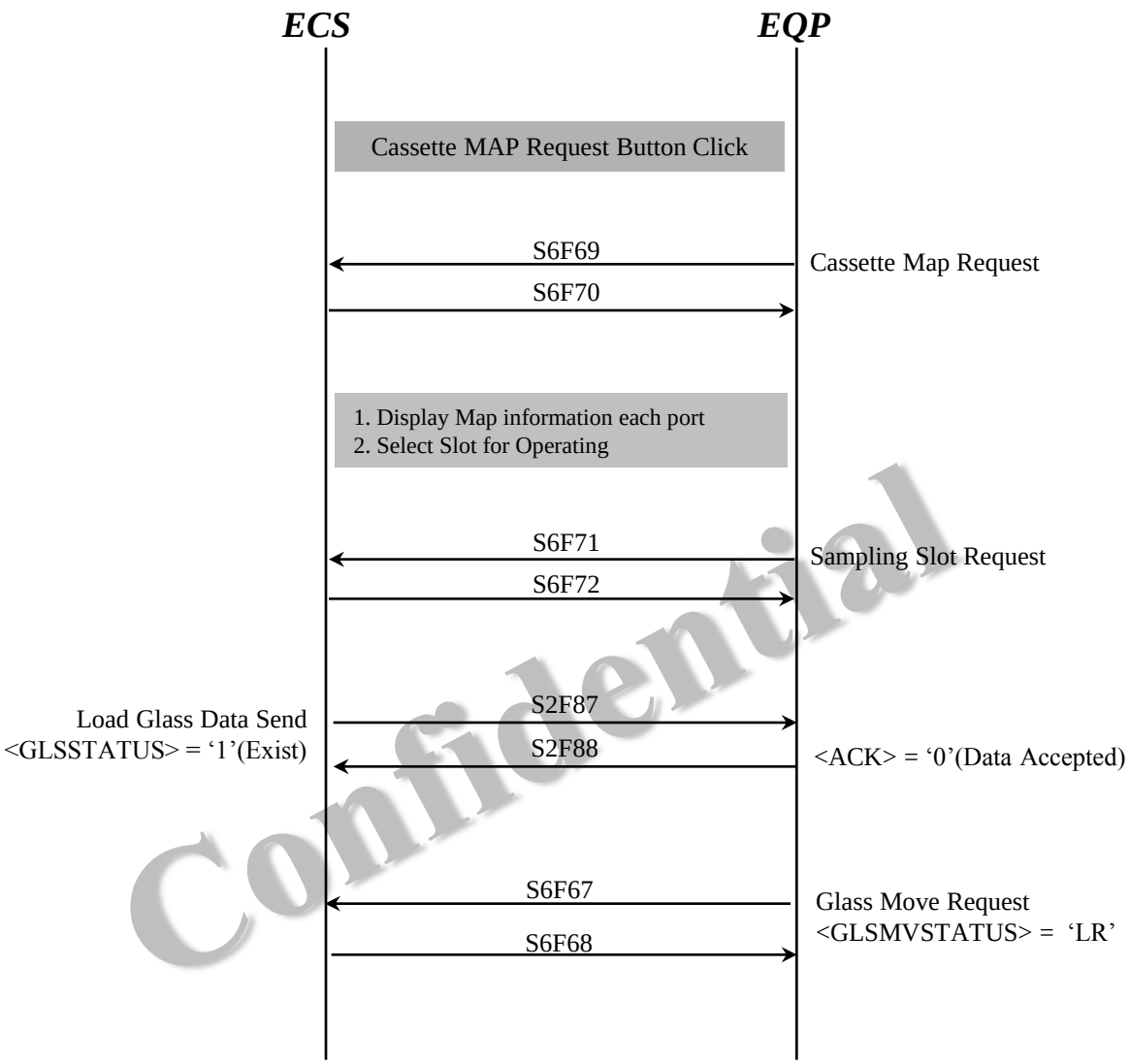
7.3.7 Glass Transfer In Online Local Control State

- 다음 Scenario는 Online Local 시 첫 번째 Glass를 장비에 투입할 때를 보여준다.



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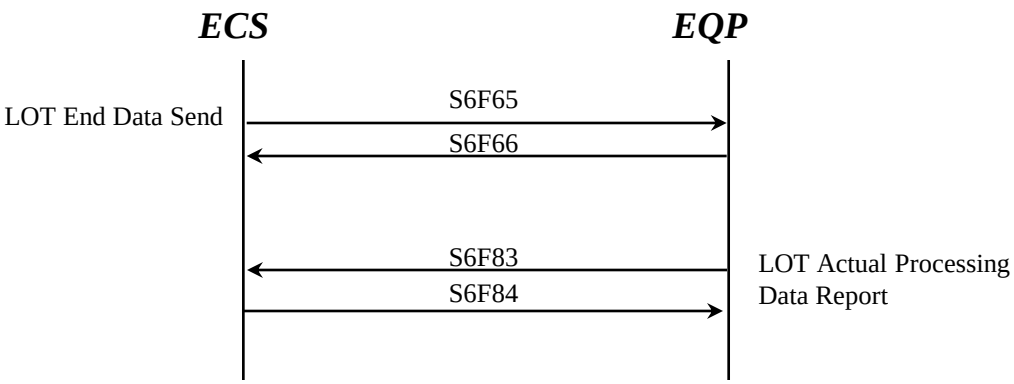


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7.4 Lot Control Sequence

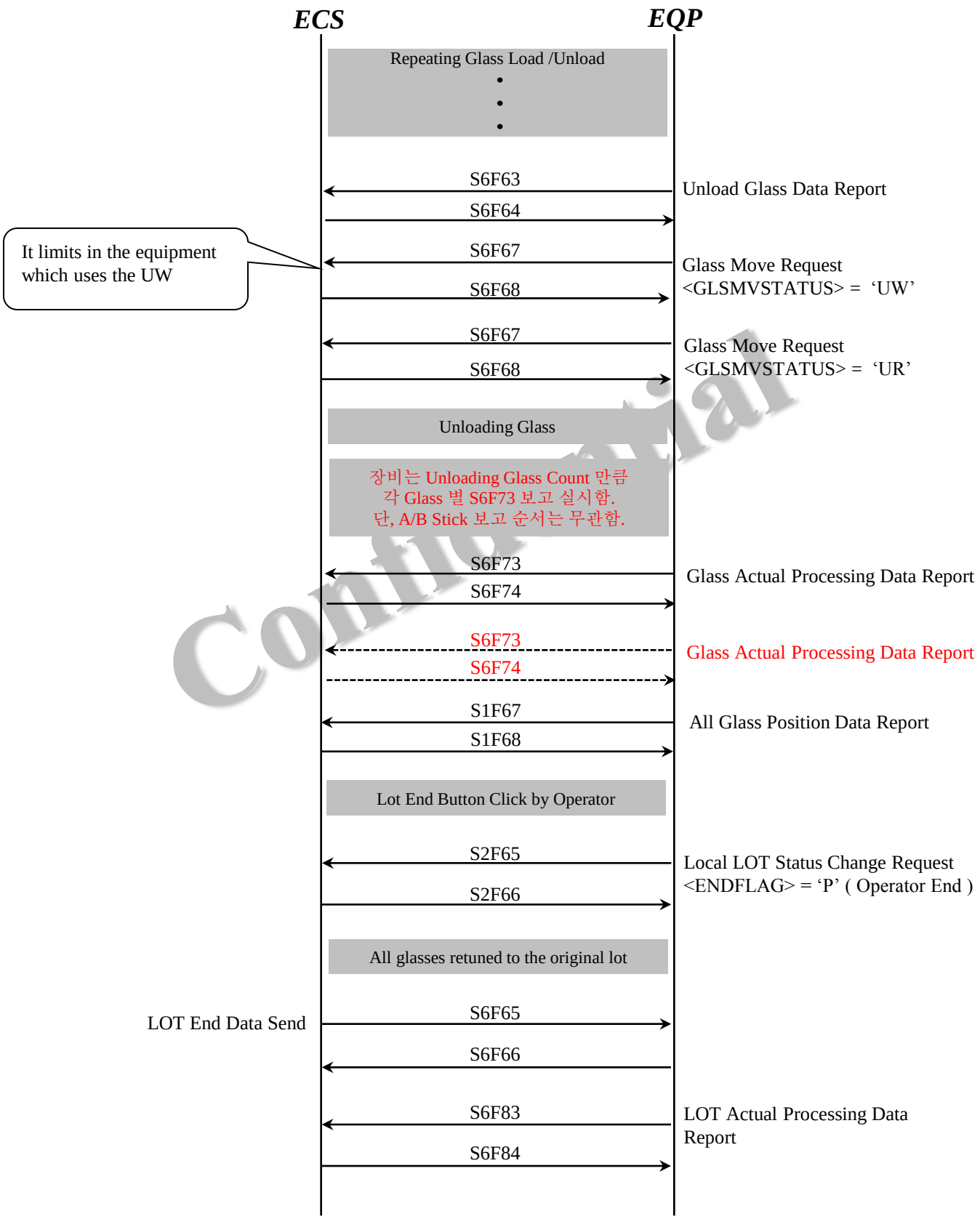
7.4.1 Lot End Data Send To Equipment



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7.4.2 Local LOT Status Change Request on Online Local Control State
- Online local state 하에서 Lot end process 는 아래의 순서를 따른다

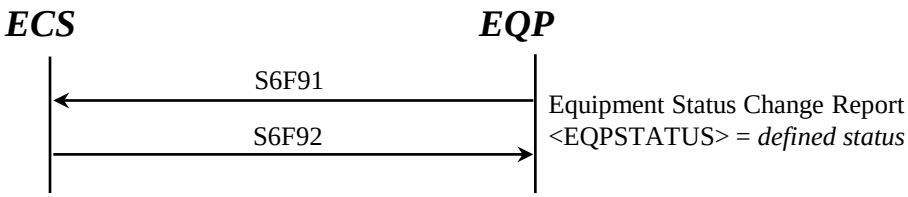


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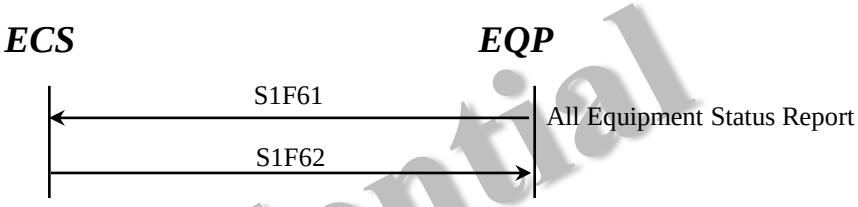
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7.5 Status Report

7.5.1 Equipment Status Change Report
- unit의 상태가 변할때마다 수행한다.

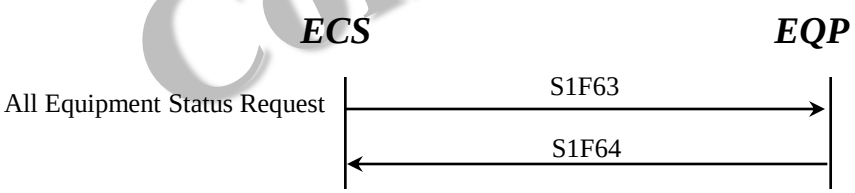


7.5.2 All Equipment Status Report
- ECS가 재시작하거나 Offline에서 Online으로 전환되는 경우에 모든 장비 상태를 보고한다.(Remote or Local).

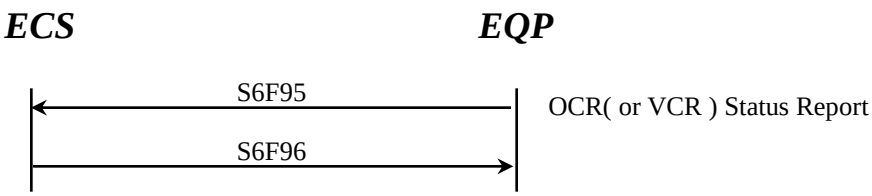


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7.5.3 All Equipment Status Request
- 모든 장비 상태를 요청 시 수행한다.



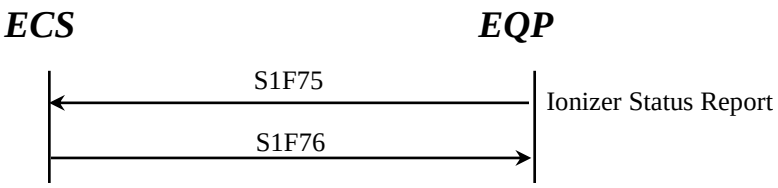
7.5.4 OCR(or VCR) Status Report
- OCR(or VCR)이 Offline에서 Online으로 전환될 때 OCR(or VCR)의 상태를 ECS로 보고한다.



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7.5.5 Ionizer Status Report

- Ionizer의 상태를 보고한다.

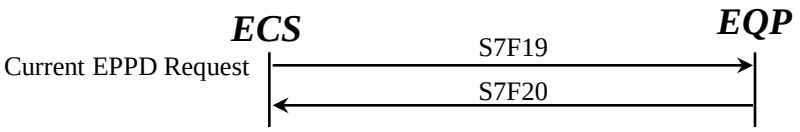


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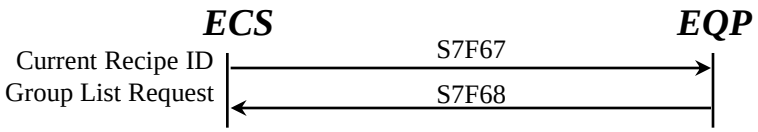
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7.6 Recipe Management

7.6.1 Recipe ID List Request



7.6.2 Current Recipe ID Group List Request



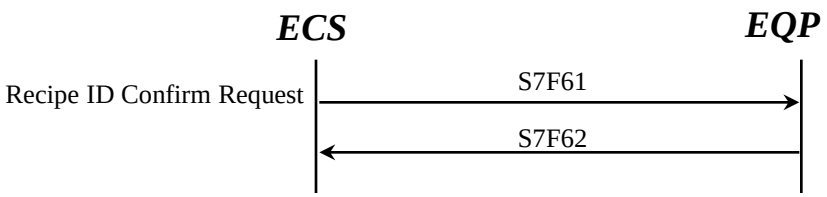
7.6.3 Current Recipe ID List Request by Group

- 특정 Group에 속하는 Recipe ID List를 요청할 시 사용한다.

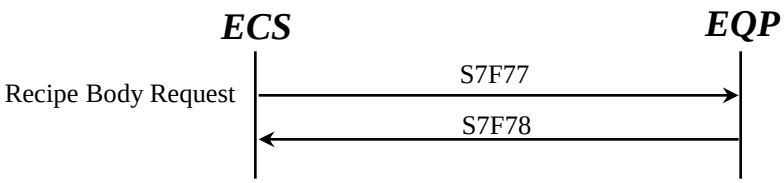


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7.6.4 Recipe ID Confirm Request



7.6.5 Recipe Body Request

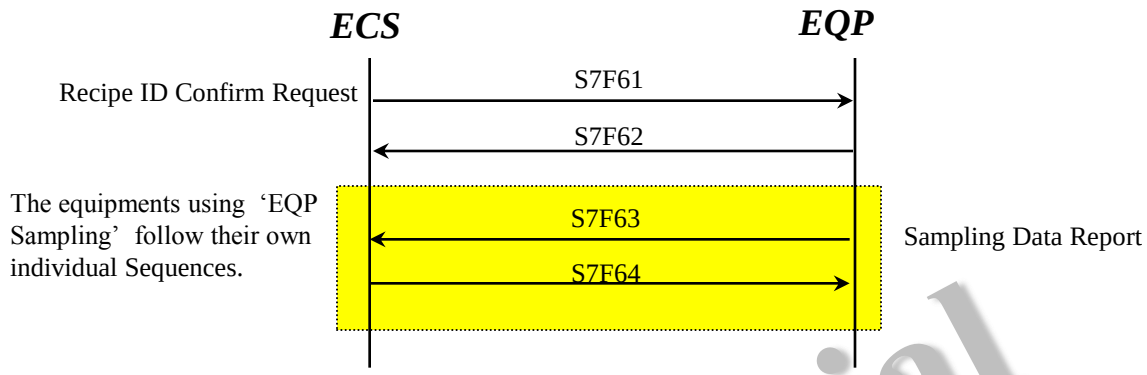


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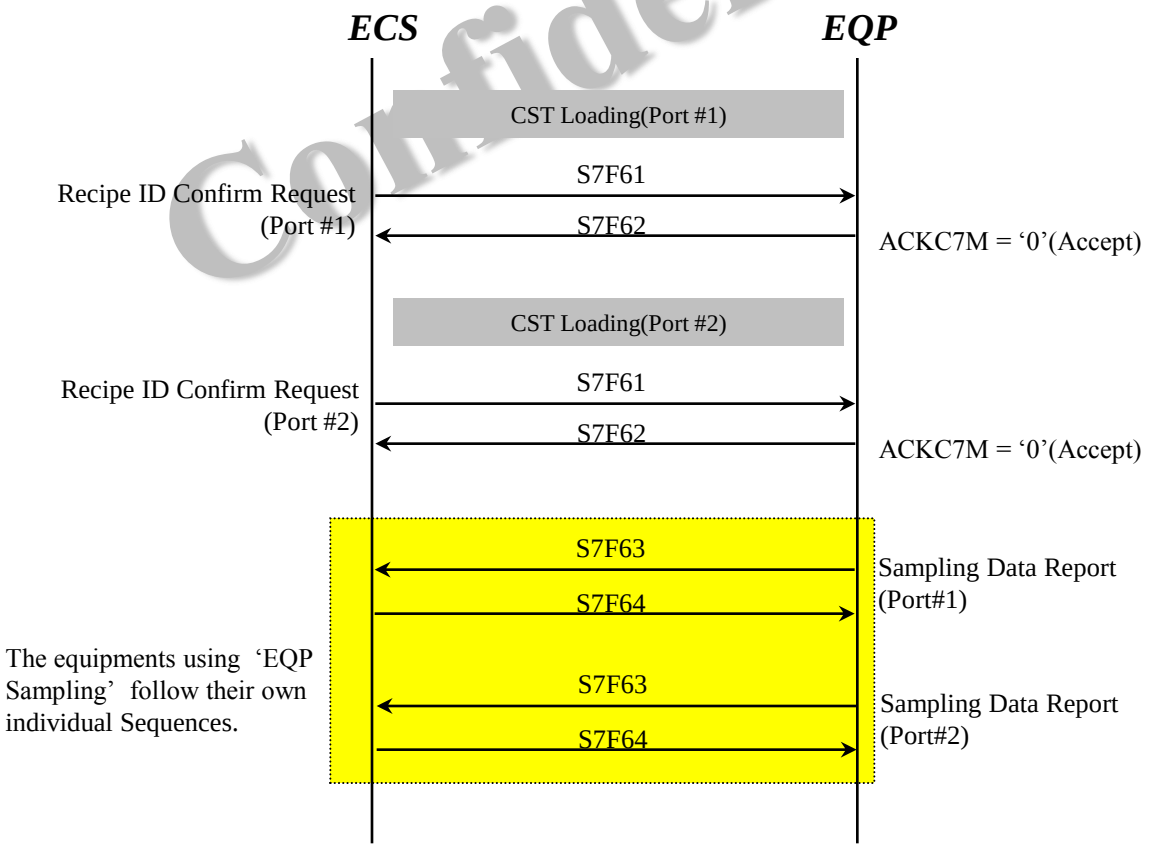
7.6.6 Recipe ID Confirm & Sampling Data Send

- 이 Scenario는 각 Slot별 모든 Recipes를 즉시 확인할 때 사용되고, 이 메시지는 Lot Recipe (cassette recipe) 인 하나의 Recipe에만 사용될 수 있다.
- 만약 EQP가 샘플링과 < Recipe ID Confirm >의 사용이 가능하다.면, 장비는 Sampling Data Report (S7F63) 를 반드시 ECS로 보내야 한다.

7.6.6.1 Recipe Accept (Single LOT)



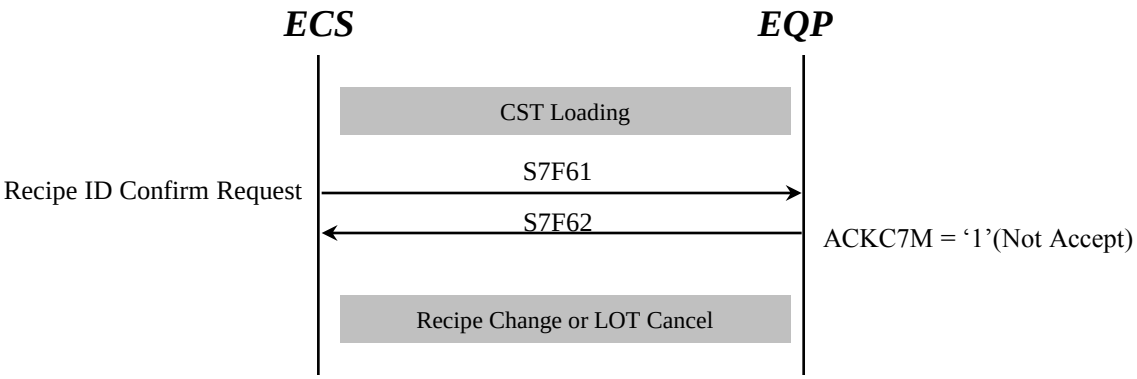
7.6.6.2 Recipe Accept(Multi LOT)



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7.6.6.3 Recipe Not Accept

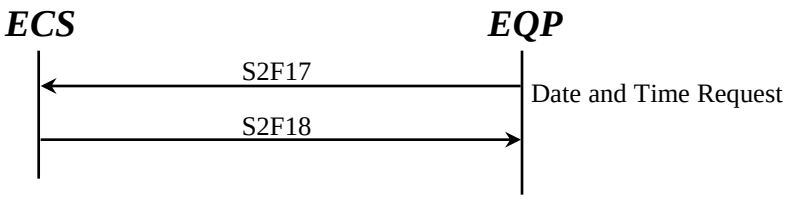


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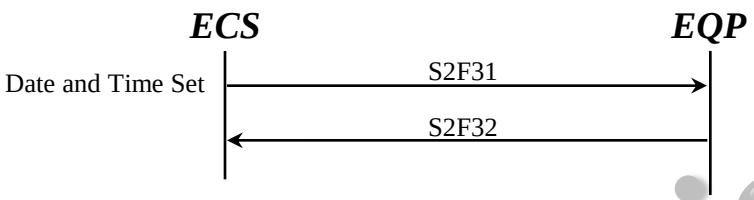
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7.7 Date and Time Data Send

7.7.1 Date and Time Request



7.7.2 Date and Time Set

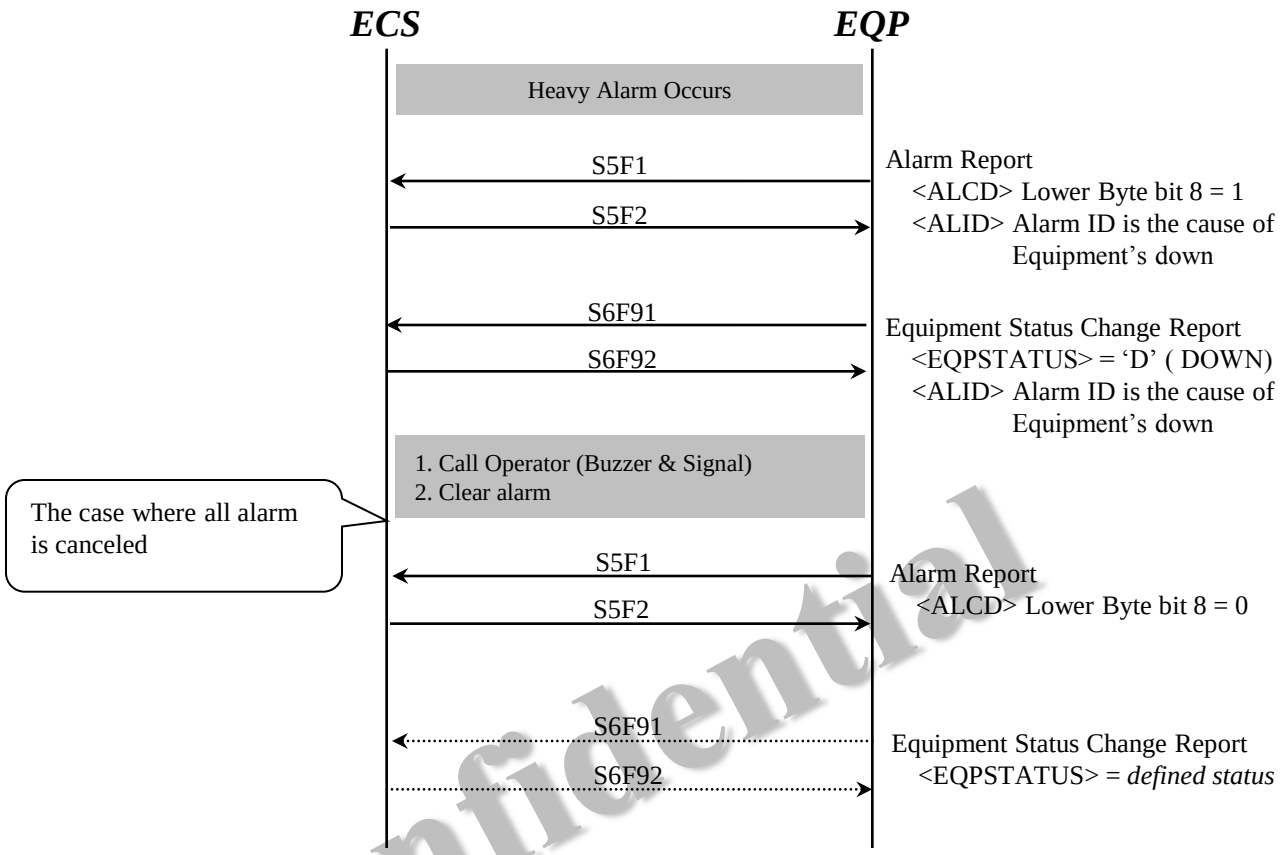


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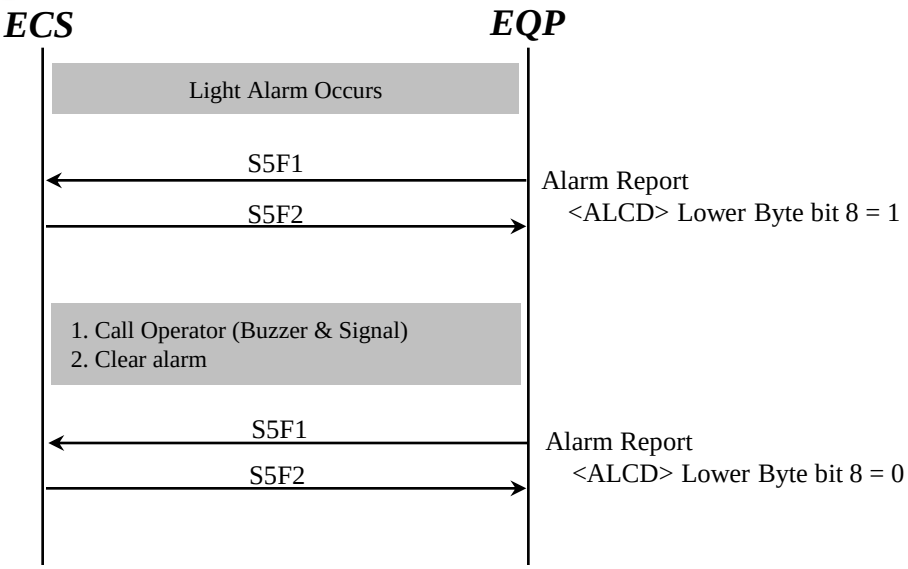
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7.8 Alarm Occurrence and Clear

7.8.1 Heavy Alarm



7.8.2 Light Alarm

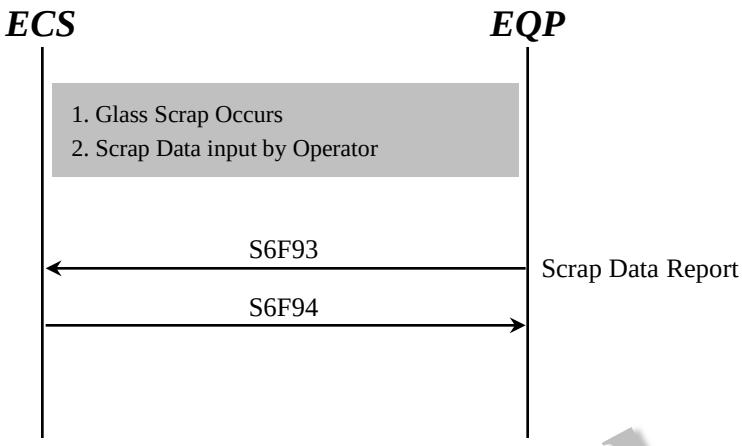


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7.9 Scrap Occurrence and Report

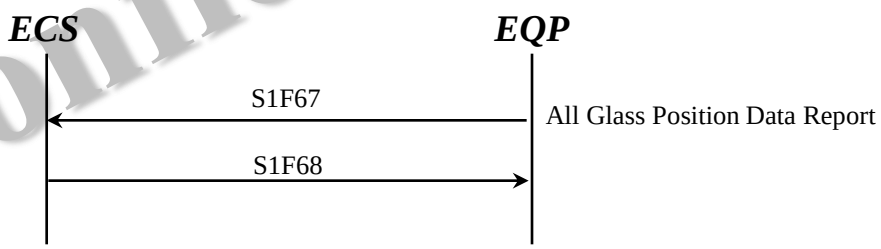
- 일반적으로 Scrap 보고는 장비에서 Glass 파손이 확인되는 즉시 ECS로 보고 해야 한다.
- 부득이한 경우는 Lot End 이전에는 보고 해야 한다.



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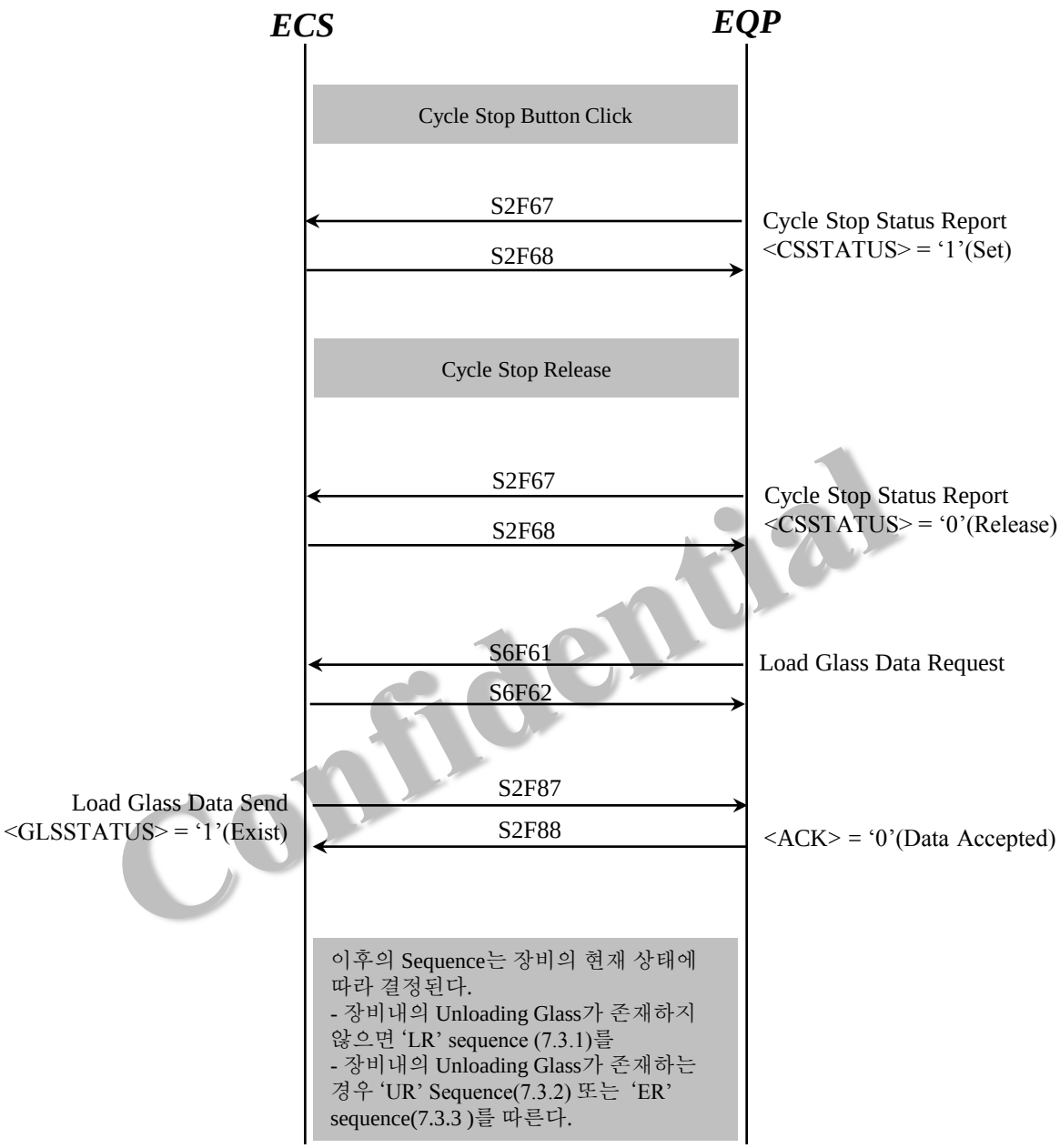
7.10 All Glass Position Data Report

-Online으로 전환되는 경우
-각 Sub unit간 glass가 이동하는 경우



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7.11 Cycle Stop Status Report

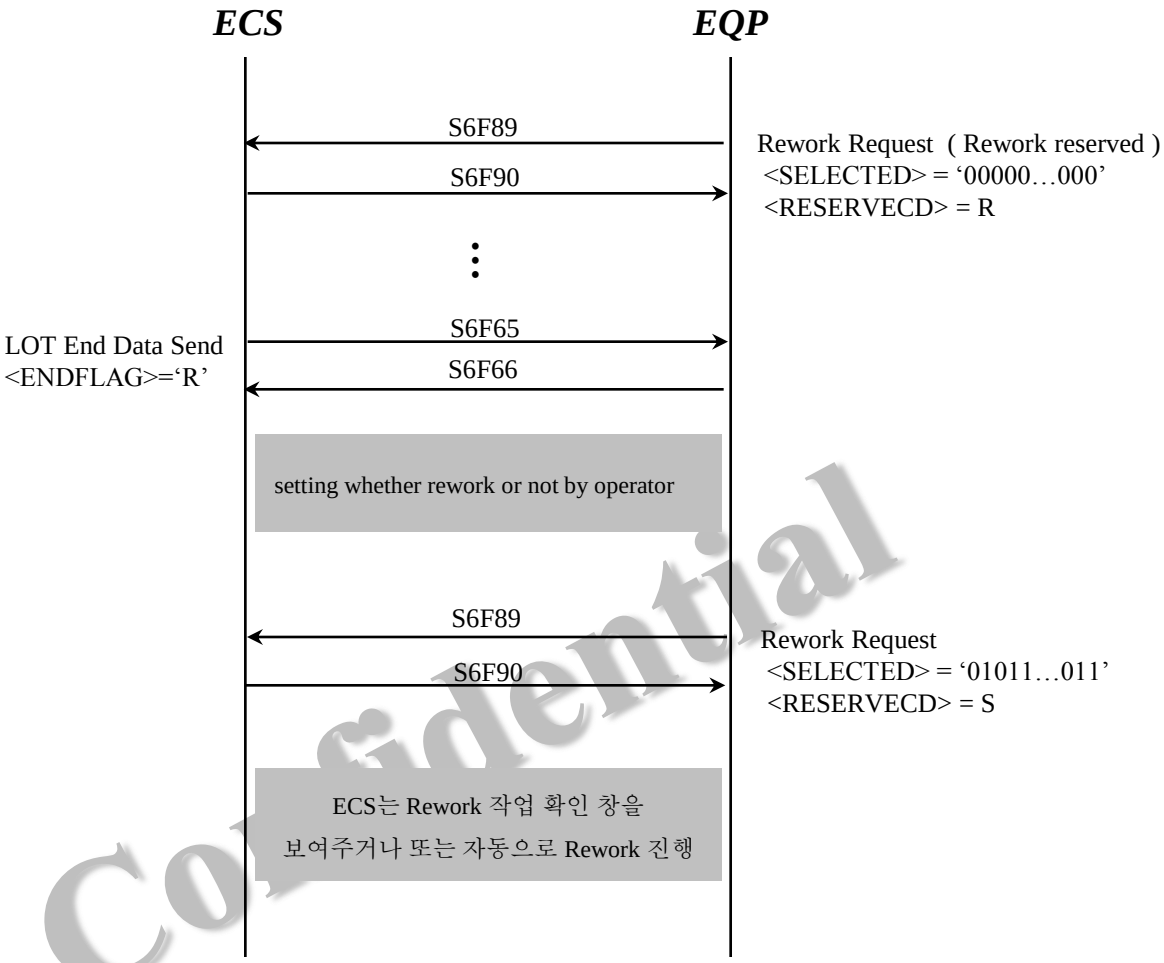


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7.12 Rework Request

-Operator가 Rework flag code를 설정하면 장비는 ECS에게 rework 요청을 한다.

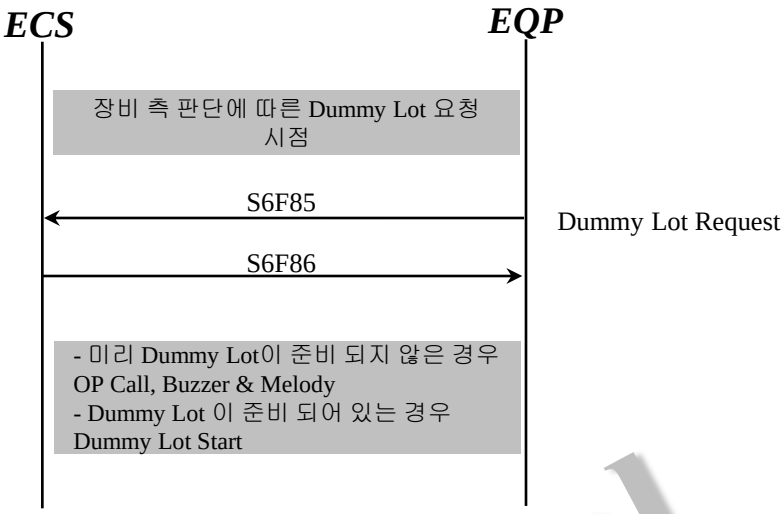


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7.13 Dummy Lot Request

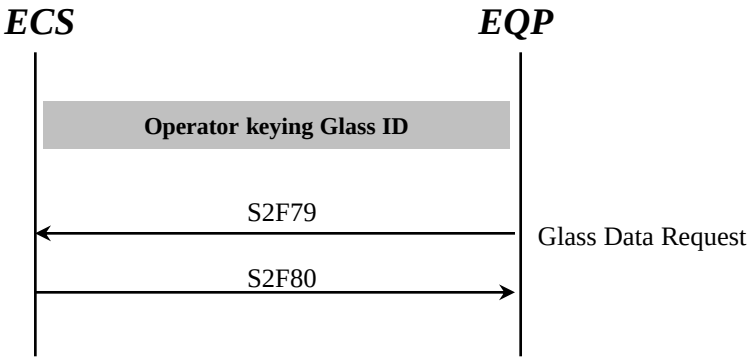
-만약 장비가 Dummy Lot을 사용해야 한다면 Dummy Lot이 필요한 시점에 ECS에게 요청한다.



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7.14 Glass Data Request

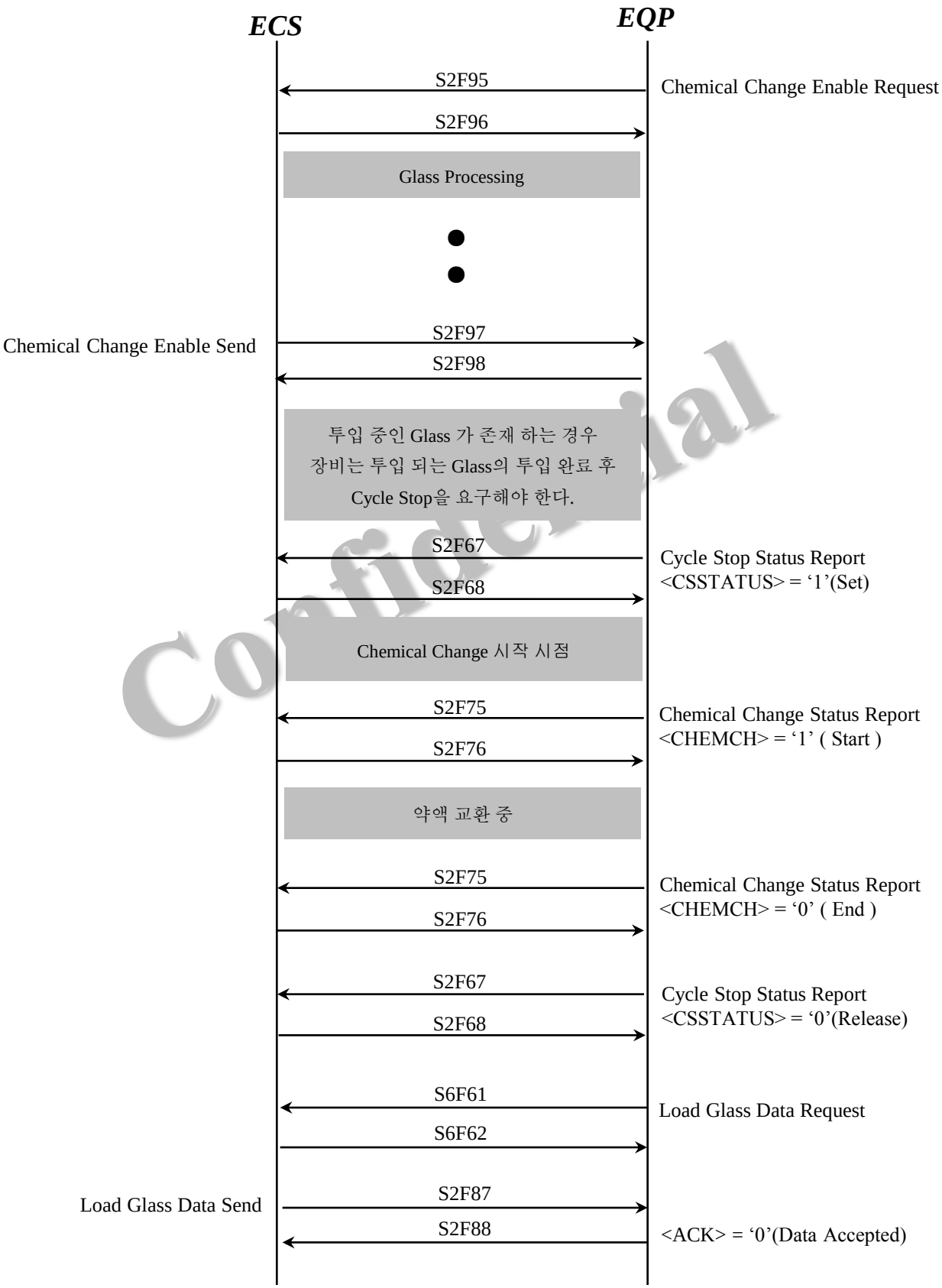
- Operator가 Glass ID 또는 Glass Code로 Glass Data를 재 요청 시 사용한다.
- 세부적인 사용에 대해서는 LG CNS, LG Display의 협의가 필요하다.



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7.15 Chemical Change

- 다음은 chemical change 시나리오이다.

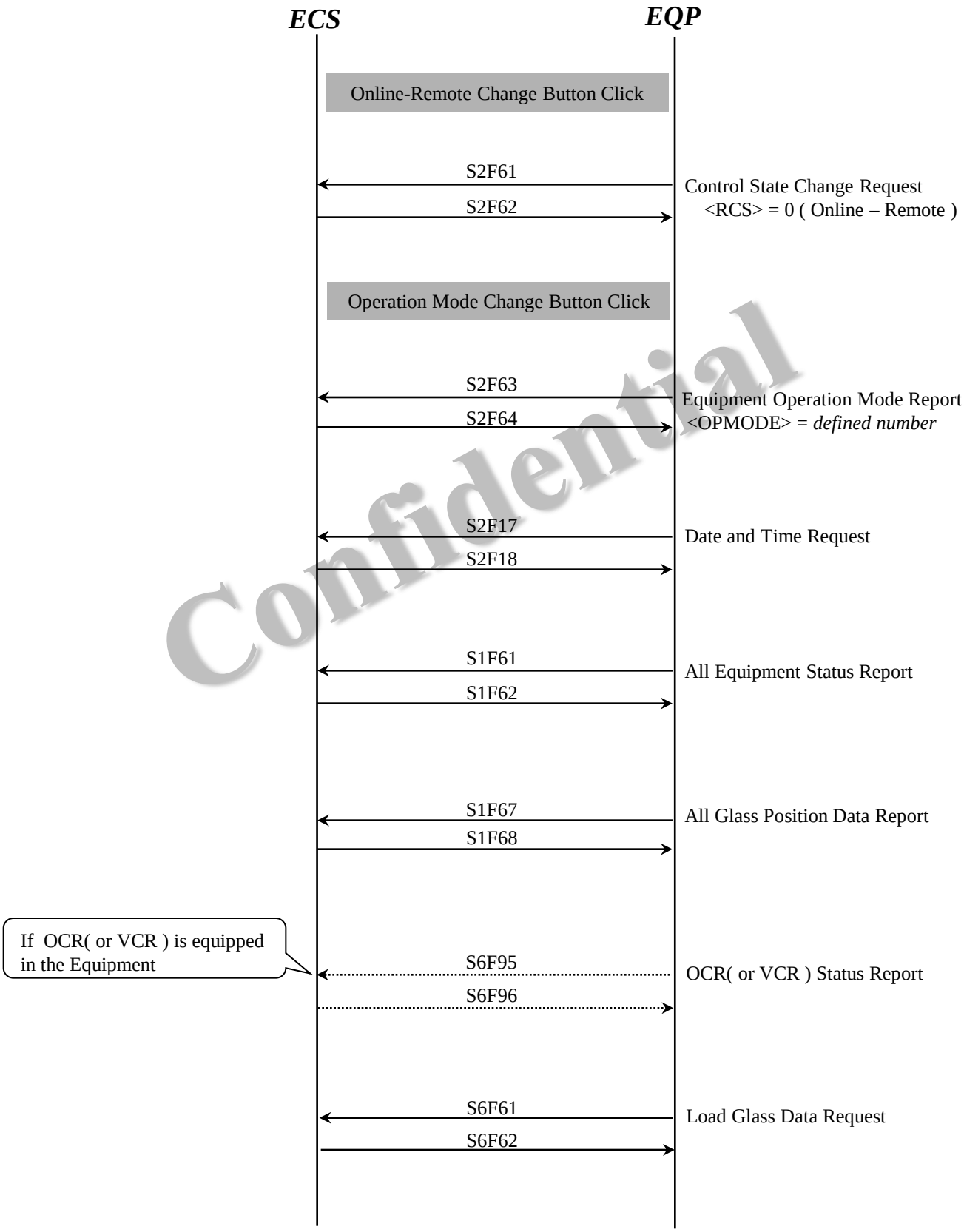


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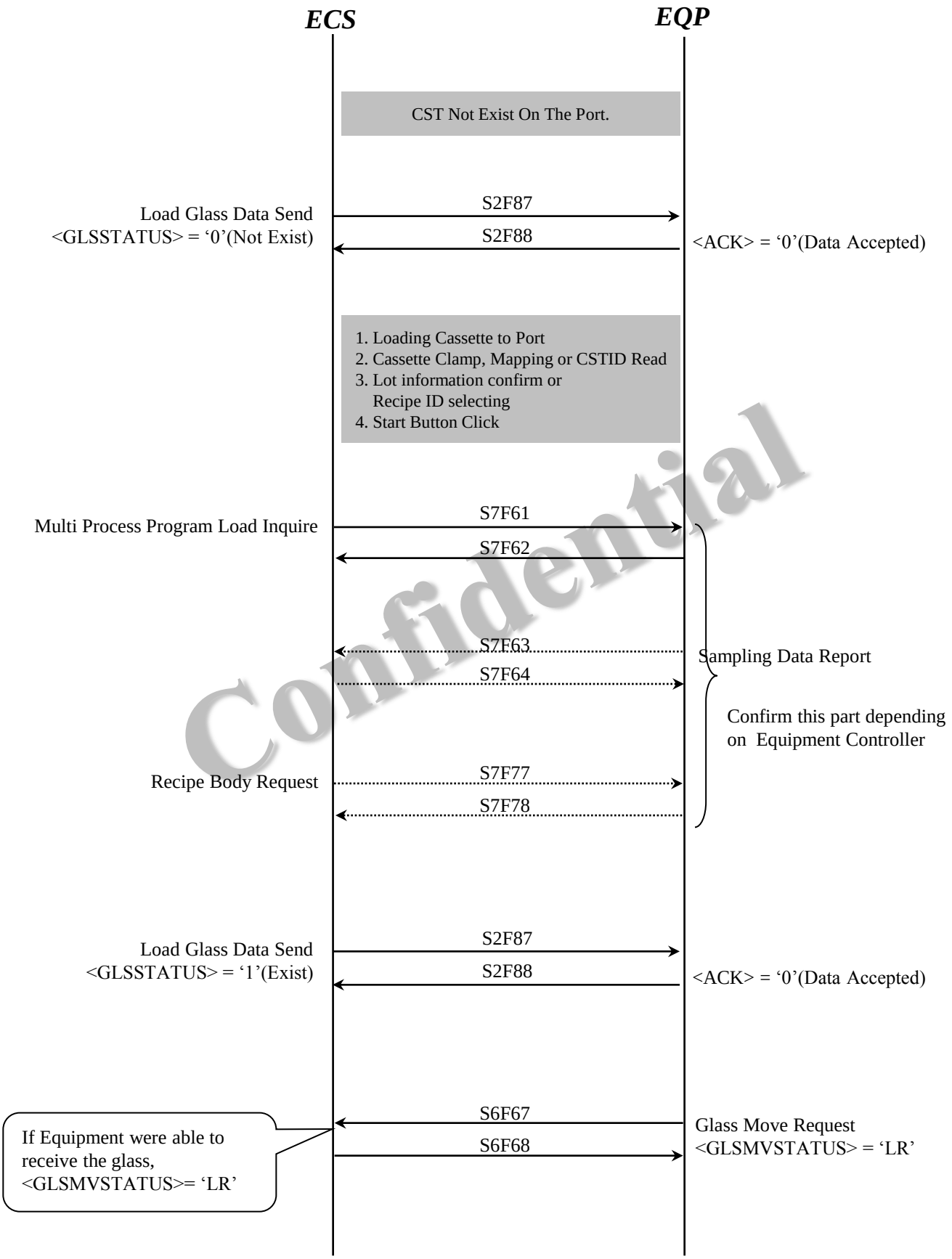
7.16 Normal Operation

- In case of 1 lot sequence (from lot start to end)



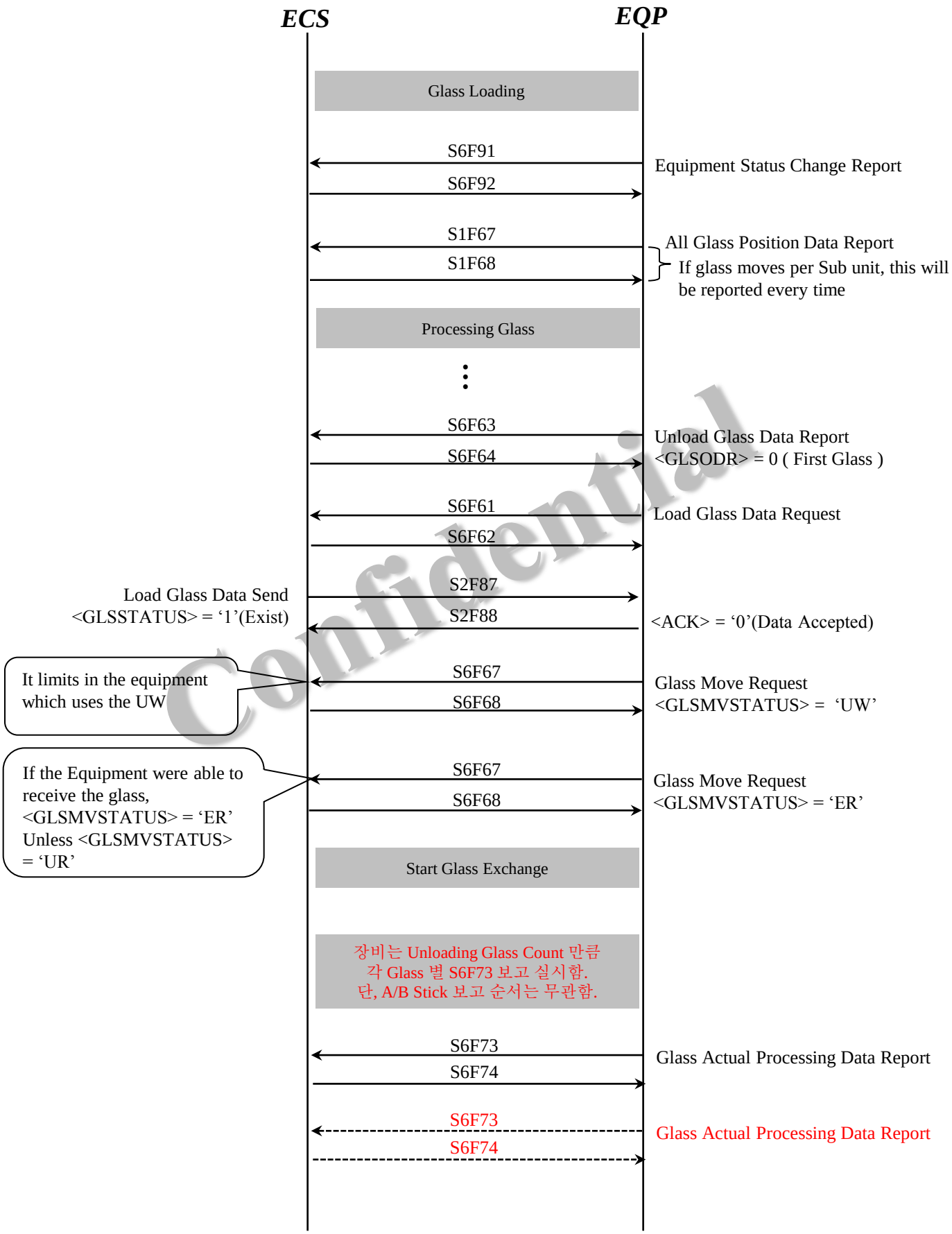
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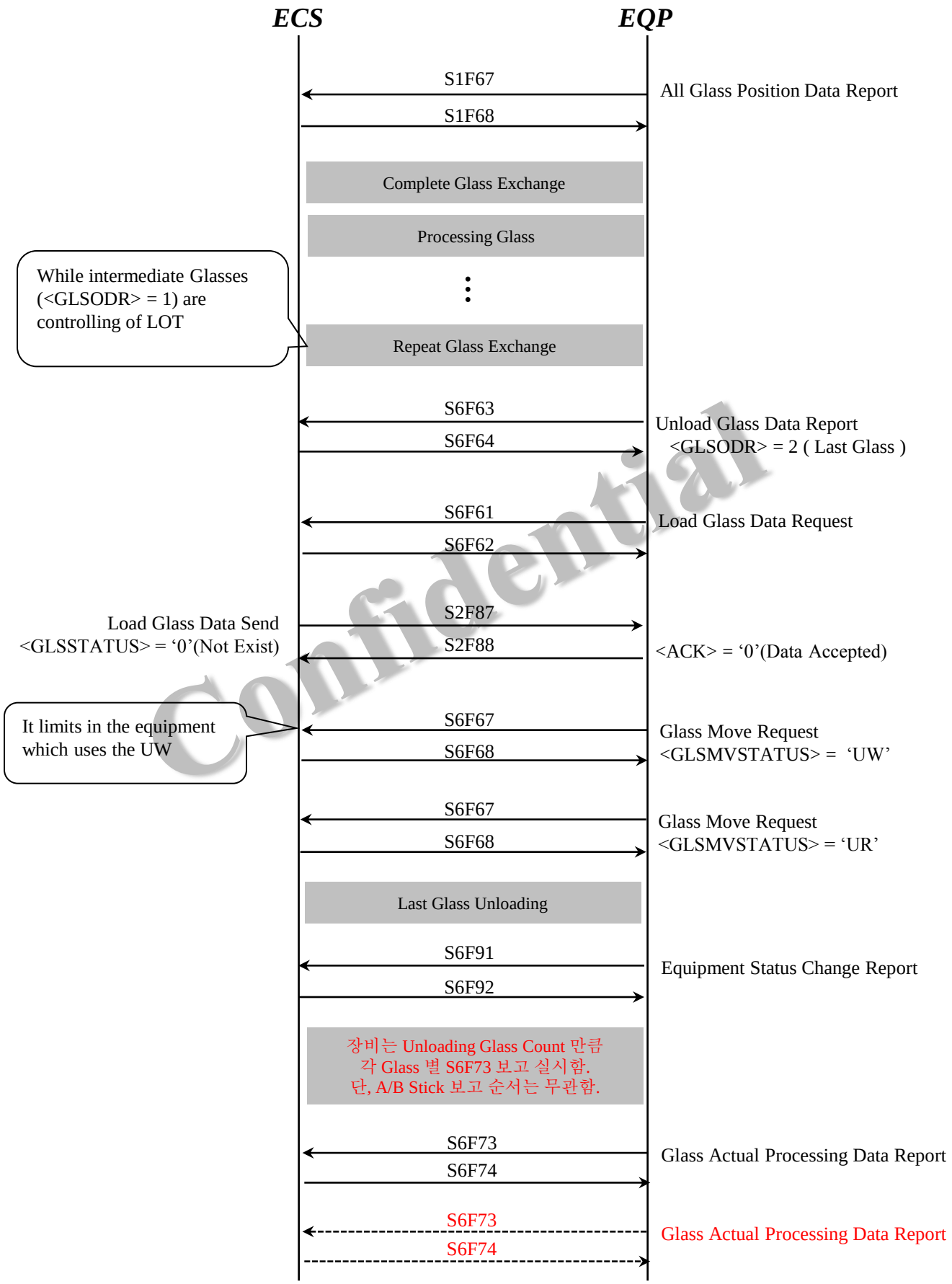
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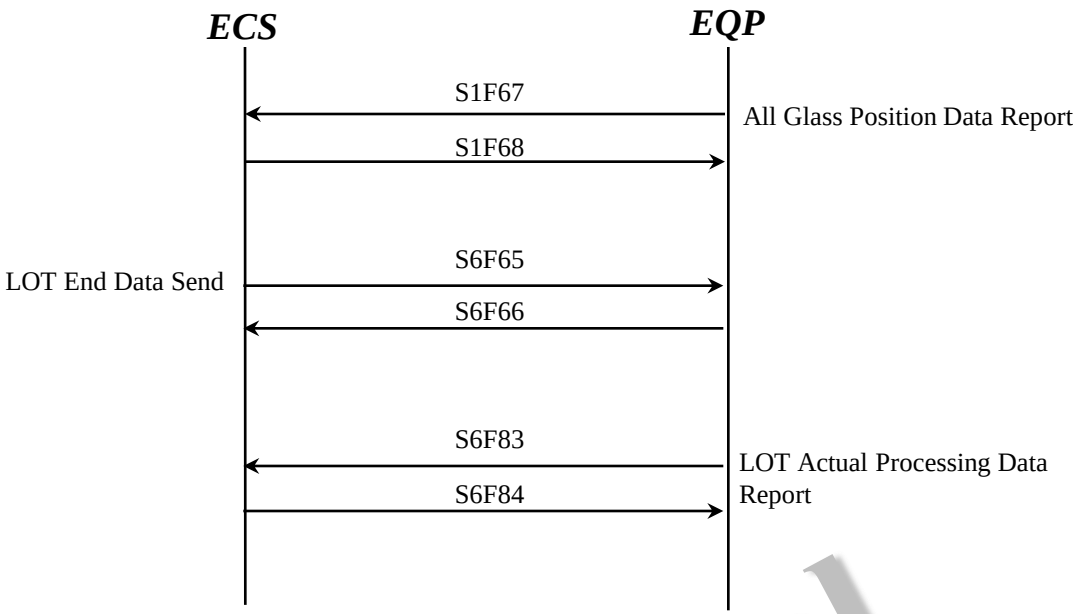
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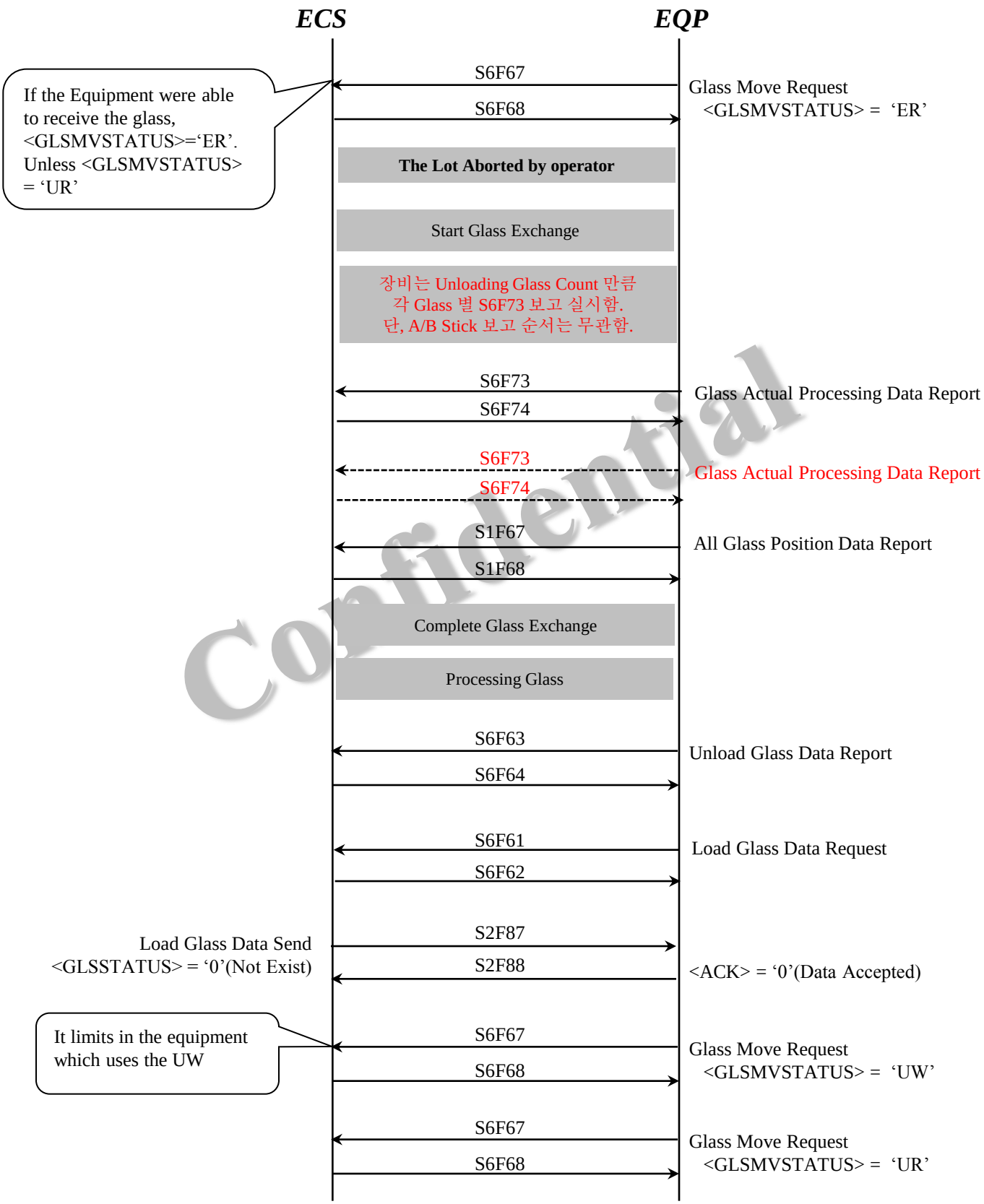


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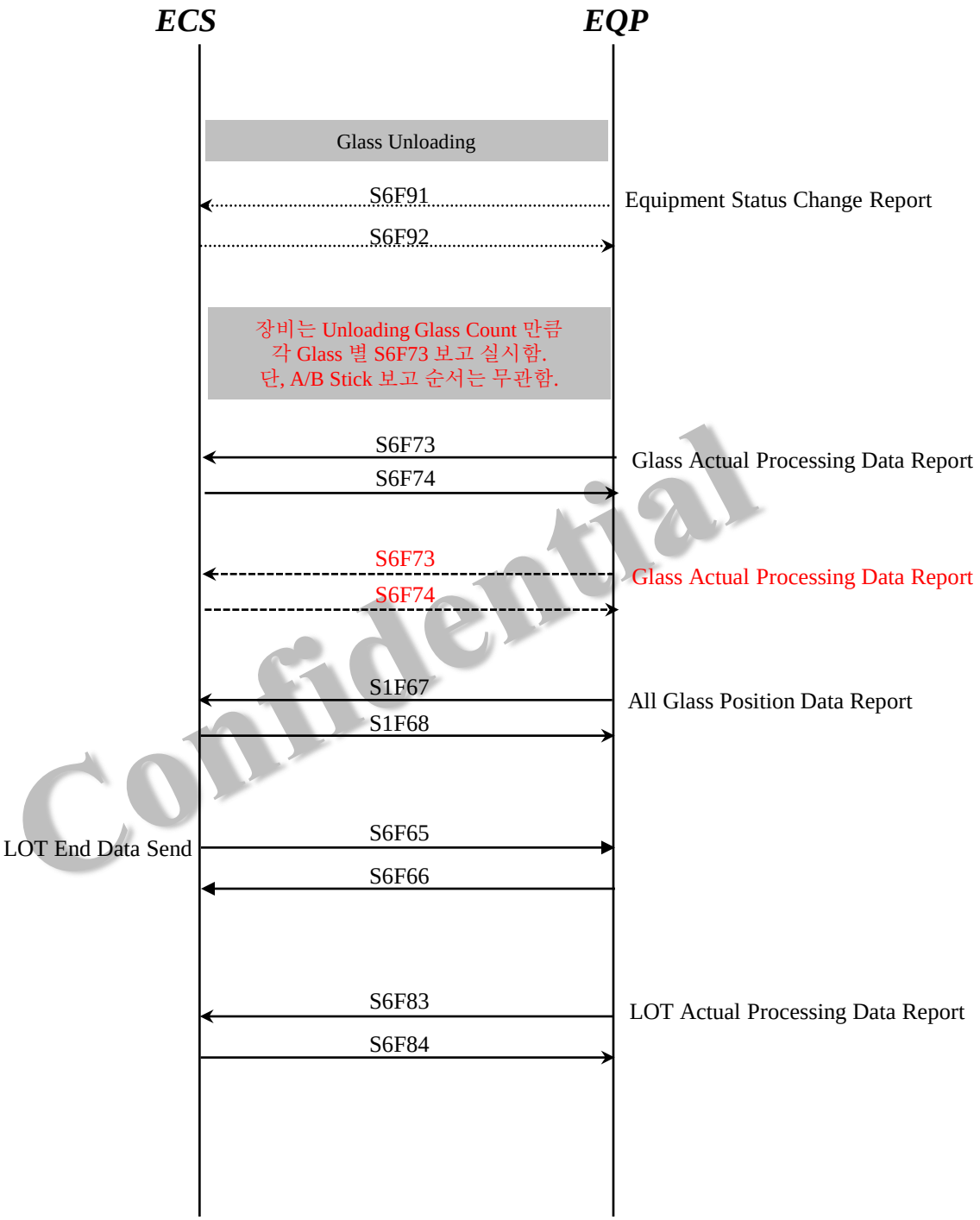
7.17 Lot Abort Scenario

7.17.1 The case where glass would be loaded is exist



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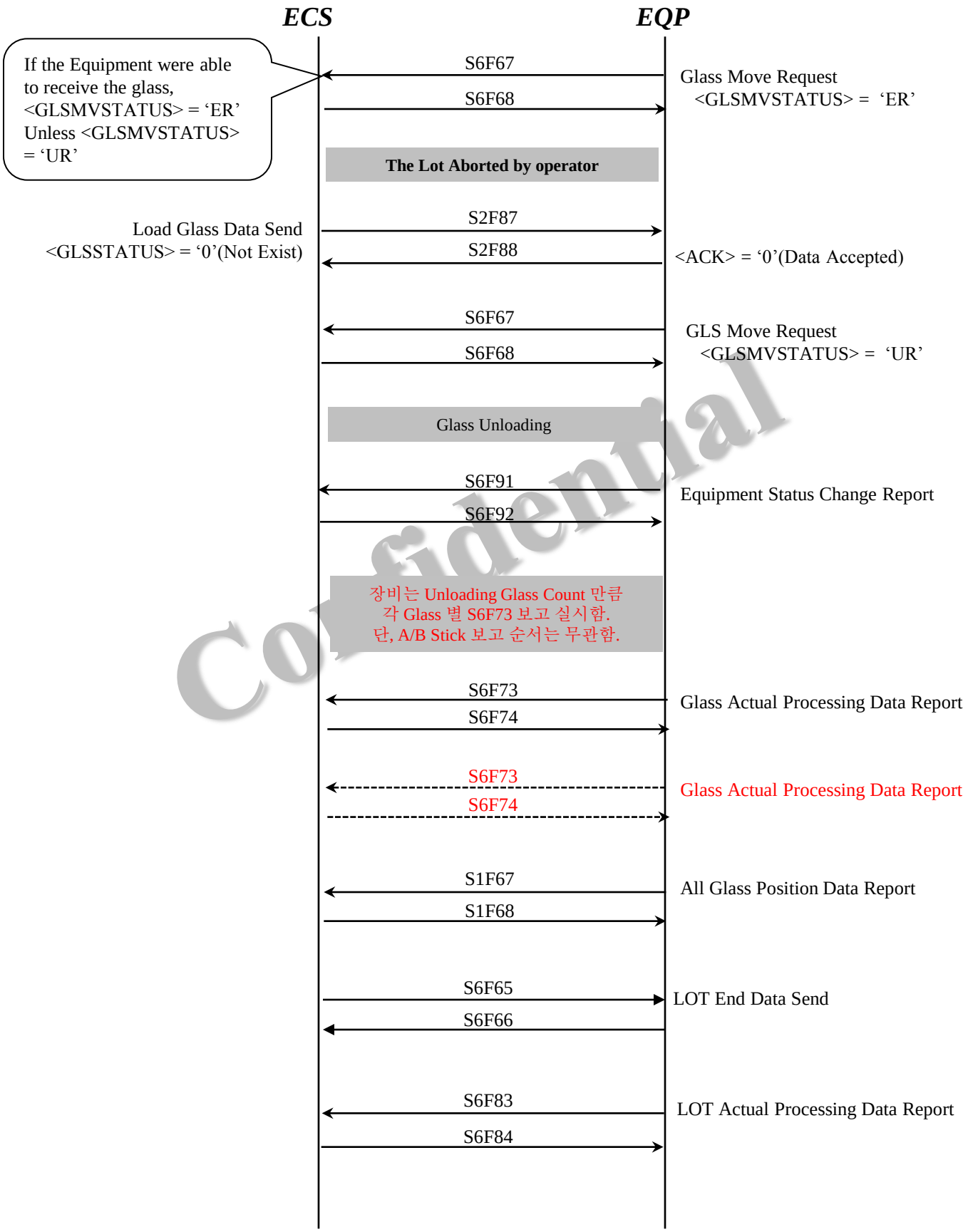
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7.17.2 The case where glass would be loaded is not exist

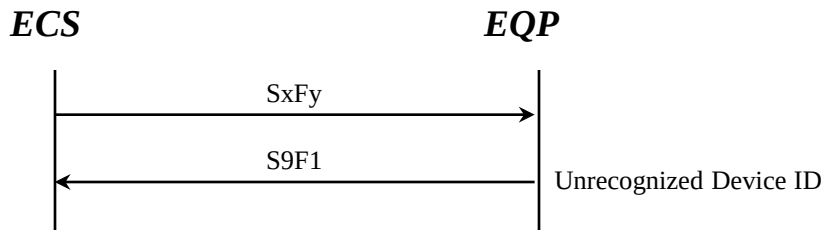


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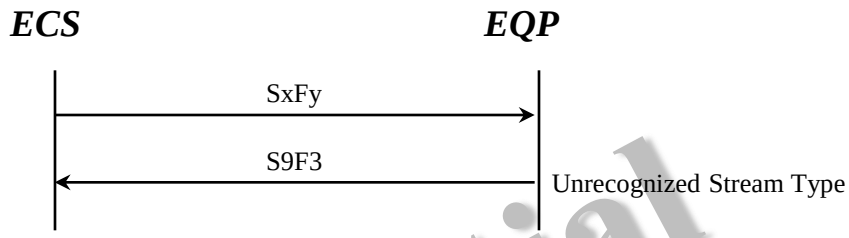
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7.18 Error Messages

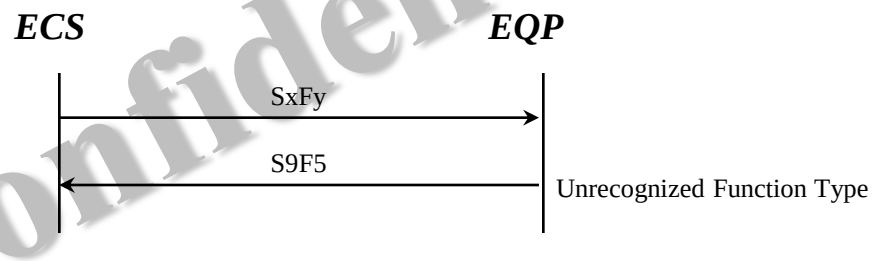
7.18.1 Message fault due to unrecognized Device ID



7.18.2 Message fault due to unrecognized Stream Type

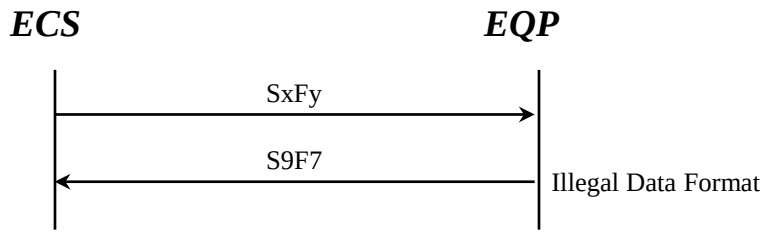


7.18.3 Message fault due to unrecognized Function Type

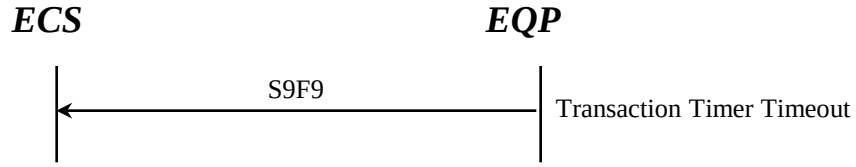


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7.18.4 Message fault due to Illegal Data Format

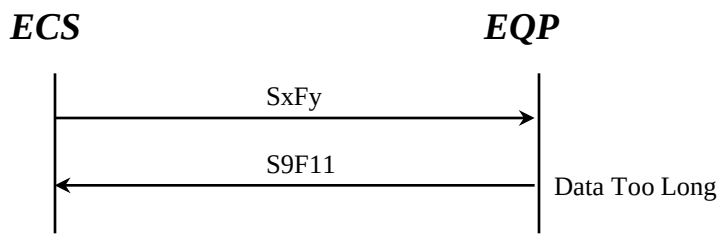


7.18.5 Message fault due to Transaction Timer Timeout

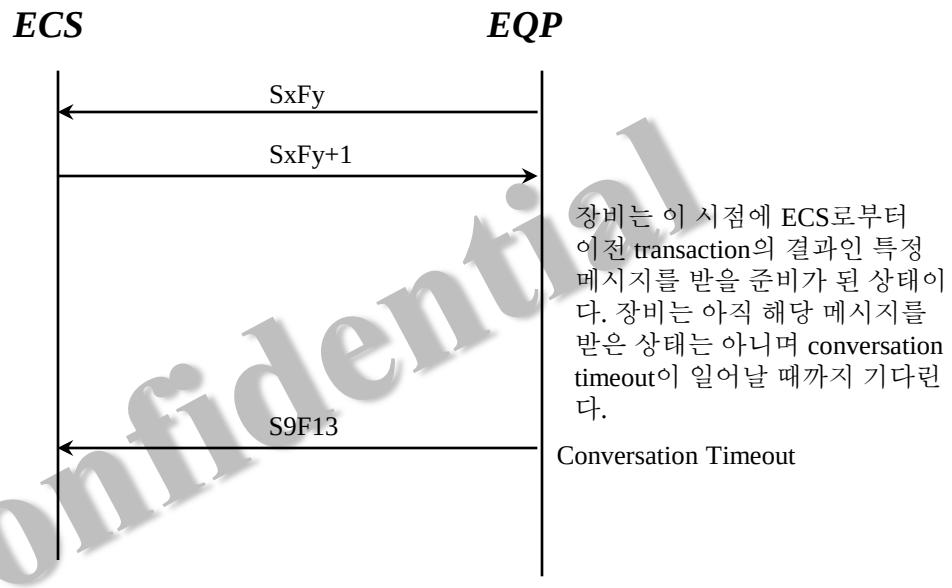


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7.18.6 Message fault due to data too long



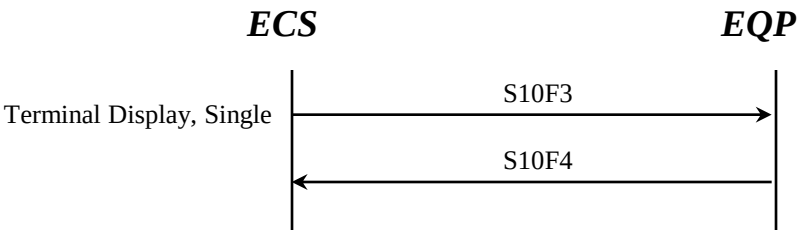
7.18.7 Message fault due to Conversation Timeout



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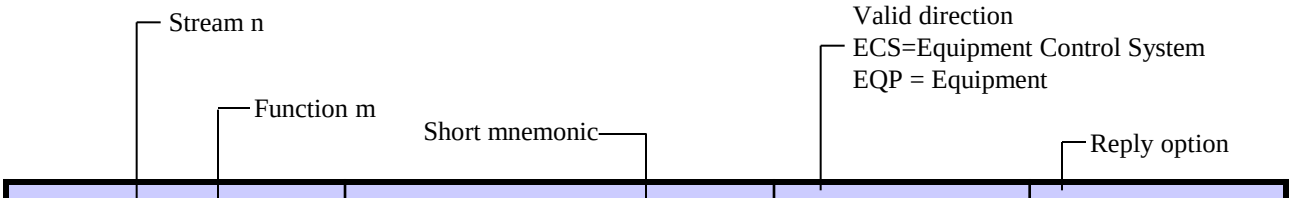
7.19 Equipment Terminal Service



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8. Standard Message Structure



Sn,Fm	Name of Function(XXX)	ECS→EQP ECS←EQP ECS↔EQP	Reply(required) [Reply](optional) "blank"(never)

Description	The description of the action generated by the function.
Structure	Detailed structure shows lists and defined items. Lists are denoted by the number separated by a comma following a capital "L". The individual elements in the list are numbered on separate lines. Nested lists are indented to emphasize the structure. The detailed form of the items is given in the define section at the beginning of the transaction. The symbols "<" and ">" are used to enclose each item in the structure data and imply that there is an item header. A detailed description of each data item as well as a list of the allowable data formats can be found in the Data Item Dictionary
Notice	A description of the data item, with the meanings of specific values.
Exception	Special cases in the structure that have a different meaning.

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Sn,F0	Abort transaction	ECS↔EQP	
Description	Function 0 exists in all streams and has the same special meaning in each stream. The function 0 message closes a transaction, so that the originator will not have to wait for a transaction timeout to proceed. Function 0 is sent in lieu of the expected secondary message when the interpreter cannot, because of a transmission error or some other reasons, respond with the expected reply. It is not a requirement that the interpreter send function 0 to close a transaction.		
Structure	Header Only		
Notice			
Exception			

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S1,F1	Are You There Request	ECS↔EQP	reply
Description	If the equipment turn to online mode, Function 0 response to this message means the communication is inoperative. In the equipment, a function 0 is equivalent to a timeout on the receive timer after issuing S1,F1 to ECS		
Structure	Header Only		
Notice			
Exception			

S1,F2	On Line Data	ECS↔EQP	
Description	Data signifying that the equipment is alive.		
Structure	L,2 1. A[8] <MDLN> 2. A[6] <SOFTREV>		
Notice	<MDLN> : Equipment Model Type <SOFTREV> : Software revision code		
Exception	The ECS sends a zero-length list to the equipment		

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S1,F61	All Equipment Status Report	ECS←EQP	reply

Description	When equipment is initialized or control state is changed to Online, the equipment reports this message
Structure	L,2 1. L,n [n = the number of Unit] 1. L,4 1. U2[1] <UNITID1> 2. A[1] <EQPSTATUS1> 3. A[10] <ALID1> 4. A[120] <ALTX1> … n. L,4 1. U2[1] <UNITIDn> 2. A[1] <EQPSTATUSn> 3. A[10] <ALIDn> 4. A[120] <ALTXn> 2. L,p [p = the number of Ionizer] 1. L,2 1. B[1] <IONIZERID1> 2. A[1] <IONIZERSTATUS1> … p. L,2 1. B[1] <IONIZERIDp> 2. A[1] <IONIZERSTATUSp>
Notice	<UNITID>Unit ID <EQPSTATUS> Equipment Status R : RUN E : Experiment V : Development C : Model Change M : Monitoring I : IDLE D : DOWN P : Process Down <ALID> Alarm Identification If <EQPSTATUS> is 'R' or 'I', then <ALID> is fixed by '0' <ALTX> Alarm Message <IONIZERID> Ionizer ID <IONIZERSTATUS> Ionizer Status R : RUN D : DOWN
Exception	

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S1,F62	All Equipment Status Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC1>		
Notice	<UNITID> Unit ID <ACKC1> 0 = Accepted 1 = Not Accepted		
Exception			

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S1,F63	All Equipment Status Request	ECS→EQP	reply
Description	ECS requests the status of all units of the Equipment.		
Structure	Header Only		
Notice			
Exception			

S1,F64	All Equipment Status Data	ECS←EQP	
Description	The Equipment reports all EQP Status Data		
Structure	L,2 1. L,n [n = the number of Unit] 1. L,4 1. U2[1] <UNITID1> 2. A[1] <EQPSTATUS1> 3. A[10] <ALID1> 4. A[120] <ALTX1> … n. L,4 1. U2[1] <UNITIDn> 2. A[1] <EQPSTATUSn> 3. A[10] <ALIDn> 4. A[120] <ALTXn> 2. L,p [p = the number of Ionizer] 1. L,2 1. B[1] <IONIZERID1> 2. A[1] <IONIZERSTATUS1> … p. L,2 1. B[1] <IONIZERIDp> 2. A[1] <IONIZERSTATUSp>		
Notice	<UNITID>Unit ID <EQPSTATUS> Equipment Status R : RUN E : Experiment V : Development C : Model Change M : Monitoring I : IDLE D : DOWN P : Process Down <ALID> Alarm Identification If <EQPSTATUS> is 'R' or 'I', then <ALID> is fixed by '0' <ALTX> Alarm Message <IONIZERID> Ionizer ID <IONIZERSTATUS> Ionizer Status R : RUN D : DOWN		
Exception			

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S1,F67	All Glass Position Data Report	ECS←EQP	reply

Description	This message is used for tracking of glasses which moved in among sub units
Structure	L,n [n=SubUnit count] 1.L,3 1.U2[1] <UNITID1> 2.B[1] <SUNITID1> 3.L,m [m = glass count] 1.U2[1] <GLSCODE1> . . m.U2[1]<GLSCODEm> . . n.L,3 1.U2[1] <UNITIDn> 2.B[1] <SUNITIDn> 3.L,m [m = glass count] 1.U2[1] <GLSCODE1> . . m.U2[1]<GLSCODEm>
Notice	<UNITID> Unit ID <SUNITID> Sub Unit ID <GLSCODE> Unique Glass Code defined by ECS. Equipment must keep this code, while glass is in equipment.
Exception	A one-to-one correspondence of UNITID and SUNITID depends on Equipment unit configuration

S1,F68	All Glass Position Data Report Acknowledge	ECS→EQP	

Description	Acknowledge
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC1>
Notice	<UNITID> Unit ID <ACKC1> 0 = Accepted 1 = Not Accepted
Exception	

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S1,F75	Ionizer Status Report	ECS←EQP	reply
Description	This message is used in Ionizer Status Report .		
Structure	L,2 1. U2[1] <UNITID> 2. L,n [n = the number of Ionizer] 1. L,2 1. B[1] <IONIZERID> 2. A[1] <IONIZERSTATUS1> … n. L,2 1. B[1] <IONIZERIDn> 2. A[1] <IONIZERSTATUSn>		
Notice	<UNITID>Unit ID <IONIZERID> Ionizer ID <IONIZERSTATUS> Ionizer Status R : RUN D : DOWN		
Exception			

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S1,F76	Ionizer Status Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC1>		
Notice	<UNITID> Unit ID <ACKC1> 0 = Accepted 1 = Not Accepted		
Exception			

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S2,F17	Date and Time Request	ECS←EQP	reply
Description	This message is Useful, when ECS checks the equipment time base or the equipment to synchronize with ECS time base.		
Structure	U2[1]<UNITID>		
Notice	<UNITID> Unit ID		
Exception			

S2,F18	Date and Time Data	ECS→EQP	
Description	Actual time data.		
Structure	A[14] <TIME>		
Notice	<TIME> : Current Time “YYYYMMDDhhmmss”		
Exception			

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S2,F25	Loopback Diagnostic Request	ECS↔EQP	reply
Description	This is the diagnostic message for checkout of protocol and communication circuits. When the binary string is sent, the same data is echoed back.		
Structure	B[10] <ABS>		
Notice	<ABS> Any binary string		
Exception			

S2,F26	Loopback Diagnostic Data	ECS↔EQP	
Description	The echoed binary string		
Structure	B[10] <ABS>		
Notice			
Exception			

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S2,F31	Date and Time Data Send	ECS→EQP	reply
Description	Actual time data Send.		
Structure	L,2 1. U2[1] <UNITID> 2. A[14] <TIME>		
Notice	<UNITID> Unit ID <TIME>: Current Time "YYYYMMDDhhmmss"		
Exception			

S2,F32	Date and Time Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC2>		
Notice	<UNITID> Unit ID <ACKC2> 0 = Accepted 1 = Not Accepted		
Exception			

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S2,F61	Control State Change Request	ECS←EQP	reply
Description	This message is used when requesting the control state change.		
Structure	L,2 1.U2[1]<UNITID> 2.A[1] <RCS>		
Notice	<UNITID> Unit ID <RCS> Remote Control State 0 = Online-Remote 1 = Online-Local 2 = Offline		
Exception			

S2,F62	Control State Change Request Acknowledge	ECS→EQP	
Description	Reply for the control state change of the equipment		
Structure	L2 1.U2[1]<UNITID> 2.B[1] <ACKC2A>		
Notice	<UNITID> Unit ID < ACKC2A > Acknowledge code, 1 byte 0 = Mode Change Accept 1 = Mode Change not Allowed 2 = Equipment Already Changed Mode 3 - 63 Reversed		
Exception			

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S2,F63	Equipment Operation Mode Report	ECS←EQP	reply
Description	Change operation mode of the equipment		
Structure	L,2 1.U2[1]<UNITID> 2.A[1] <OPMODE>		
Notice	<UNITID> Unit ID < OPMODE > Operation Mode Code *The following OPMODE values define the case of operate to Online(Remote or Local)* 1 = Normal / Load & Process / Auto / Remote / Operation / Automatic Mode 2 = Sequence Transfer / Online Local / Local / Semi-Auto Mode 3 = Cleanout / Recovery Mode 4 = Test / Atmospheric Transfer Mode 5 = Transfer / Vacuum Transfer Mode 6 = Particle Check / Pass Mode 7 = Reverse Mode 8 = Repair Mode 9 = Manual A = Restoration		
Exception	The operation mode to operate Offline dose not define OPMODE (ex. Manual / Maintenance / Offline / Local / etc…)		

S2,F64	Equipment Operation Mode Report Acknowledge	ECS→EQP	
Description	Reply for Change operation mode of the equipment		
Structure	L,2 1.U2[1]<UNITID> 2.B[1] <ACKC2A>		
Notice	<UNITID> Unit ID < ACKC2A > Acknowledge code, 1 byte 0 = Mode Change Accept 1 = Mode Change not Allowed 2 = Equipment Already Changed Mode 3 - 63 Reversed		
Exception			

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S2,F65	Local LOT Status Change Request	ECS←EQP	reply
Description	This message is used when sending Lot End Data under Online Local Control state.		
Structure	L,4 1.U2[1]<UNITID> 2. B[1]<PORTID> 3. A[16] <LOTID> 4. A[1]<ENDFLAG>		
Notice	<UNITID> Unit ID <PORTID> Port ID <LOTID> LOT ID <ENDFLAG> End Flag N : Normal End P : Operator End		
Exception			

S2,F66	Local LOT Status Change Request Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC2>		
Notice	<UNITID> Unit ID <ACKC2> 0 = Accepted 1 = Not Accepted		
Exception			

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S2,F67	Cycle Stop Status Report	ECS←EQP	reply
Description	Change cycle stop status of the equipment		
Structure	L,2 1.U2[1] <UNITID> 2.B[1]<CSSTATUS>		
Notice	<UNITID> Unit ID <CSSTATUS> 0 = Cycle Stop Release 1 = Cycle Stop Set		
Exception			

S2,F68	Cycle Stop Status Report Acknowledge	ECS→EQP	
Description	Reply for Change cycle stop status of the equipment		
Structure	L,2 1.U2[1] <UNITID> 2.B[1]<ACKC2>		
Notice	<UNITID> Unit ID <ACKC2> 0 = Accepted 1 = Not Accepted 2 – 63 Reserved		
Exception			

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S2,F69	Unload Glass Data Request	ECS→EQP	reply
Description	This message is used when ECS Request Unload Glass Data from the Equipment		
Structure	L,3 1. U2[1] <UNITID> 2. B[1] <STAGEID> 3. A[80] <HOSTMSG>		
Notice	<UNITID> Unit ID <STAGEID> Stage ID <HOSTMSG> Host message in Korean and English		
Exception			

S2,F70	Unload Glass Data Request Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,2 1.U2[1] <UNITID> 2.B[1]<ACKC2>		
Notice	<UNITID> Unit ID <ACKC2> 0 = Accepted 1 = Not Accepted 2 = Unload Glass Not Exist 3 – 63 Reserved		
Exception			

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S2,F75	Chemical Change Status Report	ECS←EQP	reply
Description	When process changes the chemical, process reports chemical change status report.		
Structure	L,2 1.U2[1] <UNITID> 2. A[1] <CHEMCH>		
Notice	<UNITID> Unit ID <CHEMCH> Chemical Change Status 1 = Start of Chemical Change 0 = End of Chemical Change		
Exception			

S2,F76	Chemical Change Status Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC2>		
Notice	<UNITID>Unit ID <ACKC2> 0 = Accepted >0 = Error, not accepted 1 – 63 Reserved		
Exception			

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S2,F79	Glass Data Request	ECS←EQP	reply
Description	The Equipment only can use this control sequence in its manual operation. It can be use when equipment lost 'Glass Data', but equipment still have Glass ID or Glass code.		
Structure	L,4 1. U2[1]<UNITID> 2. A[16] <GLSID> 3. U2[1] <GLSCODE> 4. A[1] <REQOPTION>		
Notice	<UNITID> Unit ID <GLSID> Glass ID <REQOPTION> Request Option C : Glass Code I : Glass ID A : Glass Code and Glass ID		
Exception			

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S2,F80	Glass Data	ECS→EQP	
Description	The Equipment (manual operation) sends S2F79 and then ECS sends 'Glass Load Data' as reply. 'Glass Load Data' is data about injected glass when glass inject to equipment.		
Structure	L, 3 1. U2[1] <UNITID> 2. B[1]<ACKC2> 3. L,n [n = 0(GLSSTATUS = 0) or 21 (GLSSTATUS = 1)] 1. A[16] <LOTID> 2. A[14] <CSTID> 3. A[3] <SLOTID> 4. A[24] <RECIPEID> 5. A[8] <PRCID> 6. A[1] <GLSODR> 7. A[16] <GLSID> 8. A[2] <GLSJUDGE> 9. A[2000] <PNLIF> 10. A[2000] <SUBMDLIF> 11. B[1] <PORTID> 12. U2[1] <GLSCODE> 13. B[1] <REINPUT> 14. A[5] <GLSTHICK> 15. A[14] <PARTNUM> 16. A[12] <PRODTYPE> 17. A[12] <ATTRIBUTE> 18. A[1] <GLSTYPE> 19. A[32]<KEYID> 20. L, p [p = the number of LSD] 1. L,2 1. A[24] <LSDITEM1> 2. A[40] <LSDVALUE1> : p. L,2 1. A[24] <LSDITEMp> 2. A[40] <LSDCALUEp> 21. L,q [q = the number of GSD] 1.L,2 1. A[24] <GSDITEM11> 2. A[40] <GSDVALUE11> : q. L,2 1. A[24] <GSDITEM1q> 2. A[40] <GSDVALUE1q>		

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Notice	<UNITID> Unit ID <ACKC2> 0 = Accepted 1 = Not Accepted (Unknown GLSID or GLSCODE) 2 – 63 Reserved <LOTID> LOT ID <CSTID> Cassette ID <SLOTID> Slot ID <RECIPEID> Recipe ID <PRCID> Processing Code <GLSODR> Glass Order 0 = The first glass 1 = All glasses except the first glass and the last glass 2 = The Last glass(In this case, this value may not be delivered for Scrap or reason of LOT Abort) 3 = the only glass, when there is just one in a LOT. <GLSID> Glass ID <GLSJUDGE> Glass Grade Code <PNLIF> Panel Judge Information <SUBMDLIF> Sub Panel Information <PORTID> Port ID <GLSCODE> Glass ID Code. <REINPUT> Glass Re-input Flag 0 = Not Re-input 1 = Re-input <GLSTHICK> Glass's Thickness <PARTNUM> Part Number <PRODTYPE> Type of Producing <ATTRIBUTE> LOT Attribute <GLSTYPE> Glass Type N = Normal Glass D = Dummy Glass <KEYID> Glass Key ID <LSDITEM> LOT Specific Data Item defined by LG Display <LSDVALUE> LOT Specific Data Value defined by LG Display <GSDITEM> Glass Specific Data Item defined by LG Display <GSDVALUE> Glass Specific Data Value defined by LG Display
Exception	

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S2,F87	Load Glass Data Send	ECS→EQP	reply

Description	This is the Glass data that would be loaded into the equipment.
Structure	L, 4 1. U2[1] <UNITID> 2. B[1] <STAGEID> 3. A[1] <GLSSTATUS> 4. L,n [n = 0 (GLSSTATUS = 0) or 2 (GLSSTATUS = 1)] 1. L,21 [(A Cut Glass)] 1. A[16] <LOTID> 2. A[14] <CSTID> 3. A[3] <SLOTID> 4. A[24] <RECIPEID> 5. A[8] <PRCID> 6. A[1] <GLSODR> 7. A[16] <GLSID> 8. A[2] <GLSJUDGE> 9. A[2000] <PNLIF> 10. A[2000] <SUBMDLIF> 11. B[1] <PORTID> 12. U2[1] <GLSCODE> 13. B[1] <REINPUT> 14. A[5] <GLSTHICK> 15. A[14] <PARTNUM> 16. A[12] <PRODTYPE> 17. A[12] <ATTRIBUTE> 18. A[1] <GLSTYPE> 19. A[32] <KEYID> 20. L,q [q = the number of LSD] 1. L,2 1. A[24] <LSDITEM1> 2. A[40] <LSDVALUE1> : q. L,2 1. A[24] <LSDITEMq> 2. A[40] <LSDCALUEq> 21. L,r [r = the number of GSD] 1.L,2 1. A[24] <GSDITEM11> 2. A[40] <GSDVALUE11> : r. L,2 1. A[24] <GSDITEM1r> 2. A[40] <GSDVALUE1r> 2. L,21 [(B Cut Glass)] 1. A[16] <LOTID> 2. A[14] <CSTID> 3. A[3] <SLOTID> 4. A[24] <RECIPEID> 5. A[8] <PRCID> 6. A[1] <GLSODR>

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Structure	7. A[16] <GLSID> 8. A[2] <GLSJUDGE> 9. A[2000] <PNLIF> 10. A[2000] <SUBMDLIF> 11. B[1] <PORTID> 12. U2[1] <GLSCODE> 13. B[1] <REINPUT> 14. A[5] <GLSTHICK> 15. A[14] <PARTNUM> 16. A[12] <PRODTYPE> 17. A[12] <ATTRIBUTE> 18. A[1] <GLSTYPE> 19. A[32] <KEYID> 20. L,t [t = the number of LSD] 1. L,2 1. A[24] <LSDITEM1> 2. A[40] <LSDVALUE1> : t. L,2 1. A[24] <LSDITEMt> 2. A[40] <LSDCALUEt> 21. L,u [u = the number of GSD] 1.L,2 1. A[24] <GSDITEM11> 2. A[40] <GSDVALUE11> : u. L,2 1. A[24] <GSDITEM1u> 2. A[40] <GSDVALUE1u>
Notice	<UNITID> Unit ID <STAGEID> Stage ID <GLSSTATUS> Status of the glass that would be loaded 0 = Not Exist 1 = Exist <LOTID> LOT ID <CSTID> Cassette ID <SLOTID> Slot ID <RECIPEID> Recipe ID <PRCID> Processing Code <GLSODR> Glass Order 0 = The first glass 1 = All glasses except the first glass and the last glass 2 = The Last glass (In this case, this value may not be delivered for Scrap or reason of LOT Abort) 3 = the only glass, when there is just one in a LOT. <GLSID> Glass ID <GLSJUDGE> Glass Grade Code <PNLIF> Panel Judge Information <SUBMDLIF> Sub Panel Information <PORTID> Port ID <GLSCODE> Glass ID Code. <REINPUT> Glass Re-input Flag 0 = Not Re-input 1 = Re-input <GLSTHICK> Glass's Thickness <PARTNUM> Part Number

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Notice	<PRODTYPE> Type of Producing <ATTRIBUTE> LOT Attribute <GLSTYPE> Glass Type N = Normal Glass D = Dummy Glass <KEYID>Glass Key ID <LSDITEM> LOT Specific Data Item defined by LG Display <LSDVALUE> LOT Specific Data Value defined by LG Display <GSDITEM> Glass Specific Data Item defined by LG Display <GSDVALUE> Glass Specific Data Value defined by LG Display <--GSD Item/Value Define-> <CUT_TYPE> Glass Cut Type A : A Cut DA : Dummy A Cut B : B Cut DB : Dummy B Cut <PAIRINFO> A,B Cut Glass Pair Information Y : Paired N : Unpaired <PAIR_GLASS_CODE> Matching Pair Glass Code Glass Code <ROUTE_CHAMBER> Chamber Route Information Chamber Route Information
Exception	

S2,F88	Load Glass Data Send Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,3 1. U2[1] <UNITID> 2. B[1] <STAGEID> 3. B[1] <ACKC2>		
Notice	<UNITID> Unit ID <STAGEID> Stage ID <ACKC2> 0 = Accept 1 = Process Condition is not ready 2 = The recipe is not exist in equipment's recipe list 3 = Glass data is not accepted 4 = Process is not ready to load this glass 5 = A, B Cut Recipe Mismatching 6~63 = Reserved		
Exception			

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S2,F95	Chemical Change Enable Request	ECS←EQP	reply
Description	This message is used to request chemical change enable to ECS.		
Structure	U2[1] <UNITID>		
Notice	<UNITID> Unit ID		
Exception			

S2,F96	Chemical Change Enable Request Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC2>		
Notice	<UNITID>Unit ID <ACKC2> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved		
Exception			

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S2,F97	Chemical Change Enable Send	ECS→EQP	reply
Description	This message is used to send the chemical change enable to the equipment		
Structure	L,2 1. U2[1] <UNITID> 2. A[1] <PERMITCD>		
Notice	<UNITID> Unit ID <PERMITCD> Chemical change permit code 0 : change enable 1 ~ 63 : Reserved		
Exception			

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S2,F98	Chemical Change Enable Send Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC2>		
Notice	<UNITID>Unit ID <ACKC2> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved		
Exception			

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S5,F1	Alarm Report	ECS←EQP	reply

Description	This message reports the alarm condition changes inside EQP or of the previous step. One message will be issued when the alarm is set and the other will be issued when cleared. Irrecoverable errors and attention flags may not have the clear message corresponding to them.
Structure	L,4 1.U2[1] <UNITID> 2. B[1] <ALCD> 3. A[10] <ALID> 4. A[120] <ALTX>
Notice	<UNITID> Unit ID <ALCD>Alarm Code bit 8 = 1 Alarm set bit 8 = 0 Alarm Reset bit 7~1 is Alarm category(0~127) 0 = Not Used 1 = Personal safety 2 = Equipment safety 3 = Parameter control warning 4 = Parameter control error 5 = Irrecoverable error 6 = Equipment status warning 7 = Attention flags 8 = Data integrity <ALID> Alarm Identification <ALTX>Alarm Message
Exception	

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S5,F2	Alarm Report Acknowledge	ECS→EQP	

Description	Acknowledge or error.
Structure	L,2 1. U2[1]<UNITID> 2. B[1] <ACKC5>
Notice	<ACKC5> Acknowledge code, 1 byte 0 = Accepted (Fixed)
Exception	

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S6,F61	Load Glass Data Request	ECS←EQP	reply
Description	This message is used in Load Glass Data Request		
Structure	L,2 1. U2[1] <UNITID> 2. B[1] <STAGEID>		
Notice	<UNITID> Unit ID <STAGEID> Stage ID		
Exception			

S6,F62	Load Glass Data Request Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,3 1. U2[1] <UNITID> 2. B[1] <STAGEID> 3. B[1]<ACKC6>		
Notice	<UNITID> Unit ID <STAGEID> Stage ID <ACKC2> 0 = Accept 1 = Not Accept		
Exception			

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S6,F63	Unload Glass Data Report	ECS←EQP	reply

Description	This message is used in Unload Glass Data Report.
Structure	L, 3 1. U2[1] <UNITID> 2. B[1] <STAGEID> 3. L,2 1. L,21 [(A Cut Glass)] 1. A[1] <GLSODR> 2. A[16] <LOTID> 3. A[14] <CSTID> 4. A[3] <SLOTID> 5. A[16] <GLSID> 6. A[2000] <PNLIF> 7. A[2000] <SUBMDLIF> 8. A[8] <PRCID> 9. B[1] <DICODE> 10. B[1] <PORTID> 11. A[24] <RECIPEID> 12. A[2] <GLSJUDGE> 13. A[14] <PARTNUM> 14. A[12] <PRODTYPE> 15. A[12] <ATTRIBUTE> 16. A[1] <GLSTYPE> 17. A[1] <GLSPROC> 18. U2[1] <GLSCODE> 19. U2[1] <DEST> 20. A[32] <KEYID> 21. L,p [n = the number of GSD] 1. L, 2 1. A[24] <GSDITEM1> 2. A[40] <GSDVALUE1> : p. L, 2 1. A[24] <GSDITEMp> 2. A[40] <GSDVALUEp> 2. L,21 [(B Cut Glass)] 1. A[1] <GLSODR> 2. A[16] <LOTID> 3. A[14] <CSTID> 4. A[3] <SLOTID> 5. A[16] <GLSID> 6. A[2000] <PNLIF> 7. A[2000] <SUBMDLIF> 8. A[8] <PRCID> 9. B[1] <DICODE> 10. B[1] <PORTID> 11. A[24] <RECIPEID> 12. A[2] <GLSJUDGE> 13. A[14] <PARTNUM> 14. A[12] <PRODTYPE> 15. A[12] <ATTRIBUTE>

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Structure	16. A[1] <GLSTYPE> 17. A[1] <GLSPROC> 18. U2[1] <GLSCODE> 19. U2[1] <DEST> 20. A[32] <KEYID> 21. L,r [n = the number of GSD] 1. L, 2 1. A[24] <GSDITEM1> 2. A[40] <GSDVALUE1> : r. L, 2 1. A[24] <GSDITEMr> 2. A[40] <GSDVALUEr>
Notice	<UNITID> Unit ID <STAGEID> Stage ID <GLSODR> Glass Order 0 = The first glass 1 = All glasses except the first glass and the last glass 2 = The Last glass 3 = the only glass, when there is just one in a LOT. <LOTID> LOT ID <CSTID> Cassette ID <SLOTID> Slot ID <GLSID> Glass ID <PNLIF> Panel Information <SUBMDLIF> Sub Panel Information <PRCID> Processing Code <DICODE> Different Code 0 : All GLS IDs are completely matching with GLS IDs from ECS. 1 : Any GLS ID is not matching with GLS ID from ECS 2 : When each ID of glass is not checked by VCR or OCR in the equipment has VCR or OCR <PORTID> Port ID <RECIPE ID> Recipe ID <GLSJUDGE> Glass Grade Code <PARTNUM> Part Number <PRODTYPE> Type of Producing <ATTRIBUTE> LOT Attribute <GLSTYPE> Glass Type N = Normal Glass D = Dummy Glass <GLSPROC> Glass Process Status 0 : Unprocessed 1 : Processing 2 : Processed 3 : Rejected <GLSCODE> Glass ID Code

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Notice	<DEST> Glass Next Destination 0 = ECS determine destination of glass. >0 = Equipment determine destination of glass. *The value of <DEST> will be decided later through conversation* <KEYID> Glass Key ID <GSDITEM> Glass Specific Data Item defined by LG Display <GSDVALUE> Glass Specific Data Value defined by LG Display <--GSD Item/Value Define-> <CUT_TYPE> Glass Cut Type A : A Cut DA : Dummy A Cut B : B Cut DB : Dummy B Cut <PAIRINFO> A,B Cut Glass Pair Information Y : Paired N : Unpaired <PAIR_GLASS_CODE> Matching Pair Glass Code Glass Code <ROUTE_CHAMBER> Chamber Route Information Chamber Route Information
Exception	When <GSDITEM>, <GSDVALUE> is not necessary, n is 0.

S6,F64	Unload Glass Data Report Acknowledge	ECS→EQP	
Description	Acknowledge.		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>		
Notice	<UNITID>Unit ID <ACKC6> 0 = Accepted >0 = Error, not accepted 1 – 63 Reserved		
Exception			

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S6,F65	LOT End Data Send	ECS→EQP	reply

Description	This message is used in Lot End Data Send
Structure	L,9 1. U2[1]<UNITID> 2. B[1]<PORTID> 3. A[16] <LOTID> 4. A[14] <CSTID> 5. A[24] <RECIPEID> 6. A[8] <PRCID> 7. A[1]<ENDFLAG> 8. A[1] <MULTIFLAG> 9. L,n (n = 0 or the number of slot) 1. L,10 1. A[1] <GLSODR1> 2. A[3] <SLOTID1> 3. A[16] <GLSID1> 4. A[24] <RECIPEID1> 5. B[1]<GLSEXIST1> 6. B[1]<GLSSELECTED1> 7. U2[1]<PROCUNITID1> 8. A[2]<GLSJUDGE1> 9. A[2000]<PNLIF1> 10. A[2000]<SUBMDLIF1> : n. L,10 1. A[1] <GLSODRn> 2. A[3] <SLOTIDn> 3. A[16] <GLSIDn> 4. A[24] <RECIPEIDn> 5. B[1]<GLSEXISTn> 6. B[1]<GLSSELECTEDn> 7. U2[1]<PROCUNITIDn> 8. A[2]<GLSJUDGEn> 9. A[2000]<PNLIFn> 10. A[2000]<SUBMDLIFn>
Notice	<UNITID> Unit ID <PORTID> Port ID <LOTID> LOT ID <CSTID> Cassette ID <RECIPE ID> Recipe ID <PRCID> Processing Code <ENDFLAG> End Flag N : Normal End P : Operator End E : Last processed unit R : Last processed unit for rework

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Notice	<MULTIFLAG> T : True(Multi Recipe ID) F : False(Single Recipe ID) <GLSODR> Glass Order 0 = The first glass 1 = All glasses except the first glass and the last glass 2 = The Last glass 3 = the only glass, when there is just one in a LOT <SLOTID> Slot ID <GLSID> Glass ID <GLSEXIST> Glass Existence 0 = Not exist 1 = Exist <GLSSELECTED> Glass Selection 0 = Not selected 1 = selected <PROCUNITID> Unit ID to process Glass <GLSJUDGE> Glass Grade Code <PNLIF> Panel Information <SUBMDLIF> Sub Panel Information
Exception	If n=0, this means 'Lot Judge Report' facility is not used. If n≠0, this means 'Lot Judge Report' facility is used.

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S6,F66	LOT End Data Send Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,4 1. U2[1]<UNITID> 2. B[1] <PORTID> 3. A[16] <LOTID> 4. B[1] <ACKC6>		
Notice	<UNITID> Unit ID <PORTID> Port ID <LOTID> Lot ID <ACKC6> Acknowledge code, 1 byte 0 = Accepted 1 = Wait Lot Actual Processing Data 2 = Lot End without Lot Actual Processing Data >3 = Error , not accepted 1 – 63 Reserved		
Exception			

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S6,F67	Glass Move Request	ECS←EQP	reply

Description	This message is used in Load Request, Unload Request from Equipment.
Structure	L,5 1. A[2] <GLSMVSTATUS> 2. U2[1] <UNITID> 3. B[1] <STAGEID> 4. U2[1] <LDGLSCODE_A> 5. U2[1] <LDGLSCODE_B> 6. U2[1] <UDGLSCODE_A> 7. U2[1] <UDGLSCODE_B>
Notice	<GLSMVSTATUS> Glass Move Status LW : Glass Load Wait Request UW : Glass Unload Wait Request LR : Glass Load Request UR : Glass Unload Request ER : Glass Exchange Request CR : Cancel Request <UNITID> Unit ID <STAGEID> Stage ID <LDGLSCODE_A> A Cut Load Glass Code <LDGLSCODE_B> B Cut Load Glass Code <UDGLSCODE_A> A Cut Unload Glass Code <UDGLSCODE_B> B Cut Unload Glass Code
Exception	

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S6,F68	Glass Move Request Acknowledge	ECS→EQP	

Description	Acknowledge
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>
Notice	<UNITID>Unit ID <ACKC6> 0 = Accepted 1 = Not Accepted
Exception	

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S6,F69	Cassette Map Request	ECS←EQP	reply
Description	The Equipment requests Cassette Map(Online Local Control state only)		
Structure	U2[1] <UNITID>		
Notice	<UNITID> Unit ID		
Exception			

S6,F70	Cassette Map Request Acknowledge	ECS→EQP	
Description	Cassette Map Data Send		
Structure	L,2 1. U2[1]<UNITID> 2. L,n (n = Cassette port number) 1.L,5 1. A[16] <LOTID1> 2. A[14]<CSTID1> 3. B[1] <PORTID1> 4. A[40] <MAP1> 5. A[1] <CSTREADY1> ……. n.L,5 1. A[16] <LOTIDn> 2. A[14]<CSTIDn> 3. B[1] <PORTIDn> 4. A[40] <MAPn> 5. A[1] < CSTREADYn>		
Notice	<UNITID>Unit ID <LOTID>LOT ID <CSTID>Cassette ID <PORTID> Port ID <MAP> Cassette Mapping Information e.g. In case there are 2 glassed in Slot 1 and Slot 20. Then MAP = "10000000000000000001000000000000000000000" 0 : Glass Not Exist 1 : Glass Exist <CSTREADY> Cassette State 0 : Cassette Not Exist 1 : Cassette Exist (LC)		
Exception			

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S6,F71	Sampling Slot Request	ECS←EQP	reply
Description	The Equipment requests the specific glass selected by the equipment from ECS on Online Local State.		
Structure	L,3 1.U2[1]<UNITID> 2.B[1]<STAGEID> 3.L,4 1. A[16] <LOTID> 2. A[14]<CSTID> 3. B[1] <PORTID> 4. A[3] <SLOTID>		
Notice	<UNITID> Unit ID <STAGEID> Stage ID <LOTID> LOT ID <CSTID> Cassette ID <PORTID> Port ID <SLOTID> Slot ID		
Exception			

S6,F72	Sampling Slot Request Acknowledge	ECS→EQP	
Description	Acknowledge or error		
Structure	L,2 1. U2[1]<UNITID> 2. B[1] <ACKC6A>		
Notice	<ACKC6A> Acknowledge code, 1byte 0 = Accept 1 = Glass not exist 2 = Cassette not exist 3 = Invalid LOT ID 4 = Busy, Try later 5 – 63 Reserved		
Exception			

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S6,F73	Glass Actual Processing Data Report	ECS←EQP	reply

Description	The purpose of this message is for the equipment to send the glass processing data.
Structure	L,8 1. A[16] <LOTID> 2. A[14] <CSTID> 3. A[24] <RECIPEID> 4. A[8] <PRCID> 5. A[3] <SLOTID> 6. U2[1] <UNITID> 7. L,7 1. A[16] <GLSID> 2. A[2] <GLSJUDGE> 3. A[2000] <PNLIF> 4. A[2000] <SUBMDLIF> 5. U2[1]<GLSCODE> 6. A[32]<KEYID> 7. L,p [p = the number of GSD] 1. L, 2 1. A[24] <GSDITEM1> 2. A[40] <GSDVALUE1> : p. L, 2 1. A[24] <GSDITEMp> 2. A[40] <GSDVALUEp> 8. L, q [q = number of parameter] 1. L, 2 1. A[14] <DVNAME11> 2. A[14] <DVVAL11> : q. L, 2 1. A[14] <DVNAME1q> 2. A[14] <DVVAL1q>
Notice	<LOTID> LOT ID <CSTID> Cassette ID <RECIPE ID> Recipe ID <PRCID> Processing Code <SLOTID> Slot ID <UNITID>Unit ID <GLSID> Glass ID <GLSJUDGE> Glass Judge Code <PNLIF> Panel Information <SUBMDLIF>Sub Panel Information <GLSCODE> Glass Code <KEYID> Glass Key ID

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Notice	<GSDITEM> Glass Specific Data Item defined by LG Display <GSDVALUE> Glass Specific Data Value defined by LG Display <DVNAME> APD Data Name defined by LG Display <DVVALUE> APD Data Value defined by LG Display
Exception	When <GSDITEM>, <GSDVALUE> is not necessary, p is 0.

S6,F74	Glass Actual Processing Data Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>		
Notice	<UNITID>Unit ID <ACKC6> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved		
Exception			

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S6,F83	LOT Actual Processing Data Report	ECS←EQP	reply

Description	The purpose of this message is that the equipment sends the Lot Actual processing data. Only when ECS sends Lot End Data, the equipment sends this data
Structure	L,9 1. A[16] <LOTID> 2. A[14] <CSTID> 3. A[24] <RECIPE ID> 4. A[8] <PRCID> 5. U2[1] <UNITID> 6. A[32] <KEYID> 7. A[2] <LOTJUDGE> 8. L,p [p = the number of LSD] 1. L,2 1. A[24] <LSDITEM1> 2. A[40] <LSDVALUE1> : p. L, 2 1. A[24] <LSDITEMp> 2. A[40] <LSDVALUEp> 9. L, q [q = number of parameter] 1. L, 2 1. A[14] <DVNAME11> 2. A[14] <DVVAL11> : q. L, 2 1. A[14] <DVNAME1q> 2. A[14] <DVVAL1q>
Notice	<LOTID> LOT ID <CSTID> Cassette ID <RECIPE ID> Recipe ID <PRCID> Processing Code <UNITID>Unit ID <KEYID> Glass Key ID <LOTJUDGE>Lot Judge <LSDITEM> LOT Specific Data Item defined by LG Display <LSDVALUE> LOT Specific Data Value defined by LG Display <DVNAME> APD Data Name defined by LG Display <DVVALUE> APD Data Value defined by LG Display
Exception	

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S6,F84	LOT Actual Processing Data Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>		
Notice	<UNITID>Unit ID <ACKC6> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved		
Exception			

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S6,F85	Dummy Lot Request	ECS←EQP	reply
Description	This message is used to request Dummy Lot.		
Structure	U2[1]<UNITID>		
Notice	<UNITID> Unit ID		
Exception			

S6,F86	Dummy Lot Request Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>		
Notice	<UNITID>Unit ID <ACKC6> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved		
Exception			

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S6,F89	Rework Request	ECS←EQP	reply

Description	This message is used to request rework.
Structure	L,4 1. U2[1] <UNITID> 2. A[16] <LOTID> 3. A[1] <RESERVECD> 4. L,n [n = The Number of Slot] 1.L,2 1. A[3] <SLOTID1> 2. A[1] <SELECTED1> . . n.L,2 1. A[3] <SLOTIDn> 2. A[1] <SELECTEDn>
Notice	<UNITID> Unit ID <LOTID> Lot ID <RESERVECD> Rework reservation code R : Reservation of rework S : Rework Start <SELECTED> Glass Selected Information. 0 : Not Selected 1 : Selected
Exception	

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S6,F90	Rework Request Acknowledge	ECS→EQP	

Description	Acknowledge
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>
Notice	<UNITID>Unit ID <ACKC6> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved
Exception	

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S6,F91	Equipment Status Change Report	ECS←EQP	reply
Description	When the Equipment status is changed, the Equipment reports this message		
Structure	L,3 1. U2[1] <UNITID> 2. A[1] <EQPSTATUS> 3. A[10] <ALID> 4. A[120] <ALTX>		
Notice	<UNITID>Unit ID <EQPSTATUS> Equipment status R : RUN E : Experiment V : Development C : Model Change M : Monitoring I : IDLE D : DOWN P : Process Down <ALID> Alarm Identification If <EQPSTATUS> is 'R' or 'I', then <ALID> is fixed by '0' <ALTX> Alarm Message		
Exception			

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S6,F92	Equipment Status Change Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>		
Notice	<UNITID> Unit ID <ACKC6> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved		
Exception			

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S6,F93	Scrap Data Report	ECS←EQP	reply
Description	When a glass or a panel is scrapped, the Equipment reports this message		
Structure	L,5 1. A[16] <LOTID> 2. A[16] <GLSID> 3. U2[1] <UNITID> 4. A[1] <SCRPTYPE> 5. A[4] <SCRPCODE.>		
Notice	<LOTID> LOT ID <GLSID> Glass ID <UNITID> Unit ID <SCRPTYPE> Scrap Type P = Panel Scarp G = Glass Scrap <SCRPCODE>Scrap Code defined by LG Display		
Exception			

S6,F94	Scrap Data Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>		
Notice	<UNITID> Unit ID <ACKC6> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved		
Exception			

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S6,F95	OCR(VCR) Status Report	ECS←EQP	reply
Description	This message is used in OCR Status Report .		
Structure	L,2 1. U2[1]<UNITID> 2. B[1] <OCRSTATUS>		
Notice	<UNITID> Unit ID <OCRSTATUS> Status for OCR 0 : OCR off 1 : OCR On		
Exception			

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S6,F96	OCR(VCR) Status Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>		
Notice	<UNITID> Unit ID <ACKC6> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved		
Exception			

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S7,F19	Current EPPD Request	ECS→EQP	reply
Description	This message is used to request the transmission of the current equipment process program directory. This is the list of all the RECIPE IDs of the process programs stored in the equipment		
Structure	U2[1] <UNITID>		
Notice	<UNITID> Unit ID		
Exception			

S7,F20	Current EPPD Request Acknowledge	ECS←EQP	
Description	This message is used to transmit the current EPPD.		
Structure	L,2 1. U2[1] <UNITID> 2. L,n (n = Recipe List Count) 1. A[24] <RECIPEID1> . . . n. A[24] <RECIPEIDn>		
Notice	<UNITID> Unit ID <RECIPEID> Recipe ID		
Exception			

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S7,F61	Recipe ID Confirm Request	ECS→EQP	reply

Description	This message is used to ask the equipment whether the recipe ID is or not in the list, which the equipment has.
Structure	L,14 1. A[16] <LOTID> 2. A[14] <CSTID> 3. A[24] <RECIPEID> 4. A[8] <PRCID> 5. B[1] <PORTID> 6. U2[1] <UNITID> 7. A[5] <GLSTHICK> 8. A[14] <PARTNUM> 9. A[12] <PRODTYPE> 10. A[12] <ATTRIBUTE> 11. A[1] <MODECODE> 12. A[1] <MULTIFLAG> 13. L, m [m = the number of LSD] 1. L,2 1. A[24] <LSDITEM1> 2. A[40] <LSDVALUE1> : m. L,2 1. A[24] <LSDITEMm> 2. A[40] <LSDVALUEm> 14. L,n [n = The Number of Slot] 1. L,10 1. A[3] <SLOTID1> 2. A[16] <SLTLOTID1> 3. A[16] <GLSID1> 4. A[24] <SRECIPEID1> 5. B[1] <GLSEXIST1> 6. A[2] <GLSJUDGE1> 7. A[14] <PARTNUM1> 8. A[2000] <PNLIF1> 9. A[2000] <SUBMDLIF1> 10. L,p [p = the number of GSD] 1. L, 2 1. A[24] <GSDITEM11> 2. A[40] <GSDVALUE11> : p. L, 2 1. A[24] <GSDITEM1p> 2. A[40] <GSDVALUE1p> :

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Structure	: n. L,10 1. A[3] <SLOTIDn> 2. A[16] <SLTLOTIDn> 3. A[16] <GLSIDn> 4. A[24] <SRECIPEIDn> 5. B[1]<GLSEXISTn> 6. A[2] <GLSJUDGEn> 7. A[14] <PARTNUMn> 8. A[2000] <PNLIFn> 9. A[2000] <SUBMDLIFn> 10. L,q [q = the number of GSD] 1. L, 2 1. A[24] <GSDITEMn1> 2. A[40] <GSDVALUEn1> : q. L, 2 1. A[24] <GSDITEMnq> 2. A[40] <GSDVALUEnq>
Notice	<UNITID> Unit ID <LOTID> LOT ID <CSTID> Cassette ID <SLOTID> Slot ID <RECIPEID> Recipe ID <PRCID> Processing Code <SLTLOTID> Each Slot LOT ID <SRECIPEID> Each Slot Recipe ID <GLSEXIST>Glass Existence 0 = Not exist , 1 = Exist <GLSID> Glass ID <GLSJUDGE> Glass Grade Code <PNLIF> Panel Information <SUBMDLIF> Sub Panel Information <PORTID> Port ID <GLSTHICK> Glass's Thickness <PARTNUM> Part Number <PRODTYPE> Type of Producing <ATTRIBUTE> LOT Attribute <MODECODE> Host Online / Offline Code N : Recipe has downloaded by Host (Host Online Mode) F : Recipe has inputted by Operator (Host Offline Mode) <MULTIFLAG> Multi Flag T : True(Multi Lot or Multi Recipe) F : False(Single Lot and Single Recipe) <KEYID>Glass Key ID <LSDITEM> LOT Specific Data Item defined by LG Display <LSDVALUE> LOT Specific Data Value defined by LG Display <GSDITEM> Glass Specific Data Item defined by LG Display <GSDVALUE> Glass Specific Data Value defined by LG Display
Exception	

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S7,F62	Recipe ID Confirm and Sampling Data Request Acknowledge	ECS←EQP	
Description	This message is used to transmit the confirm result of the recipe id and the sampling data of the equipment.		
Structure	L,7 1.U2[1] <UNITID> 2.B[1] <PORTID> 3.A[24] <RECIPEID> 4.A[16] <LOTID> 5.A[1] <SAMPLINGFLAG> 6.B[1] <ACKC7M> 7. L,n [n = The Number of Slot] 1.L,3 1. A[3] <SLOTID1> 2. A[24] <SRECIPEID1> 3. B[1] <ACKC7S1> . . n.L,3 1. A[3] <SLOTIDn> 2. A[24] <SRECIPEIDn> 3. B[1] <ACKC7Sn>		
Notice	<UNITID> Unit ID <PORTID> Port ID <RECIPEID> Recipe ID (This is recipe for LOT) <LOTID> Lot ID < SAMPLINGFLAG> Sampling Flag W : Wait equipment's Sampling Data S : Lot start without equipment's Sampling Data <ACKC7M> Acknowledge code, 1 byte 0 = Accept 1 = Not accept 2 = Process Condition is not ready 3 – 63 Reserved <SLOTID> Slot ID <SRECIPEID> Each Slot Recipe ID <ACKC7S> 0 = Accept 1 = Not accept 2 – 63 Reserved		
Exception			

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S7,F63	Sampling Data Report	ECS←EQP	reply

Description	This message is used to report the sampling data of the equipment
Structure	L,5 1. U2[1] <UNITID> 2. B[1] <PORTID> 3. A[24] <RECIPEID> 4. A[16] <LOTID> 5. L,n [n = The Number of Slot] 1.L,4 1. A[3] <SLOTID1> 2. A[24] <SRECIPEID1> 3. B[1] <ACKC7S1> 4. A[1] <SELECTED1> . . n.L,4 1. A[3] <SLOTIDn> 2. A[24] <SRECIPEIDn> 3. B[1] <ACKC7Sn> 4. A[1] <SELECTEDn>
Notice	<UNITID> Unit ID <PORTID> Port ID <RECIPEID> Recipe ID (This is recipe for LOT) <LOTID> Lot ID <SELECTED> 0 : Not Selected 1 : Selected

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S7,F64	Sampling Data Report Acknowledge	ECS→EQP	

Description	Acknowledge
Structure	L,4 1. U2[1] <UNITID> 2. B[1] <PORTID> 3. A[16] <LOTID> 4. B[1] <ACKC7>
Notice	<UNITID> Unit ID <PORTID> Port ID <LOTID> Lot ID <ACKC2> 0 = Accepted >0 = Error, not accepted 1 – 63 Reserved
Exception	

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S7,F67	Current Recipe Group List Request	ECS→EQP	reply
Description	This message is used to request the transmission of the current equipment process program group. This is the list of all the Groups of the process programs stored in the equipment		
Structure	U2[1] <UNITID>		
Notice	<UNITID> Unit ID		
Exception			

S7,F68	Current Recipe Group List	ECS←EQP	
Description	This message is used to transmit the current EPPG.		
Structure	L,2 1.U2[1] <UNITID> 2.L,n (n = Group List Count) 1. A[12] <GROUPID1> . . . n. A[12] < GROUPID n>		
Notice	<UNITID> Unit ID <GROUPID> Recipe Group ID		
Exception			

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S7,F69	Current Recipe ID List Request by Group	ECS→EQP	reply

Description	This message is used to request the transmission of the current equipment process program directory. This is the list of all RECIPE IDs of the specified Recipe Group IDs in the equipment
Structure	L,2 1.U2[1] <UNITID> 2.L,n (n = Group ID Count) 1. A[12] <GROUPID1> . . n. A[12] <GROUPIDn>
Notice	<UNITID> Unit ID <GROUPID> Recipe Group ID
Exception	Equipment using Recipe Group needs This Message. 'n=0' means all Recipe Group.

S7,F70	Current Recipe ID List Request by Group Acknowledge	ECS←EQP	

Description	This message is used to transmit all the RECIPE IDs of the specified Recipe Group IDs in the equipment
Structure	L,2 1.U2[1] <UNITID> 2.L,m (m = Group ID Count) 1. L,2 1. A[12] <GROUPID> 2. L,n (n = Recipe List Count) 1. A[24] <RECIPE ID1> . . n. A[24] <RECIPE IDn> . . m. L,2 1. A[12] <GROUPID> 2. L,n (n = Recipe List Count) 1. A[24] <RECIPE ID1> . . n. A[24] <RECIPE IDn>
Notice	<UNITID> Unit ID <GROUPID> Recipe Group ID <RECIPEID> Recipe ID
Exception	

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S7,F77	Recipe Body Request	ECS→EQP	reply

Description	Recipe Body List Request
Structure	L,6 1.A[16]<LOTID> 2.A[14]<CSTID > 3.A[24]<RECIPEID > 4.A[8]<PRCID > 5.B[1]<PORTID> 6.U2[1]<UNITID>
Notice	<LOTID> LOT ID <CSTID> Cassette ID <RECIPEID> Recipe ID <PRCID> Processing Code <PORTID> Port ID <UNITID> Unit ID
Exception	

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S7,F78	Recipe Body	ECS←EQP	
Description	This message reports the recipe parameter of the equipment		
Structure	L,8 1. A[16] <LOTID> 2. A[14]<CSTID> 3. A[24]<RECIPEID> 4. A[12]<RECIPEIDVER> 5. A[8]<PRCID> 6. B[1]<PORTID> 7. U2[1]<UNITID> 8. L,m [m = the number of Unit] 1.L.6 1. U2[1] <EUNITID1> 2. A[24] <LOWERRECIPEID1> 3. A[12] <LOWERRECIPEIDVER1> 4. A[1]<LEVEL1> 5. A[1]<RECIPEIDTYPE1> 6. L,p [p = Number of Step] 1. L,2 1. A[14] <STEP11> 2. L, n [n = Number of Parameter] 1. L,2 1. A[24] <RPARMNAME111> 2. A[24] <RPARMVAL111> : n.L,2 1. A[24] <RPARMNAME11n> 2. A[24] <RPARMVAL11n> : p. L,2 1. A[14] <STEP1p> 2. L, q [q = Number of Parameter] 1. L,2 1. A[24] <RPARMNAME1p1> 2. A[24] <RPARMVAL1p1> : q.L,2 1. A[24] <RPARMNAME1pq> 2. A[24] <RPARMVAL1pq> :		

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	: m.L,6 1. U2[1] <EUNITIDm> 2. A[24] <LOWERRECIPEIDm> 3. A[12] <LOWERRECIPEIDVERm> 4. A[1]<LEVELm> 5. A[1]<RECIPEIDTYPEm> 6. L,r [r = Number of Step] 1. L,2 1. A[14] <STEPm1> 2. L, s [s = Number of Parameter] 1. L,2 1. A[24] <RPARMNAMEm11> 2. A[24] <RPARMVALm11> : s.L,2 1. A[24] <RPARMNAMEm1s> 2. A[24] <RPARMVALm1s> : r. L,2 1. A[14] <STEPmr> 2. L, t [t = Number of Parameter] 1. L,2 1. A[24] <RPARMNAMEmr1> 2. A[24] <RPARMVALmr1> : t.L,2 1. A[24] <RPARMNAMEmrt> 2. A[24] <RPARMVALmrt>
Notice	<LOTID> LOT ID <CSTID> Cassette ID <PRCID> Processing Code <RECIPEID> Recipe ID <PORTID> Port ID <UNITID> Unit ID <EUNITID> Each Unit ID <LOWERRECIPEID> Lower Recipe ID <LOWERRECIPEIDVER> Lower Recipe ID Version <LEVEL> Level of RECIPEID 1 : Level 1 RECIPEID 2 : Level 2 RECIPEID 3 : Level 3 RECIPEID < RPARMNAME> Recipe Parameter Name < RPARMVAL> Recipe Parameter Value <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <STEP> Recipe Step 0 = Step Not Used, Another Number is Recipe Step
Exception	

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S9,F1	Unrecognized Device ID	ECS←EQP	
Description	The device ID in the message block header did not correspond to any known device ID in the node detecting the error.		
Structure	<MHEAD>		
Notice			
Exception			

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S9,F3	Unrecognized Stream Type	ECS←EQP	
Description	The equipment does not recognize the stream type in the message block header.		
Structure	<MHEAD>		
Notice			
Exception			

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S9,F5	Unrecognized Function Type	ECS←EQP	
Description	This message indicates that the function in the message ID is not recognized by the receiver.		
Structure	<MHEAD>		
Notice			
Exception			

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S9,F7	Illegal Data	ECS←EQP	
Description	This message indicates that the stream and function were recognized, but the associated data format could not be interpreted.		
Structure	<MHEAD>		
Notice			
Exception			

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S9,F9	Transaction Timer Timeout	ECS←EQP	
Description	This message indicates that a transaction(receive) timer has timed out and that the corresponding transaction has been aborted. It is up to the host to respond to this error in an appropriate manner to keep the system operational.		
Structure	<SHEAD>		
Notice			
Exception			

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S9,F11	Data Too Long	ECS←EQP	
Description	This message to the host indicates that the equipment has been sent more data than it can handle.		
Structure	<MHEAD>		
Notice			
Exception			

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S9,F13	Conversation Timeout	ECS←EQP	
Description	Data were expected but none was received within a reasonable length of time. Resources have been cleared.		
Structure	L,2 1. A[6] <MEXP> 2. A[14] <EDID>		
Notice	<MEXP> Message Expected The format is “SxxFyy” Ex) S07F83, S02F41 etc. <EDID> Expected Data ID MAX 14 Byte Case S7F83 : <EDID>= ‘<CSTID>’ Case S2F41 : <EDID>= ‘<CSTID>’		
Exception			

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S10,F3	Terminal Display, Single	ECS→EQP	
Description	Data to be displayed		
Structure	L,2 1. B[1] <TID> 2. A[80] <TEXT>		
Notice	<TID> Terminal Number Fixed as '0' <TEXT> A single line of characters.		
Exception			

S10,F4	Terminal Display, Single Acknowledge	ECS←EQP	
Description	Acknowledge or Error		
Structure	B[1] <ACKC10>		
Notice	<ACKC10> Acknowledge Code, 1 byte 0 = Accepted for display 1 = Message will not be displayed 2 = Terminal not available 3 – 63 Reserved		
Exception			

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Chapter Ⅱ. ECS-Equipment HSMS Exception Specification

For LG Display

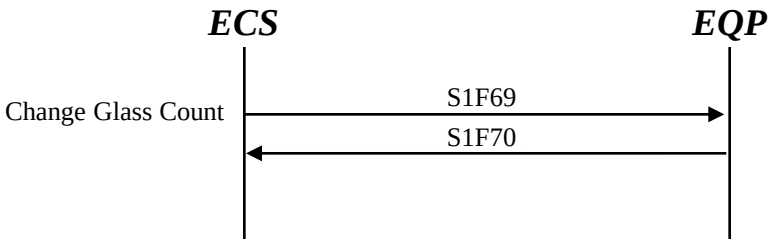
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1. Scenario

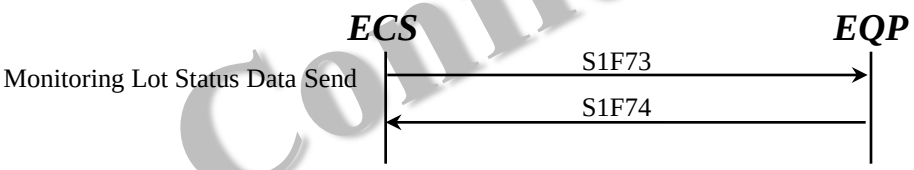
1.1. Inform processing glass count change scenario

장비로 투입될 Glass 수량이 변경 되었을 경우 ECS는 해당 메시지를 장비로 송신한다.
- Cassette Loading 시
- Offline에서 Online 전환 시
- Cassette가 port에 로딩되고 작업준비가 된 후, 해당 Cassette가 cancel 되었을 시



1.2. Monitoring Lot Status Send

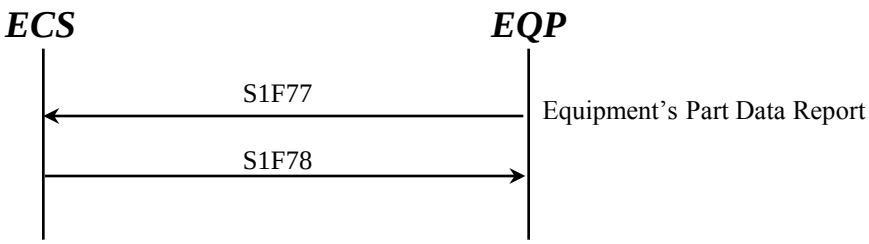
- 모니터링 Lot의 시작과 종료를 장비로 전달한다.
- 상세 사용은 LG CNS와 LG Display와 협의한다.



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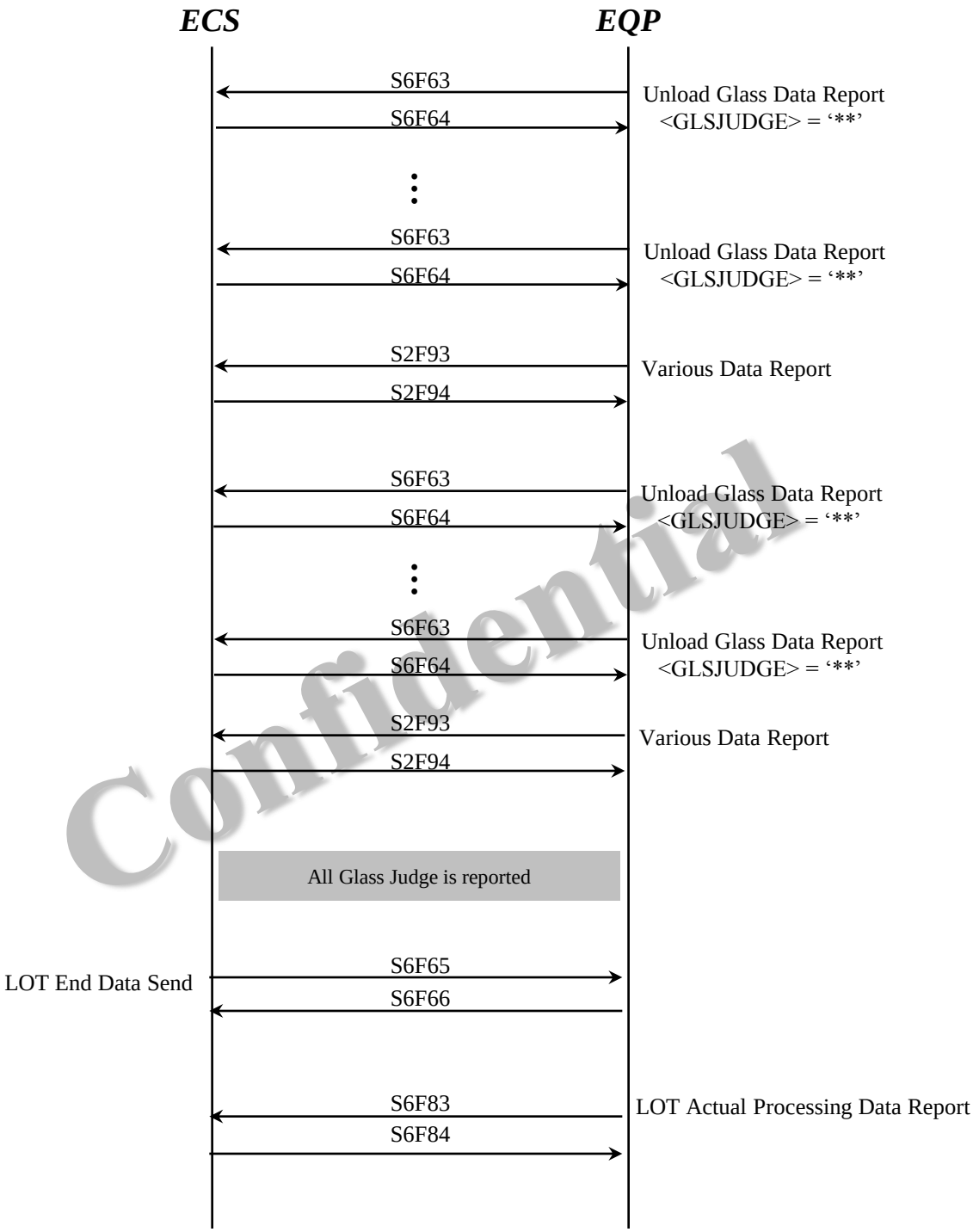
1.3. Equipment's Part Data Report

- 온라인 전환 시 사용
-Glass 또는 장비의 Part가 Unit 또는 Sub Unit간에 이동이 있을 경우 장비는 보고한다.



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1.4. Special Case of Glass Judge Report



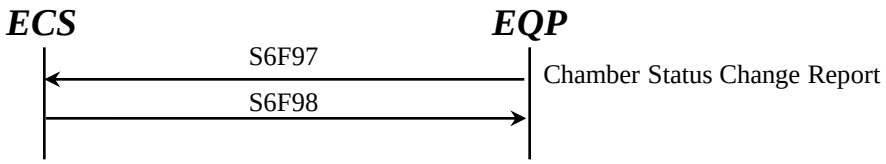
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1.5. Chamber Status Change Report

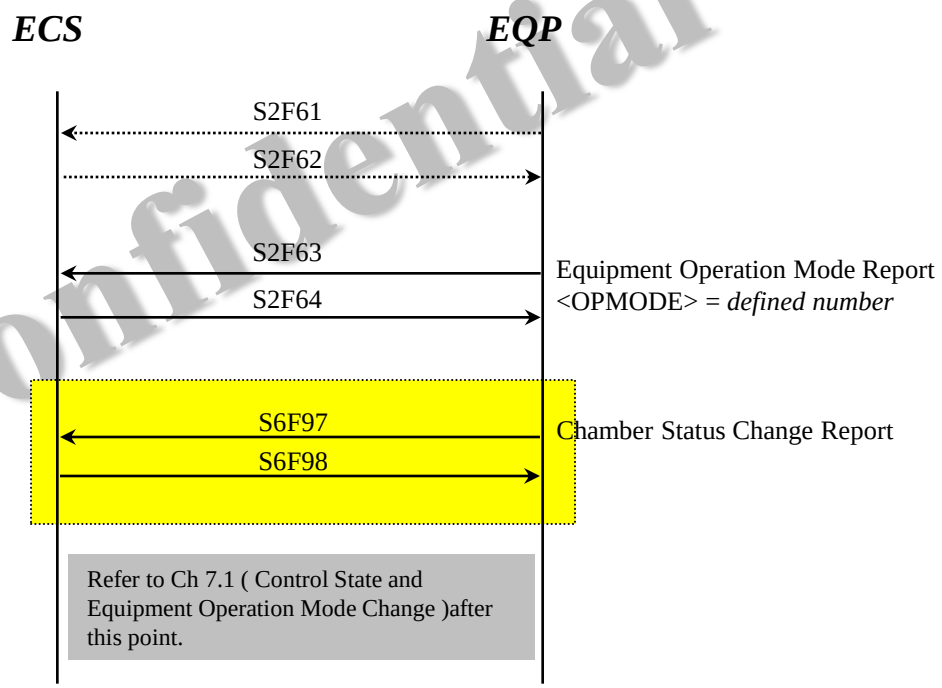
1.5.1 Chamber Status Change Report when chamber status is changed.

- 장비 Chamber의 Processing 상태가 변경 되었을 경우 장비는 ECS로 해당 메시지를 보고한다.
-만약 응답이 없거나 통신에러가 발생하면 기존 상태를 유지한다.
- 상세 사용은 LG CNS와 LG Display와 협의한다.



1.5.2 Chamber Status Change Report (In Online Sequence or In ECS Restart Sequence)

-장비는 장비 동작 모드 변경 보고 메시지(equipment operation mode change report message, S2F63)을 보낸 후에 S6F97을 보고해야 한다.
-이 경우 모든 Chamber 상태를 보고해야 한다.



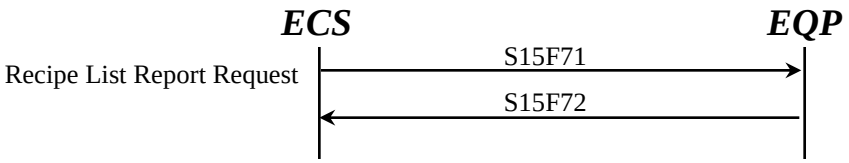
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1.6. Recipe Management Sequence

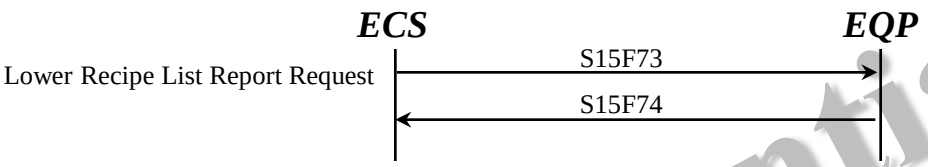
1.6.1 Equipment Recipe List Request

-ECS가 장비의 recipe list를 조회 시 사용한다.



1.6.2 Equipment Lower Recipe List Request

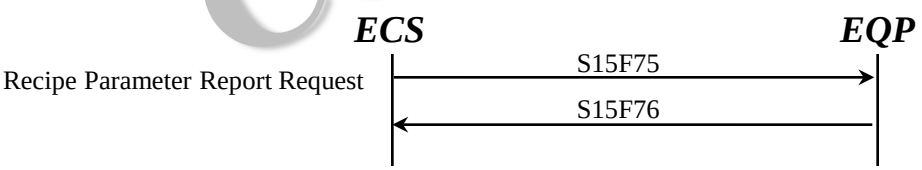
-ECS가 장비의 하위 recipe list를 조회 시 사용한다.



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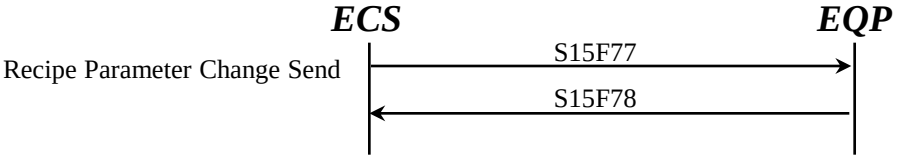
1.6.3 Equipment Recipe Parameter Request

-장비의 Recipe Parameter를 조회 시 사용한다.



1.6.4 Equipment Recipe Parameter Change Send

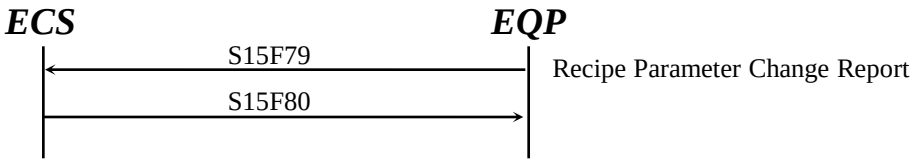
- 장비의 Recipe Parameter를 변경할 경우 사용한다.



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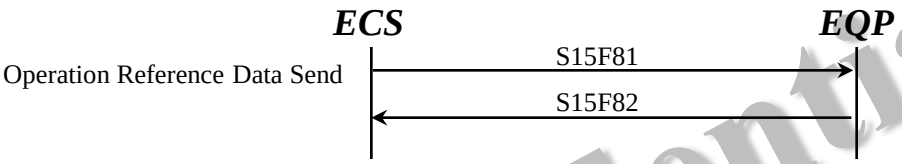
1.6.5 Equipment Recipe Change Report

- 장비의 Recipe Parameter를 변경할 경우 사용한다.



1.6.6 Equipment Operation Reference Data Send

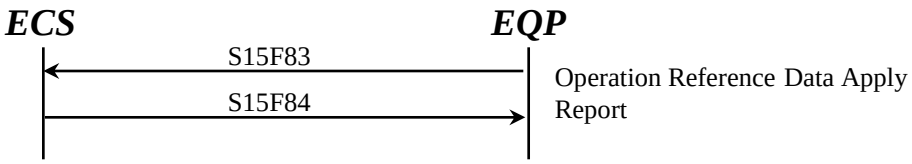
- 장비로 Operation Reference Data를 전송할 경우 사용한다.



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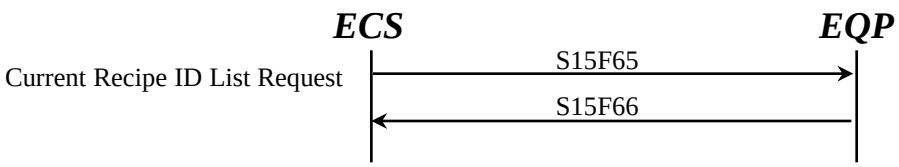
1.6.7 Equipment Operation Reference Data Apply Report

-장비가 operation reference data를 적용하면 장비는 이 메시지를 전송한다.



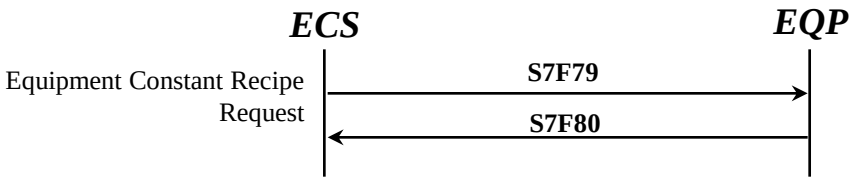
1.6.8 Current Recipe ID List Request

- 현재 장비 process program directory의 전송을 요청 시
- 장비에 등록된 process program의 모든 Recipe ID 와 Recipe ID Version 목록이다.

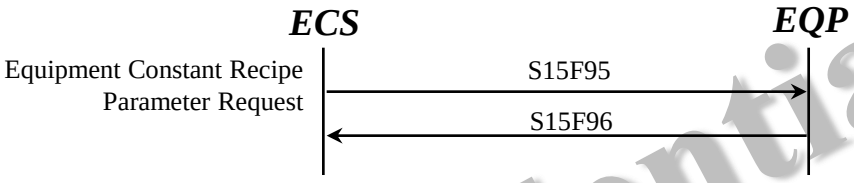


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1.6.9. Equipment Constant Recipe Request

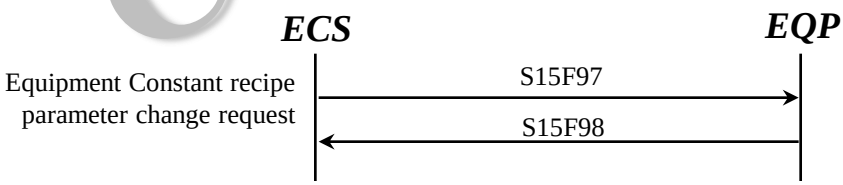


1.6.10. Equipment Constant Recipe Parameter Request

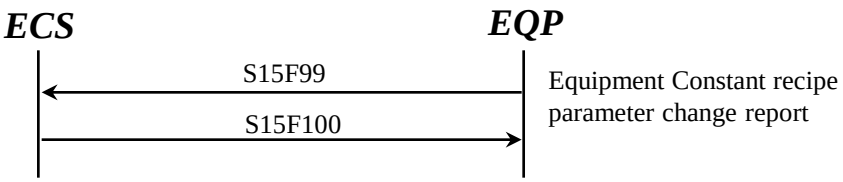


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1.6.11. Equipment Constant Recipe Parameter Change Command



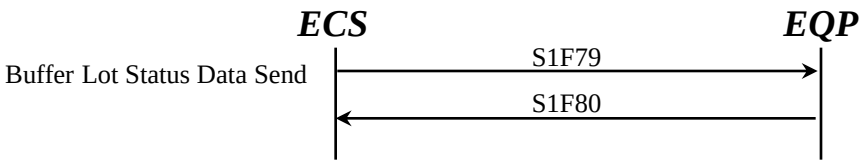
1.6.12. Equipment Constant Recipe Parameter Change Report



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1.7. Buffer Lot Status Send

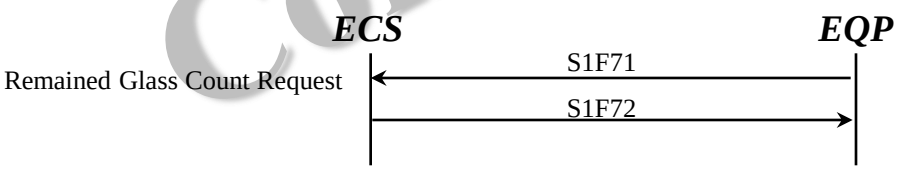
- Buffer Lot 상태를 전달할 경우 사용한다.
- 상세 사용은 LG CNS와 LG Display와 협의한다.



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1.8. Remained Glass Count Request

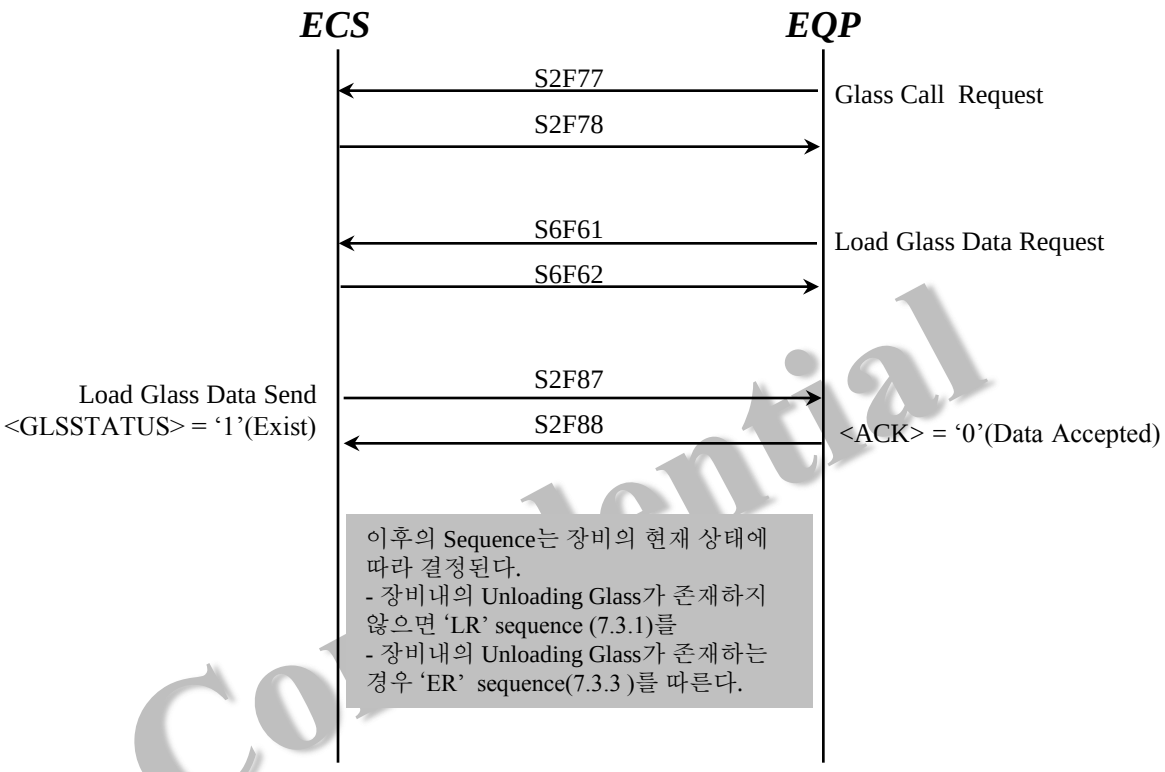
- 장비가 해당 Glass가 속한 Lot의 투입 대기 Glass의 수량을 요청할 경우 사용한다.
- 상세 사용은 LG CNS와 LG Display와 협의한다.



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1.9. Glass Call Request

-수동으로 장비에 glass를 투입할 때 사용한다.
-만약 장비에서 이를 사용하려면 추후 ECS와 상세 협의가 필요하다.



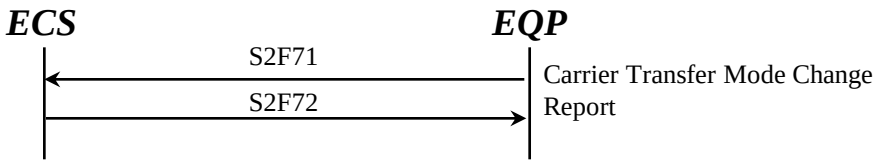
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1.10. Carrier Transfer Mode Change Report

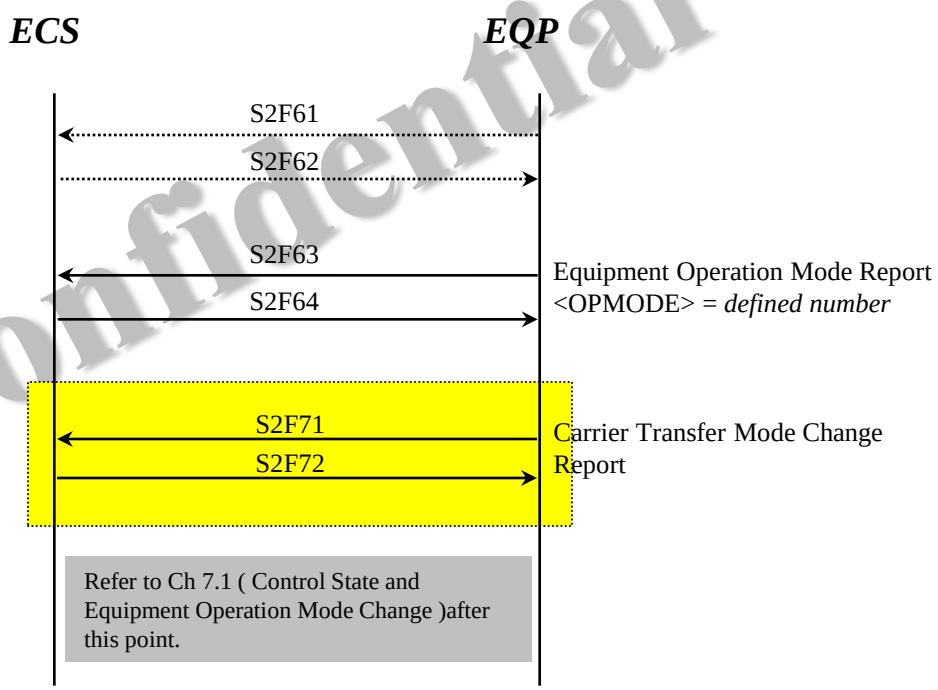
1.10.1 장비의 Carrier 반송 모드가 변경 보고 용도로 사용한다.

- 본 메시지는 장비의 Carrier 반송 모드가 변경이 되었을 경우 장비에서 보고한다.
- 장비의 보고에 대한 ECS 응답이 없는 경우 현재 상태를 유지한다.
- 상세 사용 및 용도는 LG CNS 및 LG Display 와 협의한다.



1.10.2 장비의 온라인 상태로 변경 (ECS Restart 포함)중에 Carrier 반송 모드 변경 보고

- 장비는 Online 변경 중 Operation Mode 변경 보고 이후에 Carrier 반송 모드 상태를 보고한다.

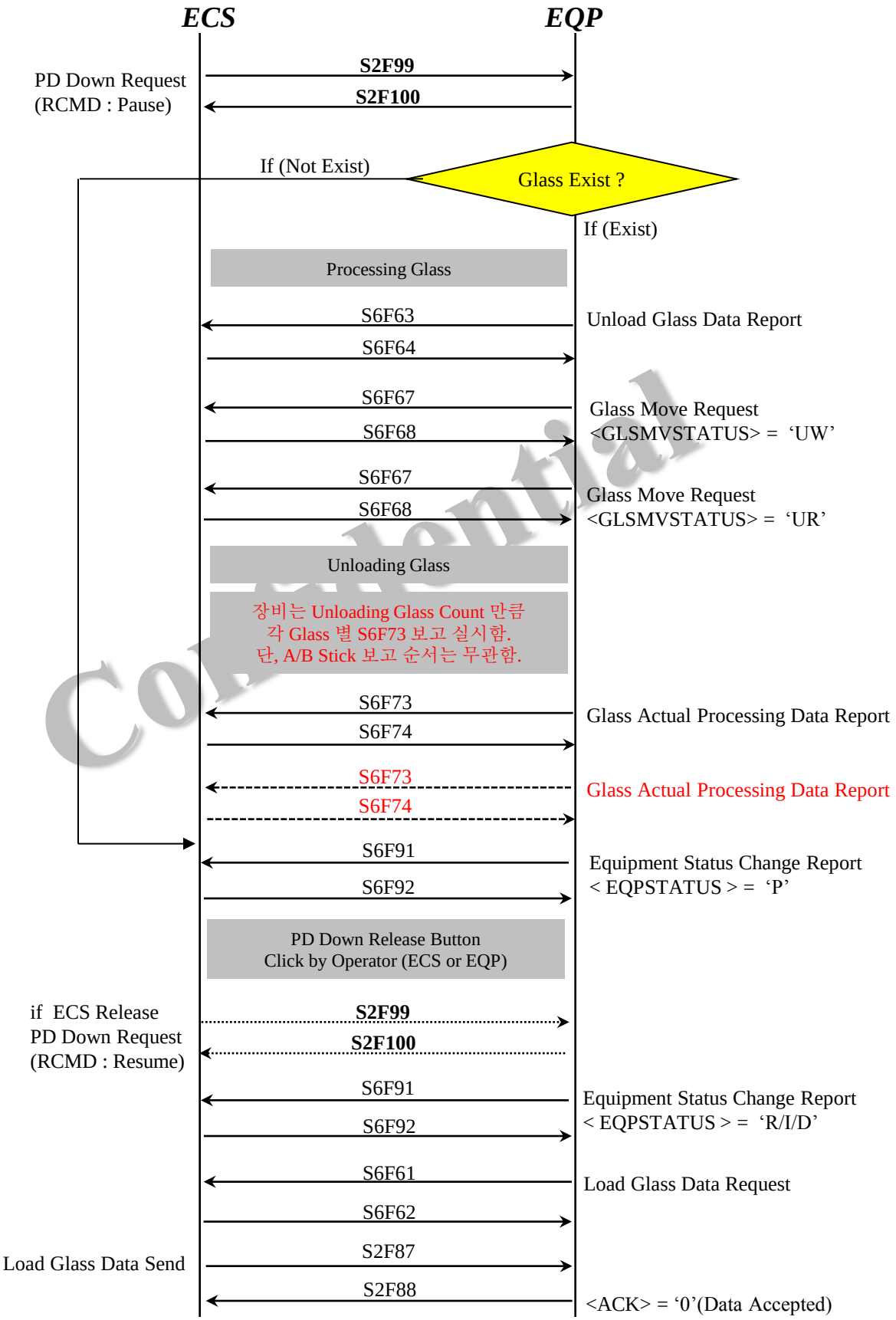


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1.11. PD Down Request

-장비에 Pause를 요청할 때 사용한다.
-PD Request 시나리오는 Machine/Unit 단에서 Glass Hold 가 가능한 장비에 한해서 적용한다.



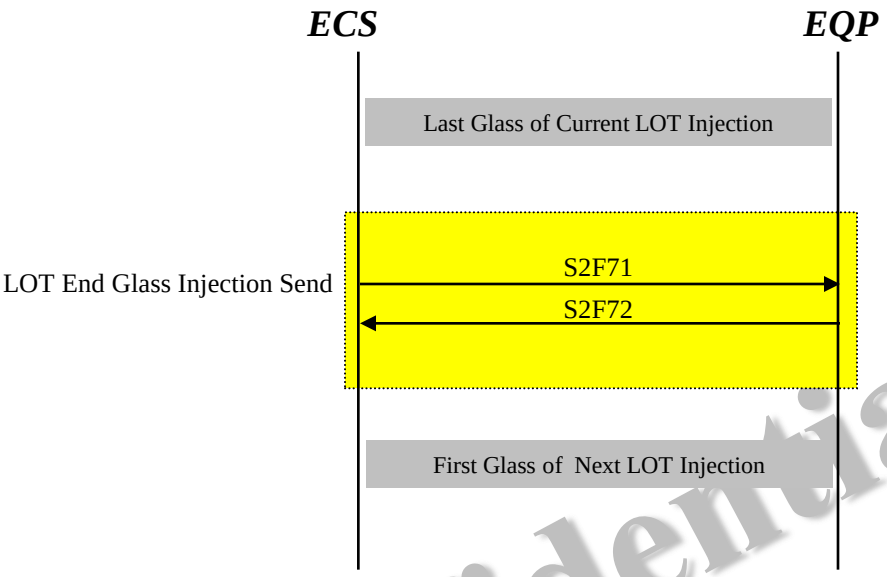
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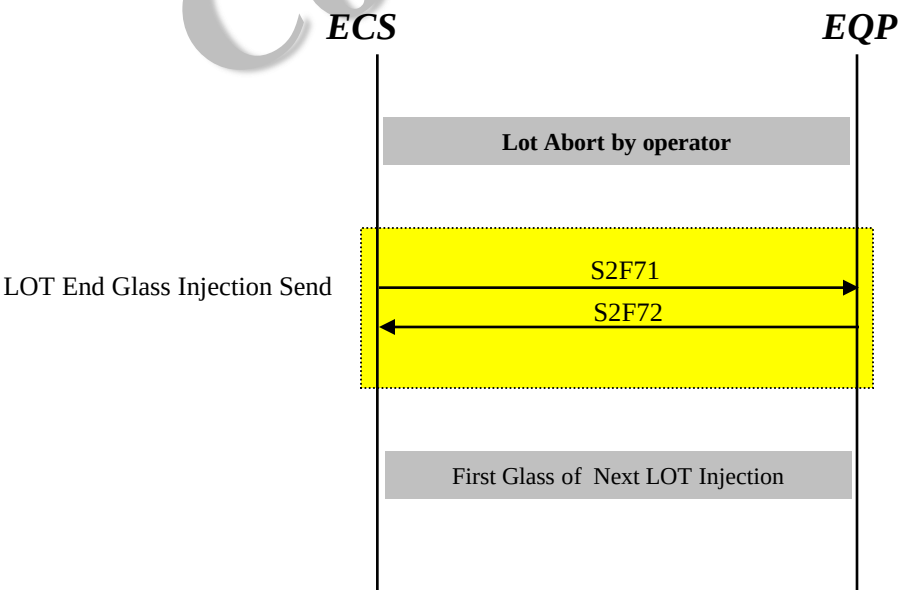
1.12. LOT End Glass Injection Send

- This Message is used when the last glass of LOT was injected in Equipment

1.12.1 In case the normal sequence



10.2 In case the abnormal sequence

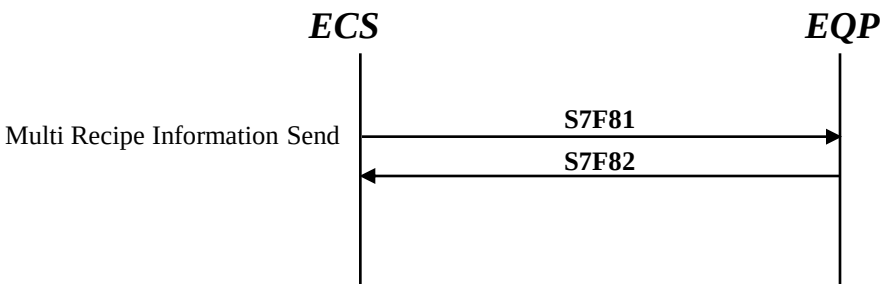


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1.13. Multi Recipe Information Send

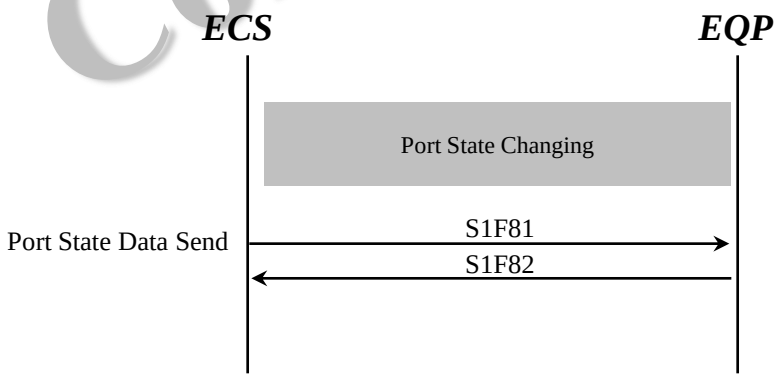
- This message is used when equipment requires recipe and sampling codes for each slots
- If equipment want to use this sequence, it needs detailed conference later with ECS.



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1.14. Port State and Cassette Map Data Send Scenario

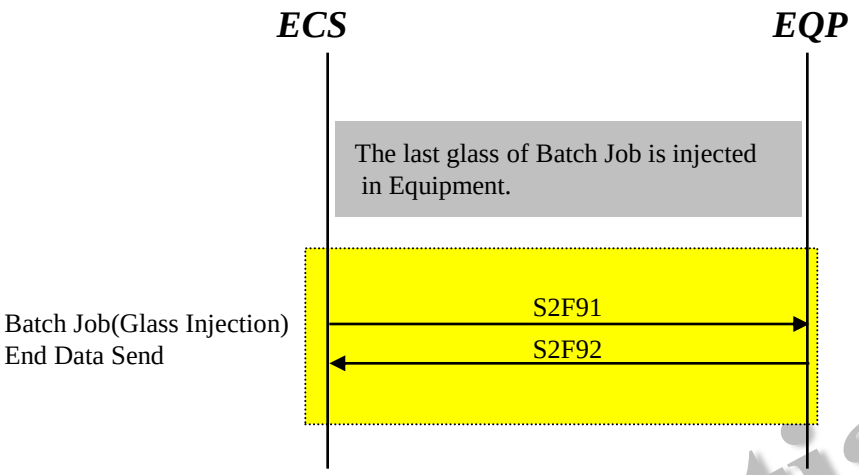
- This message is used when equipment requires the map data for specific port.
- ECS informs the map data to equipment when Port state is changed



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1.15. Batch Job (Glass Injection) End Data Send

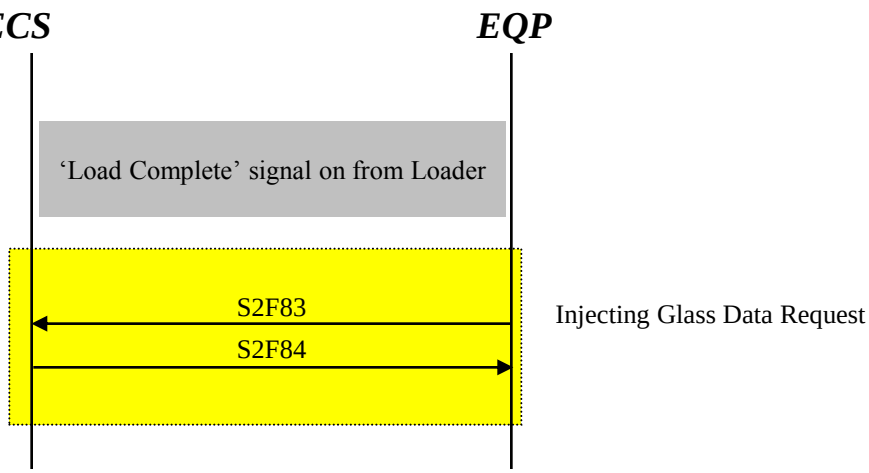
- This message is used when the last glass of Batch Job is injected in Equipment.
- If equipment want to use this sequence, it needs detailed conference later with ECS.



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1.16. Injecting Glass Data Request

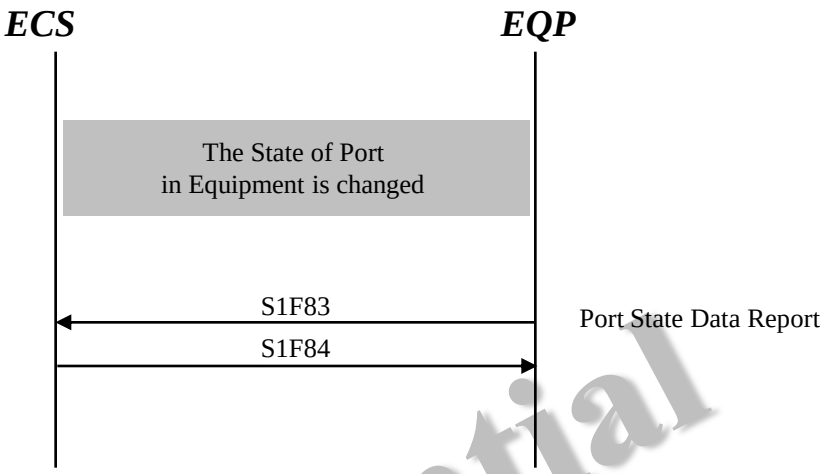
- This message is used when the glass is injecting in Equipment.
- If equipment want to use this sequence, it needs detailed conference later with ECS.



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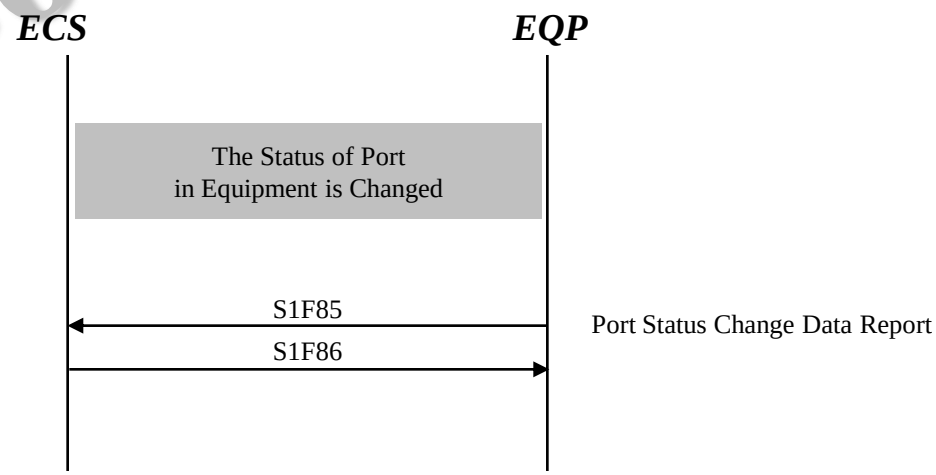
1.17. Port State Data Report

- This message is used when the State (LR, LC, UR, UC) of Port in Equipment is changed.
- If equipment want to use this sequence, it needs detailed conference later with ECS.



1.18. Port Status Change Data Report

- This message is used when the Status (RUN, IDLE, DOWN) of Port in Equipment is changed.
- If equipment want to use this sequence, it needs detailed conference later with ECS.

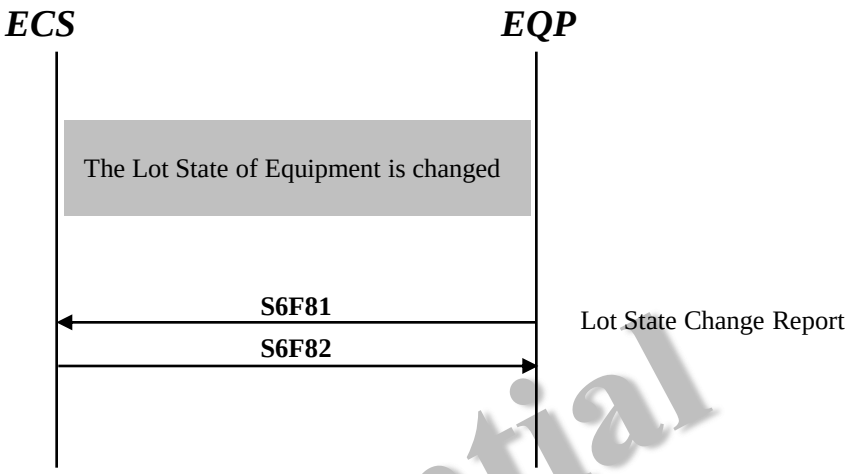


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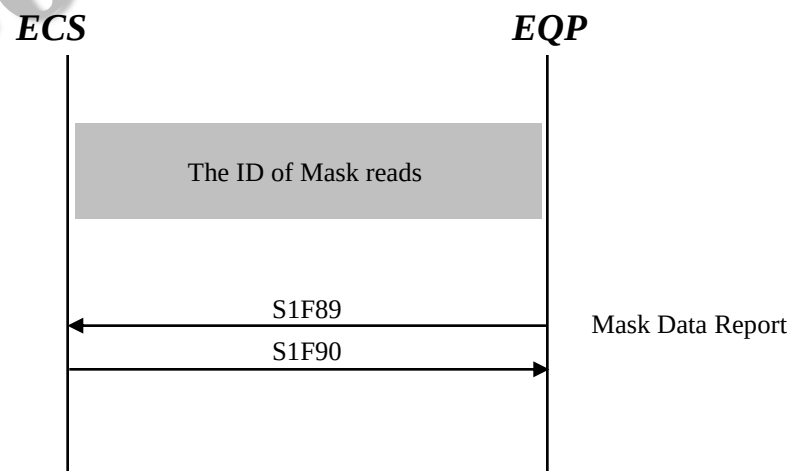
1.19. Lot State Change Report

- This message is used when the Lot State (R, W, P, E) of port in Equipment is changed.
- If equipment want to use this sequence, it needs detailed conference later with ECS.



1.20. Mask Data Report

- This message is used after the ID of Mask reading in EQP .
- If equipment want to use this sequence, it needs detailed conference later with ECS.

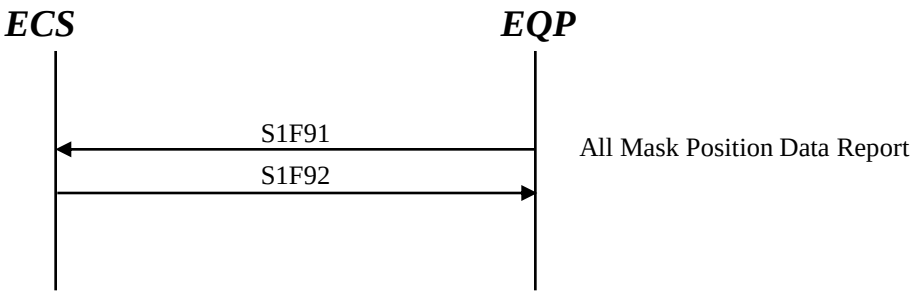


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1.21. All Mask Position Data Report

- In case of changing to Online
- In case of mask moving between each sub unit.



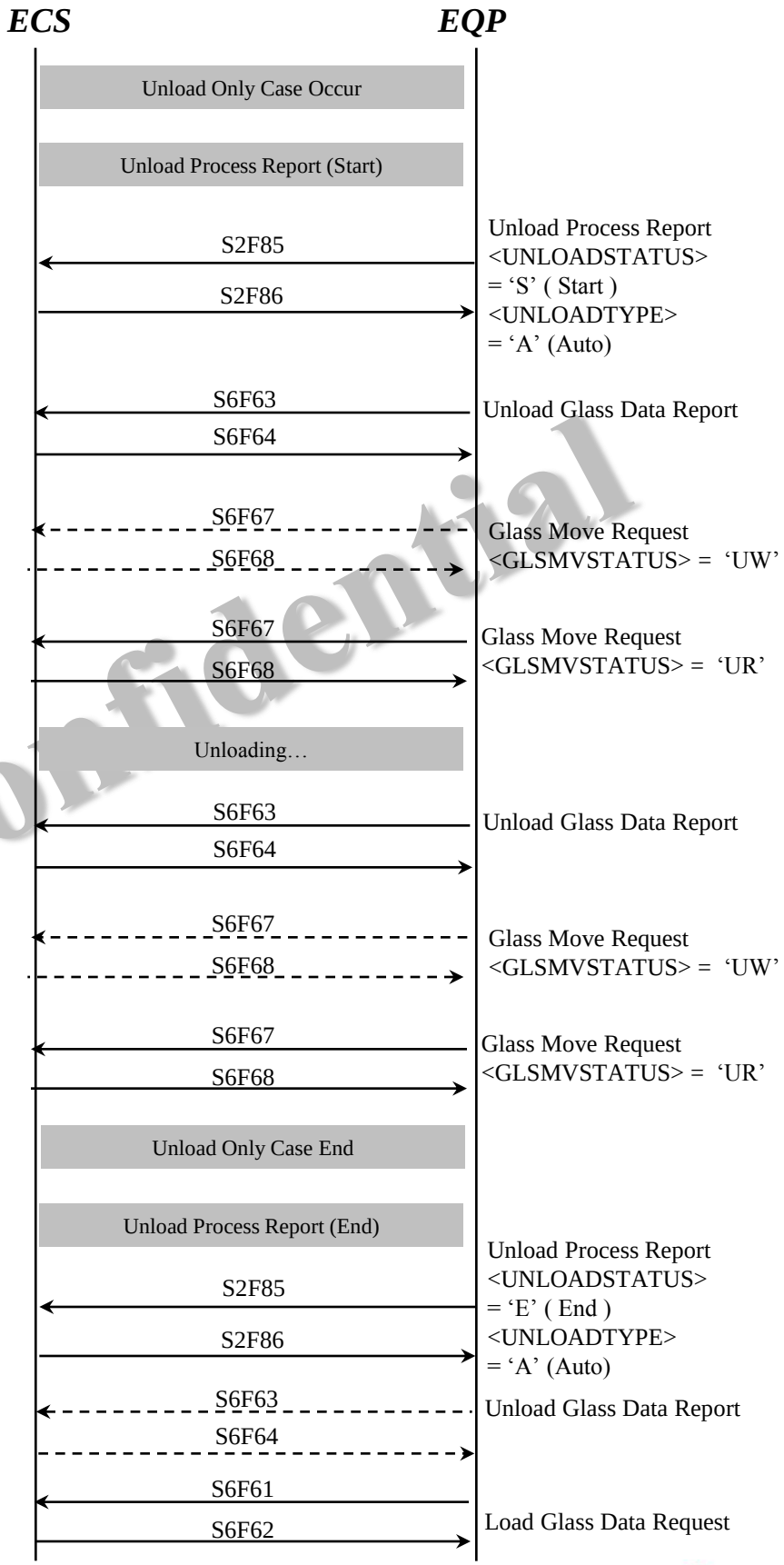
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1.22. Unload Process Report

- Equipment performs this scenario when unit's status is changed.

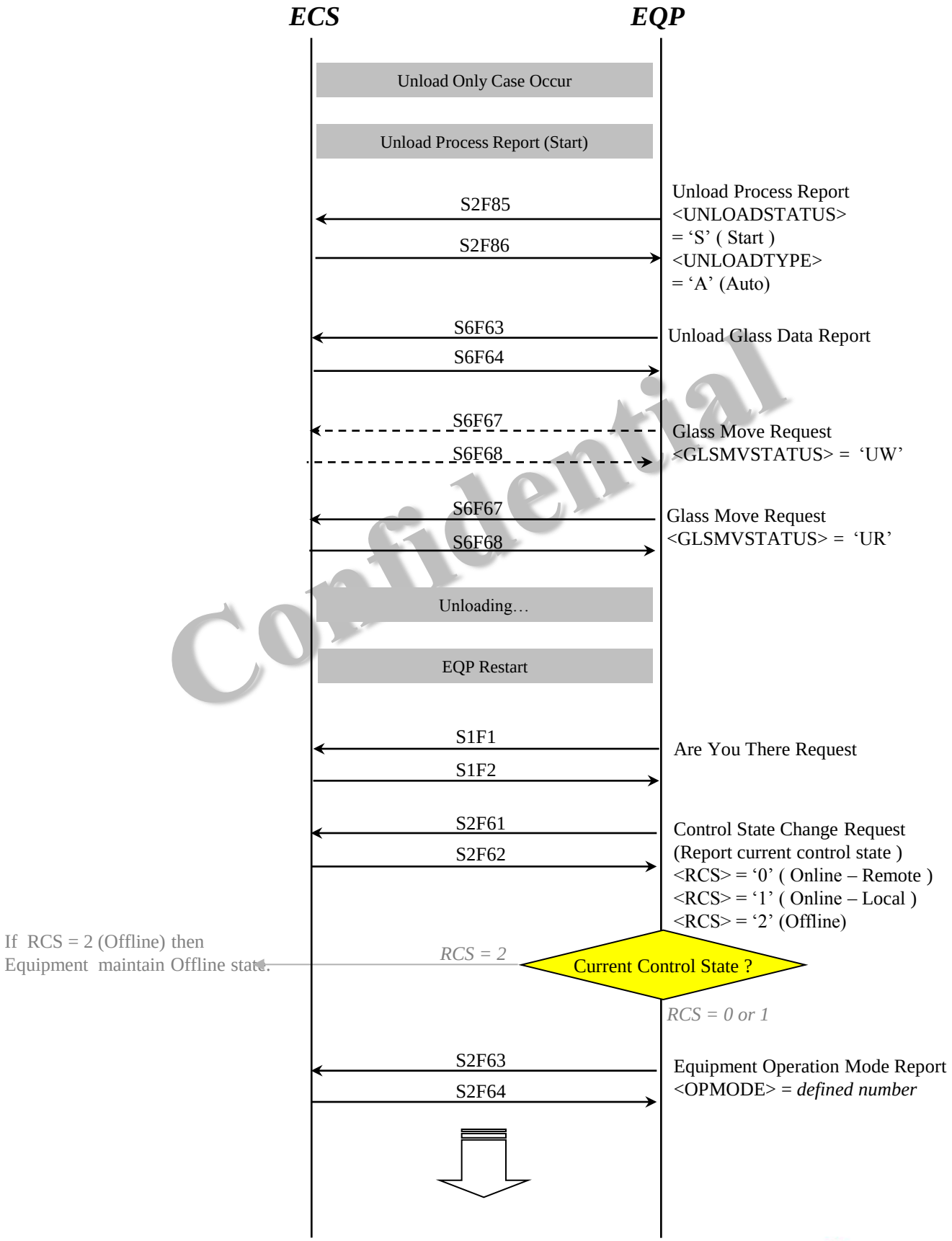
1. 22.1 Unload Start & End



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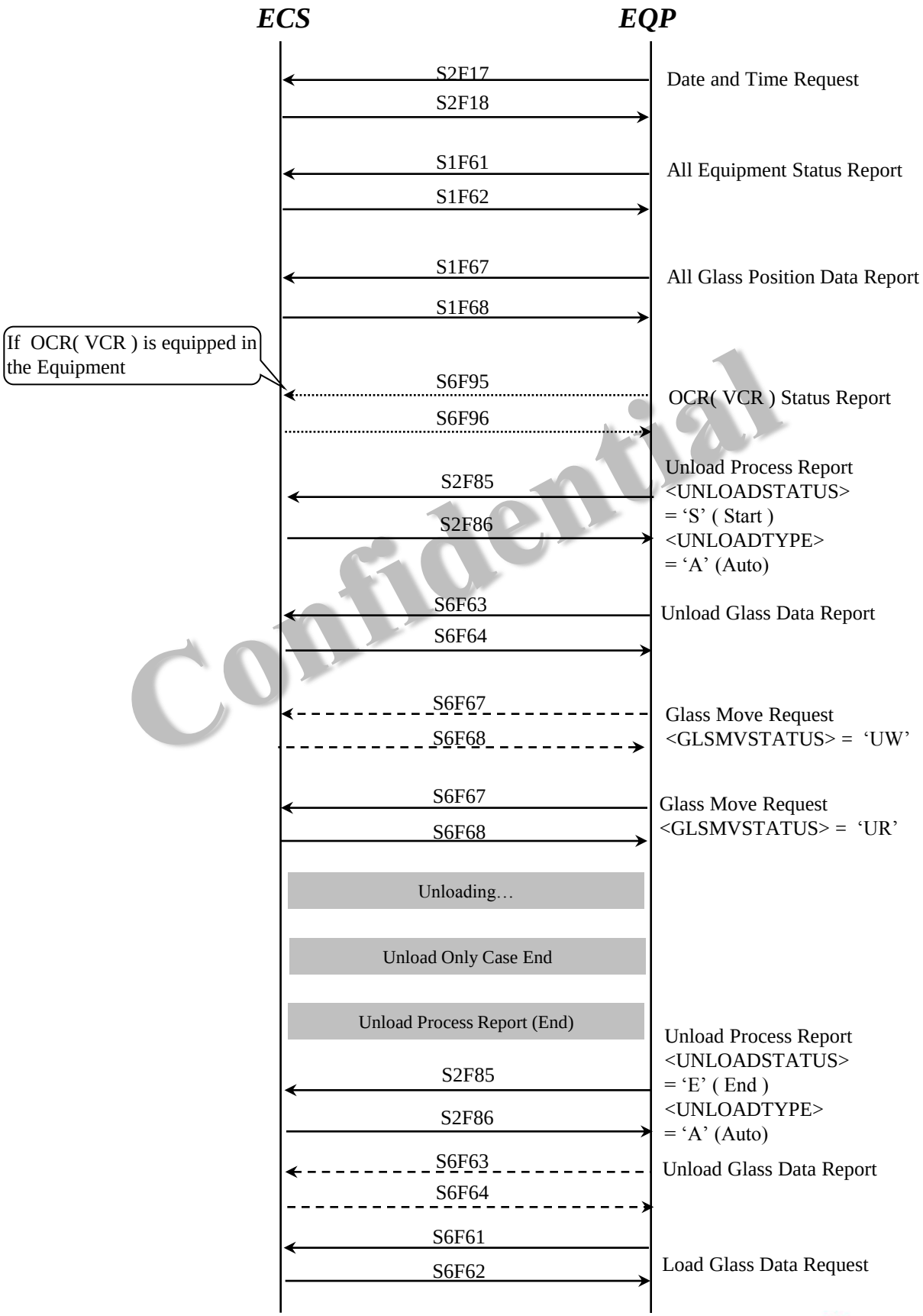
1. 22.2 Unload Start & End (EQP Restart After Unload Start) 1/2



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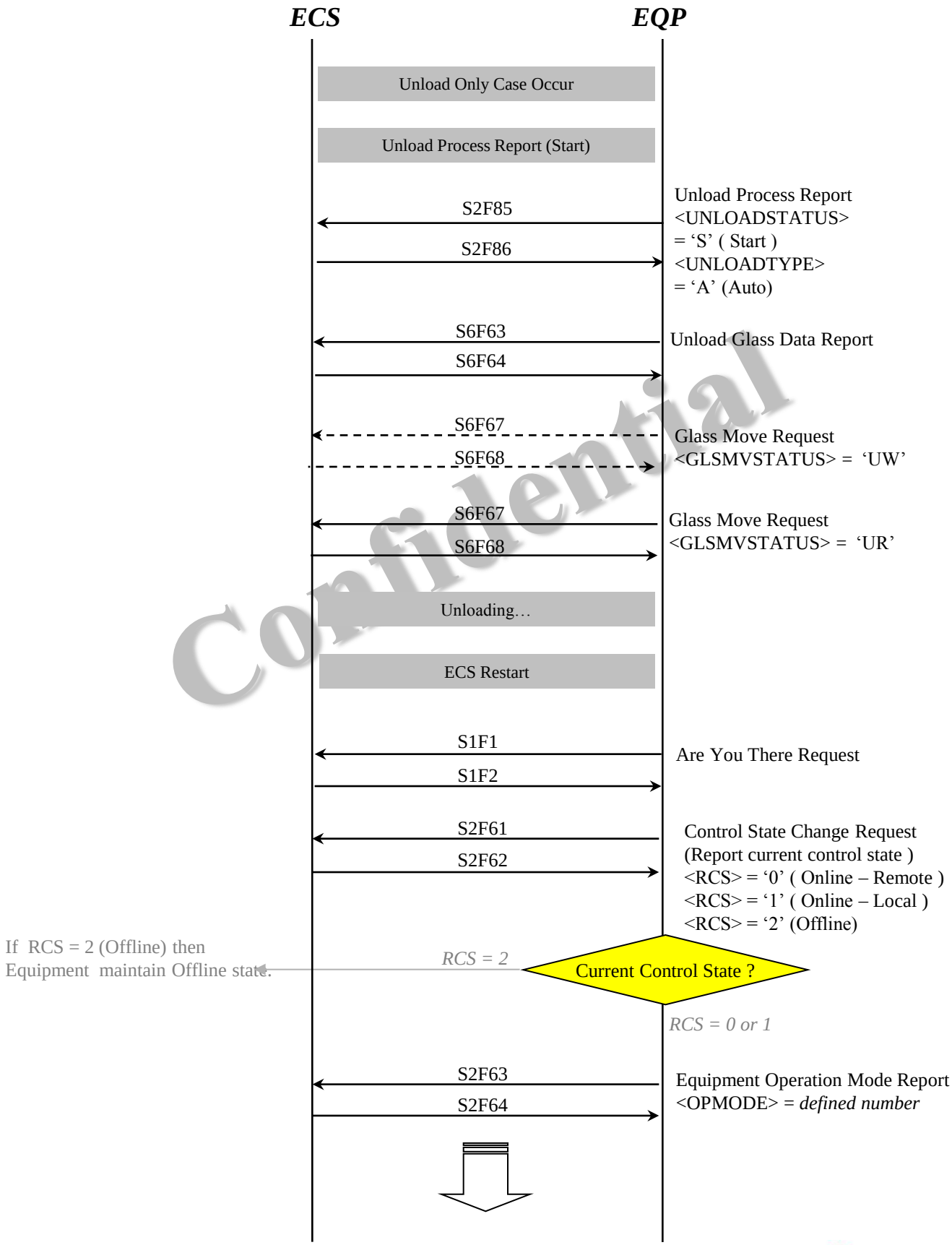
1. 22.2 Unload Start & End (EQP Restart After Unload Start) 2/2



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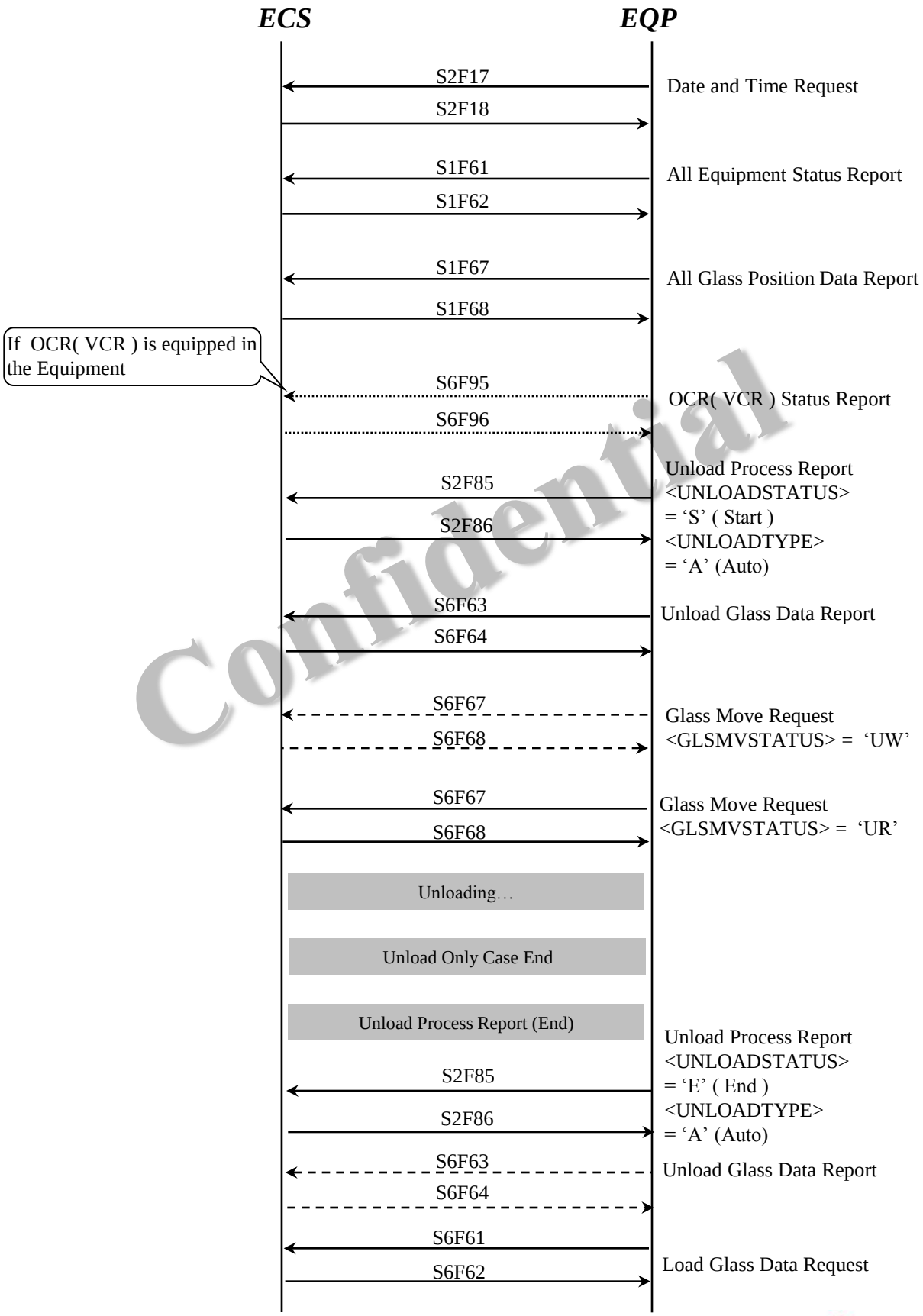
1. 22.3 Unload Start & End (ECS Restart After Unload Start) 1/2



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1. 22.3 Unload Start & End (ECS Restart After Unload Start) 2/2



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2. Exception Message Structure

S1,F69	Change Glass Count Send	ECS→EQP	reply
Description	This message is used when ECS informs engaged glass' quantity data to the Equipment.		
Structure	L,2 1. U2[1] <UNITID> 2. B[1] <GLSCNT>		
Notice	<GLSCNT> Total glasses count for processing <UNITID> Unit ID		
Exception			

S1,F70	Change Glass Count Send Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC1>		
Notice	<UNITID> Unit ID <ACKC1> 0 = Accepted 1 = Not Accepted		
Exception			

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S1,F71	Remained Glass Count Request	ECS←EQP	reply
Description	The Equipment requests remained glass count from the lot.		
Structure	L,3 1.U2[1] <UNITID> 2.B[1] <PORTID> 3.A[16] <LOTID>		
Notice	<UNITID> Unit ID <PORTID> Port ID <LOTID> Lot ID		
Exception			

S1,F72	Remained Glass Count Send	ECS→EQP	
Description	Acknowledge		
Structure	L,4 1.U2[1] <UNITID> 2.B[1] <PORTID> 3.A[16] <LOTID> 4. B[1] <REMAINGLSCNT>		
Notice	<UNITID> Unit ID <PORTID> Port ID <LOTID> Lot ID <REMAINGLSCNT> Remained Glass Count of Requested Lot		
Exception			

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S1,F73	Monitoring Lot Status Data Send	ECS→EQP	reply
Description	This message sends monitoring lot status data.		
Structure	L,2 1. U2[1] <UNITID> 2. A[1] <MLOTSTATUS>		
Notice	<UNITID> Unit ID <MLOTSTATUS> Monitoring Lot Status S = Start / E = End		
Exception			

S1,F74	Monitoring Lot Status Data Send Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC1>		
Notice	<UNITID> Unit ID <ACKC1> 0 = Accepted 1 = Not Accepted		
Exception			

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S1,F77	Equipment's Part Data Report	ECS←EQP	reply

Description	This message is used for tracking of glasses which moved in among sub units
Structure	L,n (n=SubUnit count) 1.L,3 1.U2[1] <UNITID1> 2.B[1] <SUNITID1> 3.L,m (m = glass count) 1.L,2 1.A[6] <CRRID1> 2.U2[1] <GLSCODE1> . . m.L,2 1.A[6] <CRRIDm> 2.U2[1] <GLSCODEm> . . n.L,3 1.U2[1] <UNITIDn> 2.B[1] <SUNITIDn> 3.L,p (p = glass count) 1.L,2 1.A[6] <CRRIDn1> 2.U2[1] <GLSCODEn1> . . p.L,2 1.A[6] <CRRIDnp> 2.U2[1]<GLSCODEnp>
Notice	<UNITID> Unit ID <SUNITID> Sub Unit ID <CRRID> Carrier ID <GLSCODE> Unique Glass Code defined by ECS. Equipment must keep this code, while glass is in equipment.
Exception	A one-to-one correspondence of UNITID and SUNITID depends on Equipment unit configuration

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S1,F78	Equipment's Part Data Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC1>		
Notice	<UNITID> Unit ID <ACKC1> 0 = Accepted 1 = Not Accepted		
Exception			

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S1,F79	Buffer Lot Status Data Send	ECS→EQP	reply
Description	This message sends Buffer lot status data.		
Structure	L,3 1. U2[1] <UNITID> 2. A[1] <BLOTTYPE> 3. A[1] <BLOTSTATUS>		
Notice	<UNITID> Unit ID <BLOTSTATUS> Buffer Lot Status S = Start / E = End <BLOTTYPE> Buffer Lot Type L = Load to Buffer / U = Unload from Buffer		
Exception			

S1,F80	Buffer Lot Status Data Send Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC1>		
Notice	<UNITID> Unit ID <ACKC1> 0 = Accepted 1 = Not Accepted		
Exception			

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S1,F81	Port State Data Send	ECS→EQP	reply
Description	This message is used to transfer Port State Data and Map Data		
Structure	L,7 1.U2[1]<UNITID> 2. A[14] <CSTID> 3. B[1] <PORTID> 4. A[2] <PORTSTATE> 5. A[1] <LOTSTATE> 6. A[40] <CSTMAP> 7. A[40] <SELECTEDMAP>		
Notice	<UNITID> Controller ID <CSTID> CST ID <PORTID> Port ID <PORTSTATE> Port State LC : Cassette Load Complete UC : Cassette Unload Complete LR : Cassette Load Request UR : Cassette Unload Request <LOTSTATE> LOT State R : Ready W : Wait P : Processing E : End G : Going <CSTMAP> Cassette Mapping Information <SELECTEDMAP> Selected glass in the Cassette Map e.g. In case there are 2 glassed in Slot 1 and Slot 40. Then MAP = "1000000000000000000000000000000000000001" 0 : Glass Not Exist 1 : Glass Exist		
Exception	In case of PORTSTATE is UC, MAP is space.		

S1,F82	Port State Data Send Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,2 1.U2[1]<UNITID> 2. B[1] <ACKC1>		
Notice	<UNITID> Controller ID <ACKC1> 0 = Accepted 1 = Not Accepted		
Exception			

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S1,F83	Port State Data Report	ECS←EQP	reply
Description	This message is used when the State (LR, LC, UR, UC) of Port in Equipment is changed.		
Structure	L,5 1. U2[1] <UNITID> 2. A[14] <CSTID> 3. B[1] <PORTID> 4. A[2] <PORTSTATE> 5. A[1] <TRSMODE>		
Notice	<UNITID> Controller ID <CSTID> CST ID <PORTID> Port ID <PORTSTATE> Port State LC : Cassette Load Complete UC : Cassette Unload Complete LR : Cassette Load Request UR : Cassette Unload Request <TRSMODE> Transfer Mode E : PIO Enable D : PIO Disable		
Exception			

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S1,F84	Port State Data Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1.U2[1]<UNITID> 2. B[1]<ACKC1>		
Notice	<UNITID> Controller ID <ACKC1> 0 = Accepted 1 = Not Accepted		
Exception			

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S1,F85	Port Status Change Data Report	ECS←EQP	reply

Description	This message is used when the Status (RUN, IDLE, DOWN) of Port in Equipment is changed.
Structure	L,2 1. U2[1] <UNITID> 2. L,n [n = the number of Port] 1. L,2 1. B[1] <PORTID1> 2. A[1] <PORTSTATUS1> . . n. L,2 1. B[1] <PORTIDn> 2. A[1] <PORTSTATUSn>
Notice	<UNITID> Controller ID <PORTID> Port ID <PORTSTATUS> Port Status R : RUN I : IDLE D : DOWN
Exception	

S1,F86	Port Status Change Data Report Acknowledge	ECS→EQP	

Description	Acknowledge
Structure	L,2 1.U2[1]<UNITID> 2. B[1]<ACKC1>
Notice	<UNITID> Controller ID <ACKC1> 0 = Accepted 1 = Not Accepted
Exception	

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S1,F89	Mask Data Report	ECS←EQP	reply
Description	This message is used after the RFID of Mask reading in EQP		
Structure	L,4 1. U2[1] <UNITID> 2. A[14] <MASKID> 3. B[1] <PORTID> 4. L,n [n = the number of MaskSheet] 1. A[20] <MASKSHEETID1> . . n. A[20] <MASKSHEETIDn>		
Notice	<UNITID> Controller ID <MASKID> MASK ID <PORTID> Port ID <MASKSHEETID> MaskSheet ID		
Exception			

S1,F90	Mask Data Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1.U2[1]<UNITID> 2. B[1]<ACKC1>		
Notice	<UNITID> Controller ID <ACKC1> 0 = Accepted 1 = Not Accepted		
Exception			

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S1,F91	All Mask Position Data Report	ECS←EQP	reply

Description	This message is used for tracking of glasses which moved in among sub units
Structure	L,n [n=SubUnit count] 1.L,4 1.U2[1] <UNITID1> 2.B[1] <SUNITID1> 3.A[14]<MASKID1> 4.L,m [m = masksheet count] 1.A[20] <MASKSHEETID1> . . m.A[20]<MASKSHEETIDm> . . n.L,4 1.U2[1] <UNITID1> 2.B[1] <SUNITID1> 3.A[14]<MASKID1> 4.L,m [m = masksheet count] 1.A[20] <MASKSHEETID1> . . m.A[20]<MASKSHEETIDm>
Notice	<UNITID> Unit ID <SUNITID> Sub Unit ID <MASKID> Mask ID <MASKSHEETID>Masksheet ID
Exception	A one-to-one correspondence of UNITID and SUNITID depends on Equipment unit configuration

S1,F92	All Mask Position Data Report Acknowledge	ECS→EQP	

Description	Acknowledge
Structure	B[1]<ACKC1>
Notice	<ACKC1> 0 = Accepted 1 = Not Accepted
Exception	

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S2,F71	Carrier Transfer Mode Change Report	ECS←EQP	reply
Description	This message is reported by equipment, when carrier transfer mode is changed		
Structure	L,2 1. U2[1] <UNITID> 2. A[1] <CTMODE>		
Notice	<UNITID> Unit ID <CTMODE> Carrier Transfer Mode *The following CTMODE values are defined.* 1 = Normal Transferring 2 = Batch Transferring		
Exception			

S2,F72	Carrier Transfer Mode Change Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC2>		
Notice	<UNITID>Unit ID <ACKC2> 0 = Accepted >0 = Error, not accepted 1 − 63 Reserved		
Exception			

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S2,F77	Glass Call Request	ECS←EQP	reply
Description	This message is used when operator wants to inject glass into the process.		
Structure	L,3 1. U2[1] <UNITID> 2. B[1] <STAGEID> 3. U2[1] <TARGETID>		
Notice	<UNITID> Unit ID <STAGEID>Stage ID <TARGETID> Target Unit ID		
Exception			

S2,F78	Glass Call Request Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC2>		
Notice	<UNITID>Unit ID <ACKC2> 0 = Accepted >0 = Error, not accepted 1 – 63 Reserved		
Exception			

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S2,F81	Lost Glass Data Request	ECS←EQP	reply

Description	The Equipment only can use this control sequence in its manual operation. It can be use when equipment lost glass data, but equipment still have Glass ID or Glass code.
Structure	L,2 1. U2[1]<UNITID> 2. A[16] <GLSID> 3. U2[1] <GLSCODE> 4. A[1] <REQOPTION>
Notice	<UNITID> Unit ID <GLSID> Glass ID <REQOPTION> Request Option C : Glass Code I : Glass ID A : Glass Code and Glass ID
Exception	

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S2,F82	Glass Data Send	ECS→EQP	
Description	The Equipment (manual operation) sends S2F81 and then ECS sends 'Glass Load Data' as reply. 'Glass Load Data' is data about injected glass when glass inject to equipment.		
Structure	L, 3 1. U2[1] <UNITID> 2. B[1]<ACKC2> 3. L,n [n = 0 (Glass Data Not Exist) or 1 (Glass Data Exist)] 1. L,21 1. A[16] <LOTID> 2. A[14] <CSTID> 3. A[3] <SLOTID> 4. A[24] <RECIPEID> 5. A[8] <PRCID> 6. A[1] <GLSODR> 7. A[16] <GLSID> 8. A[2] <GLSJUDGE> 9. A[2000] <PNLIF> 10. A[2000] <SUBMDLIF> 11. B[1] <PORTID> 12. U2[1] <GLSCODE> 13. B[1] <REINPUT> 14. A[5] <GLSTHICK> 15. A[14] <PARTNUM> 16. A[12] <PRODTYPE> 17. A[12] <ATTRIBUTE> 18. A[1] <GLSTYPE> 19. A[32]<KEYID> 20. L, p [p = the number of LSD] 1. L,2 1. A[24] <LSDITEM1> 2. A[40] <LSDVALUE1> : p. L,2 1. A[24] <LSDITEMp> 2. A[40] <LSDCALUEp> 21. L,q [q = the number of GSD] 1.L,2 1. A[24] <GSDITEM11> 2. A[40] <GSDVALUE11> : q. L,2 1. A[24] <GSDITEM1q> 2. A[40] <GSDVALUE1q>		
Exception			

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S2,F83	Injecting Glass Data Request	ECS←EQP	reply
Description	This message is used		
Structure	L,2 1.U2[1] <UNITID> 2. B[1] <STAGEID>		
Notice	<UNITID> Unit ID <STAGEID> Stage ID		
Exception			

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S2,F84	Injecting Glass Data Request Acknowledge	ECS→EQP	
Description	Data of Injecting Glass		
Structure	L, 4 1. U2[1] <UNITID> 2. B[1] <STAGEID> 3. A[1]<GLSSTATUS> 4. L,n [n = 0 (GLSSTATUS = 0) or 1 (GLSSTATUS = 1)] 1. L,23 1. A[16] <LOTID> 2. U2[1] <LOTNUMBER> 3. A[14] <CSTID> 4. A[3] <SLOTID> 5. A[24] <RECIPEID> 6. A[8] <PRCID> 7. A[1] <GLSODR> 8. A[16] <GLSID> 9. A[2] <GLSJUDGE> 10. A[2000] <PNLIF> 11. A[2000] <SUBMDLIF> 12. B[1] <PORTID> 13. U2[1] <GLSCODE> 14. B[1] <REINPUT> 15. A[5] <GLSTHICK> 16. A[14] <PARTNUM> 17. A[16] <PRODTYPE> 18. A[12] <ATTRIBUTE> 19. A[1] <GLSTYPE> 20. A[32]<KEYID> 21. L,p [p = the number of SUBGLS] 1. L, 3 1. A[16] <SUBGLSID1> 2. A[2] <SUBGLSJUDGE1> 3. A[2000] <SUBPNLIF1> : p. L, 3 1. A[16] <SUBGLSIDp> 2. A[2] <SUBGLSJUDGEp> 3. A[2000] <SUBPNLIFp> 22. L, q [q = the number of LSD] 1. L,2 1. A[24] <LSDITEM1> 2. A[40] <LSDVALUE1> :		

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Structure	: q. L,2 1. A[24] <LSDITEMq> 2. A[40] <LSDCALUEq> 23. L,r [r = the number of GSD] 1.L,2 1. A[24] <GSDITEM1> 2. A[40] <GSDVALUE1> : r. L,2 1. A[24] <GSDITEMr> 2. A[40] <GSDVALUEr>
Notice	<UNITID> Unit ID <STAGEID> Stage ID <GLSSTATUS> Status of the glass that would be loaded 0 = Not Exist 1 = Exist <LOTID> LOT ID <LOTNUMBER> Lot Unique Number <CSTID> Cassette ID <SLOTID> Slot ID <RECIPEID> Recipe ID <PRCID> Processing Code <GLSODR> Glass Order 0 = The first glass 1 = All glasses except the first glass and the last glass 2 = The Last glass(In this case, this value may not be delivered for Scrap or reason of LOT Abort) 3 = the only glass, when there is just one in a LOT. <GLSID> Glass ID <GLSJUDGE> Glass Grade Code <PNLIF> Panel Judge Information <SUBMDLIF> Sub Panel Information <SUBGLSID> Sub Glass ID <SUBGLSJUDGE> Sub Glass Grade Code <SUBPNLIF> Sub Panel Information <PORTID> Port ID <GLSCODE> Glass ID Code. <REINPUT> Glass Re-input Flag 0 = Not Re-input, 1 = Re-input <GLSTHICK> Glass's Thickness <PARTNUM> Part Number <PRODTYPE> Type of Producing <ATTRIBUTE> LOT Attribute <GLSTYPE> Glass Type N = Normal Glass D = Dummy Glass <KEYID>Glass Key ID <LSDITEM> LOT Specific Data Item defined by LG Display <LSDVALUE> LOT Specific Data Value defined by LG Display <GSDITEM> Glass Specific Data Item defined by LG Display <GSDVALUE> Glass Specific Data Value defined by LG Display
Exception	

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S2,F85	Unload Process Report	ECS←EQP	reply
Description	Change Unload Process status of the equipment		
Structure	L,3 1. U2[1] <UNITID> 2. A[1] <UNLOADSTATUS> 3. A[1] <UNLOADTYPE>		
Notice	<UNITID>Controller ID <UNLOADSTATUS> Unload Status S : Start E : End <UNLOADTYPE> Unload Type A : EQP Auto Report M : Operator Manual Report		
Exception			

S2,F86	Unload Process Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>		
Notice	<UNITID> Controller ID <ACKC2> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved		
Exception			

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S2,F91	Batch Job(Glass Injection) End Data Send	ECS→EQP	reply
Description	This Message is used when the last glass of Batch Job was injected in Equipment.		
Structure	L,6 1.U2[1] <UNITID> 2. A[1] <BATCHFLAG> 3. A[16] <LOTID> 4. U2[1] <LOTNUMBER> 5. A[14] <CSTID> 6. B[1]<PORTID>		
Notice	<UNITID> Unit ID <BATCHFLAG> Batch Flag I : Injection End A : Lot Aborted C : Lot Cancel <LOTID> LOT ID <LOTNUMBER> Lot Unique Number <PORTID> Port ID <CSTID> Cassette ID <RECIPEID> Recipe ID <SLOTID> Slot ID <GLSID> Glass ID <PORTID> Port ID		
Exception			

S2,F92	Batch Job(Glass Injection) End Data Send Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,2 1.U2[1]<UNITID> 2.B[1]<ACKC2>		
Notice	<UNITID> Unit ID <ACKC2> 0 = Accepted 1 = Not Accepted 2 – 63 Reserved		
Exception			

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S2,F93	Various Data Report	ECS←EQP	reply

Description	This message is used in various data report to ECS.
Structure	L,3 1. U2[1] <UNITID> 2. U2[1] <CEID> 3. L,n [n = Number of slot] 1. L,7 1. A[48] <SLOTID1> 2. A[48] <GLSID1> 3. A[48] <GLSJUDGE1> 4. A[2000] <PNLIF11> 5. A[2000] <PNLIF21> 6. A[2000] <SUBMDLIF11> 7. A[2000] <SUBMDLIF21> . . . n. L,7 1. A[48] <SLOTIDn> 2. A[48] <GLSIDn> 3. A[48] <GLSJUDGEn> 4. A[2000] <PNLIF1n> 5. A[2000] <PNLIF2n> 6. A[2000] <SUBMDLIF1n> 7. A[2000] <SUBMDLIF2n>
Notice	<UNITID> Unit ID <CEID> Collected Event ID <SLOTID> Slot ID <GLSID> Glass ID <GLSJUDGE> Glass Judge <PNLIF1> Panel Judge of panle1 ~ panel2000 <PNLIF2> Panel Judge of panle2001 ~ panel4000 <SUBMDLIF1> Sub Panel Judge of sub panle1 ~ sub pane2000 <SUBMDLIF2> Sub Panel Judge of sub panle2001 ~ sub pane4000
Exception	

S2,F94	Various Data Report Acknowledge	ECS→EQP	

Description	Acknowledge
Structure	L,3 1. U2[1] <UNITID> 2. U2[1] <CEID> 3. B[1] <ACKC2>
Notice	<UNITID> Unit ID <CEID> Collected Event ID <ACKC2> Acknowledge code 0 : Accepted 1 : Not Accepted > 1 : Error
Exception	

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S2,F99	Equipment PD Down Request	ECS→EQP	reply
Description	This message is used when requesting the PD Down state change.		
Structure	L,3 1. U2[1] <UNITID> 2. A[6] <RCMD> 3. A[120] <HOSTMSG>		
Notice	<UNITID>Unit ID <RCMD>Remote command code or string – PAUSE / RESUME <HOSTMSG> Host message in Korean and English		
Exception			

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S2,F100	Equipment PD Down Request Acknowledge	ECS←EQP	
Description	Acknowledge		
Structure	L,2 1. U2[1]<UNITID> 2. B[1]<ACKC6>		
Notice	<UNITID> Unit ID <ACKC6> 0 = Accepted 1 = Not Accepted 2 = Already Release 3~ 63 Reserved		
Exception			

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S6,F81	Lot State Change Report	ECS←EQP	reply
Description	This message is used when the Lot State (R, W, P, E, G) of port in Equipment is changed.		
Structure	L,5 1. U2[1] <UNITID> 2. A[14] <CSTID> 3. A[24] <RECIPEID> 4. B[1] <PORTID> 5. A[1] <LOTSTATE>		
Notice	<UNITID> Controller ID <CSTID> CST ID <RECIPEID> Recipe ID <PORTID> Port ID <LOTSTATE> LOT State R : Ready W : Wait P : Processing E : End		
Exception			

S6,F82	Lot State Change Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,2 1.U2[1]<UNITID> 2. B[1]<ACKC1>		
Notice	<UNITID> Controller ID <ACKC1> 0 = Accepted 1 = Not Accepted		
Exception			

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S6,F97	Chamber Status Change Report	ECS←EQP	reply
Description	This message is used in Chamber Status Change Report.		
Structure	L,n [n = the number of Chamber] 1. L,2 1. U2[1] <UNITID1> 2. A[1] <CMSTATUS1> . . n. L,2 1. U2[1] <UNITIDn> 2. A[1] <CMSTATUSn>		
Notice	<UNITID> Unit ID <CMSTATUS> Chamber Status N : Normal / D : Dummy / S : Self Cleaning / E : Empty Discharge / X : No Use		
Exception			

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S6,F98	Chamber Status Change Report Acknowledge	ECS→EQP	
Description	Acknowledge		
Structure	L,n [n = the number of Chamber] 1. L,2 1. U2[1] <UNITID1> 2. B[1] < ACKC6 > . . n. L,2 1. U2[1] <UNITIDn> 2. B[1] < ACKC6 >		
Notice	<UNITID>Unit ID <ACKC6> 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved		
Exception			

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S7,F79	Equipment Constant Recipe Request	ECS→EQP	reply
Description	Equipment Constant Recipe Request		
Structure	L, 6 1.A[16]<LOTID> 2.A[14]<CSTID > 3.A [24]<RECIPEID> 4.A [8]<PRCID > 5.B [1]<PORTID> 6.U2[1]<UNITID>		
Notice	<LOTID> LOT ID <CSTID> Cassette ID <RECIPEID> Recipe ID <PRCID> Processing Code <PORTID> Port ID <UNITID> Unit ID		
Exception			

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S7,F80	Equipment Constant Recipe	ECS←EQP	
Description	This message reports the Equipment Constant Recipe parameter of the equipment		

Structure

L,8
1. A[16] <LOTID>
2. A[14]<CSTID>
3. A[24]<RECIPEID>
4. A[12]<RECIPEIDVER>
5. A[8]<PRCID>
6. B[1]<PORTID>
7. U2[1]<UNITID>
8. L,m [m = the number of Unit]
   1.L.6
   1. U2[1] <EUNITID1>
   2. A[24] <LOWERRECIPEID1>
   3. A[12] <LOWERRECIPEIDVER1>
   4. A[1]<LEVEL1>
   5. A[1]<RECIPEIDTYPE1>
   6. L,p [p = Number of Step]
      1. L,2
         1. A[14] <STEP11>
         2. L, n [n = Number of Parameter]
            1. L,2
               1. A[24] <RPARMNAME111>
               2. A[24] <RPARMVAL111>
            :
            n.L,2
               1. A[24] <RPARMNAME11n>
               2. A[24] <RPARMVAL11n>
      :
      p. L,2
         1. A[14] <STEP1p>
         2. L, q [q = Number of Parameter]
            1. L,2
               1. A[24] <RPARMNAME1p1>
               2. A[24] <RPARMVAL1p1>
            :
            q.L,2
               1. A[24] <RPARMNAME1pq>
               2. A[24] <RPARMVAL1pq>
   :

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	: m.L,6 1. U2[1] <EUNITIDm> 2. A[24] <LOWERRECIPEIDm> 3. A[12] <LOWERRECIPEIDVERm> 4. A[1]<LEVELm> 5. A[1]<RECIPEIDTYPEm> 6. L,r [r = Number of Step] 1. L,2 1. A[14] <STEPm1> 2. L, s [s = Number of Parameter] 1. L,2 1. A[24] <RPARMNAMEm11> 2. A[24] <RPARMVALm11> : s.L,2 1. A[24] <RPARMNAMEm1s> 2. A[24] <RPARMVALm1s> : r. L,2 1. A[14] <STEPmr> 2. L, t [t = Number of Parameter] 1. L,2 1. A[24] <RPARMNAMEmr1> 2. A[24] <RPARMVALmr1> : t.L,2 1. A[24] <RPARMNAMEmrt> 2. A[24] <RPARMVALmrt>
Notice	<LOTID> LOT ID <CSTID> Cassette ID <PRCID> Processing Code <RECIPEID> Recipe ID <PORTID> Port ID <UNITID> Unit ID <EUNITID> Each Unit ID <LOWERRECIPEID> Lower Recipe ID <LOWERRECIPEIDVER> Lower Recipe ID Version <LEVEL> Level of RECIPEID 1 : Level 1 RECIPEID 2 : Level 2 RECIPEID 3 : Level 3 RECIPEID < RPARMNAME> Recipe Parameter Name < RPARMVAL> Recipe Parameter Value <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <STEP> Recipe Step 0 = Step Not Used, Another Number is Recipe Step
Exception	

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S7,F81	Multi Recipe Information Send	ECS → EQP	reply

Description	This message is used to send each slot recipe ID to EQP.
Structure	L,5 1.U2[1]<UNITID> 2. B[1]<PORTID> 3. A[16]<LOTID> 4. U2[1] <LOTNUMBER> 5.L,n (n= The Number of slot) 1.L,4 1.A[3]<SLOTID> 2.B[1]<SELECTED> 3.A[24]<RECIPEID> 4.B[1]<SAMPLECODE> : n.L,4 1.A[3]<SLOTID> 2.B[1]<SELECTED> 3.A[24]<RECIPEID> 4.B[1]<SAMPLECODE>
Notice	<UNITID> Controller ID <PORTID> Port ID <LOTID> Lot ID <LOTNUMBER> Lot Unique Number <SLOTID> Slot ID <SELECTED> Glass Injection Flag 0 : Not Injection 1 : Injection <RECIPEID> Recipe ID <SAMPLECODE> Glass Inspection flag 0 : Not Inspection 1 : Inspection
Exception	

S7,F82	Multi Recipe Information Send Acknowledge	ECS ← EQP	

Description	Acknowledge
Structure	L,2 1.U2[1]<UNITID> 2.B[1]<ACKC7>
Notice	<UNITID> Controller ID <ACKC7> Acknowledge code, 1byte 0 = Accepted 1 = Not Accepted 2 ~ 63 Reserved
Exception	

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S15,F65	Current Recipe ID List Request	ECS→EQP	reply
Description	This message is used to request the transmission of the current equipment process program directory. This is the list of all the RECIPE IDs & Recipe ID Version of the process programs stored in the equipment		
Structure	U2[1] <UNITID>		
Notice	<UNITID> Unit ID		
Exception			

S15,F66	Current Recipe ID List Request Acknowledge	ECS←EQP	
Description	This message is used to transmit the current Recipe ID List		
Structure	L,2 1. U2[1] <UNITID> 2. L,n [n = Recipe List Count] 1.L,3 1. A[24] <RECIPEID1> 2. A[1] <RECIPEIDTYPE1> 3. L,p [p = Number of Recipe ID Version] 1. A[12]<RECIPEIDVER11> : n. A[12]<RECIPEIDVER1p> : n.L,3 1. A[24] <RECIPEIDn> 2. A[1] <RECIPEIDTYPEn> 3. L,q [q = Number of Recipe ID Version] 1. A[12]<RECIPEIDVERn1> . . q. A[12]<RECIPEIDVERnq>		
Notice	<UNITID> Unit ID <RECIPEID> Recipe ID <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <RECIPEIDVER> Recipe ID Version		
Exception			

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S15,F71	Recipe List Report Request	ECS→EQP	reply
Description	This message is used when ECS requests recipe list of the equipment		
Structure	L,4 1. U2[1] <UNITID> 2. A[1] <LEVEL> 3. A[1] <RECIPEIDTYPE> 4. L,m [m = the number of LSD] 1. L,2 1. A[24] <ESDITEM1> 2. A[40] <ESDVALUE1> : m. L,2 1.A[24] <ESDITEMm> 2.A[40] <ESDVALUEm>		
Notice	<UNITID> Unit ID <LEVEL> Level of RECIPEID 1 : Level 1 RECIPEID 2 : Level 2 RECIPEID 3 : Level 3 RECIPEID <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <ESDITEM> EQP Specific Data Item defined by LG Display <ESDVALUE> EQP Specific Data Value defined by LG Display		
Exception			

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S15,F72	Equipment Recipe List Report	ECS ←EQP	

Description	Acknowledge
Structure	L,2 1. L, m [m = Number of RECIPEID] 1. L,5 1. U2[1] <UNITID1> 2. A[24] <RECIPEID1> 3. A[1] <LEVEL1> 4. A[1] <RECIPEIDTYPE1> 5. L, p [p = Number of version] 1. L, 2 1. A[12] <RECIPEIDVER11> 2. A[1] <RECIPEIDSTATUS11> : p. L, 2 1. A[12] <RECIPEIDVER1p> 2. A[1] <RECIPEIDSTATUS1p> : m. L,5 1. U2[1] <UNITIDm> 2. A[24] <RECIPEIDm> 3. A[1] <LEVELm> 4. A[1] <RECIPEIDTYPEm> 5. L, q [q = Number of version] 1. L, 2 1. A[12] <RECIPEIDVERm1> 2. A[1] <RECIPEIDSTATUSm1> : q. L, 2 1. A[12] <RECIPEIDVERmq> 2. A[1] <RECIPEIDSTATUSmq> 2. L, r [r = Number of ESD] 1. L,2 1. A[24] <ESDITEM1> 2. A[40] <ESDVALUE1> : r. L,2 1. A[24] <ESDITEMr> 2. A[40] <ESDVALUEr>
Notice	<UNITID> Unit ID <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <RECIPEID> Recipe ID <LEVEL> Level of Recipe ID <RECIPEIDVER> Recipe ID Version <RECIPEIDSTATUS> Recipe ID Status R : Running (using) I : Idle (Editable) D : Disable (Locked)
Exception	

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Notice	<ESDITEM> EQP Specific Data Item defined by LG Display <ESDVALUE> EQP Specific Data Value defined by LG Display
Exception	

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S15,F73	Lower Recipe List Report Request	ECS→EQP	reply

Description	This message is used when ECS requests lower recipe list of the equipment
Structure	L,2 1. L, 5 1. U2[1] <UNITID> 2. A[24] <RECIPEID> 3. A[1] <LEVEL> 4. A[1] <RECIPEIDTYPE> 5. A[12] <RECIPEIDVER> 2. L,m [m = the number of ESD] 1. L,2 1. A[24] <ESDITEM1> 2. A[40] <ESDVALUE1> : m. L,2 1.A[24] <ESDITEMm> 2.A[40] <ESDVALUEm>
Notice	<UNITID> Unit ID <RECIPEID> Recipe ID <LEVEL> Level of Recipe ID <RECIPEIDVER> Recipe ID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <ESDITEM> EQP Specific Data Item defined by LG. Display <ESDVALUE> EQP Specific Data Value defined by LG. Display
Exception	

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S15,F74	Lower Recipe List Report	ECS ←EQP	

Description	Acknowledge
Structure	L,2 1. L, m [m = Number of RECIPEID] 1. L, 5 1. U2[1] <UNITID1> 2. A[24] <RECIPEID 1> 3. A[1] <LEVEL1> 4. A[1] <RECIPEIDTYPE> 5. L, p [p = Number of version] 1. A[12] <RECIPEIDVER11> : p. A[12] <RECIPEIDVER1p> : m. L, 5 1. U2[1] <UNITIDm> 2. A[24] <RECIPEIDm> 3. A[1] <LEVELm> 4. A[1] <RECIPEIDTYPEm> 5. L, q [q = Number of version] 1. A[12] <RECIPEIDVERm1> : q. A[12] <RECIPEIDVER1mq> 2. L, r [r = Number of ESD] 1. L,2 1. A[24] <ESDITEM1> 2. A[40] <ESDVALUE1> : r. L,2 1. A[24] <ESDITEMr> 2. A[40] <ESDVALUEr>
Notice	<UNITID> Unit ID <RECIPEID> Recipe ID <LEVEL> Level of Recipe ID <RECIPEIDVER> Recipe ID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <ESDITEM> LOT Specific Data Item defined by LG. Display <ESDVALUE> LOT Specific Data Value defined by LG. Display
Exception	

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S15,F75	Recipe Parameter Report Request	ECS→EQP	reply

Description	-This message is used when ECS requests recipe parameter of the equipment
Structure	L,6 1. U2[1] <UNITID> 2. A[24] <RECIPEID> 3. A[1] <LEVEL> 4. A[1] <RECIPEIDTYPE> 5. A[12] <RECIPEIDVER> 6. L,m [m = the number of ESD] 1. L,2 1. A[24] <ESDITEM1> 2. A[40] <ESDVALUE1> : m. L,2 1.A[24] <ESDITEMm> 2.A[40] <ESDVALUEm>
Notice	<UNITID> Unit ID <RECIPEID> Recipe ID <LEVEL> Level of Recipe ID <RECIPEIDVER> Recipe ID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <ESDITEM> EQP Specific Data Item defined by LG. Display <ESDVALUE> EQP Specific Data Value defined by LG. Display
Exception	

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S15,F76	Recipe Parameter Report	ECS ←EQP	
Description	This message reports the Recipe Parameter of the equipment		

Structure

L,2
1. L, 6
   1. U2[1] <UNITID1>
   2. A[24] <RECIPEID1>
   3. A[1] <LEVEL1>
   4. A[1] <RECIPEIDTYPE1>
   5. A[12] <RECIPEIDVER1>
   6. L,p [p = Number of Step]
      1. L,2
         1. A[14] <STEP1>
         2. L, n [n = Number of Parameter]
            1. L,2
               1. A[24] <RPARMNAME11>
               2. A[24] <RPARMVAL11>
            :
            n.L,2
               1. A[24] <RPARMNAME1n>
               2. A[24] <RPARMVAL1n>
      :
      p. L,2
         1. A[14] <STEPp>
         2. L, q [q = Number of Parameter]
            1. L,2
               1. A[24] <RPARMNAMEp1>
               2. A[24] <RPARMVALp1>
            :
            q.L,2
               1. A[24] <RPARMNAMEpq>
               2. A[24] <RPARMVALpq>
2. L, s [s = Number of ESD]
   1. L,2
      1. A[24] <ESDITEM1>
      2. A[40] <ESDVALUE1>
   :
   s. L,2
      1. A[24] <ESDITEMs>
      2. A[40] <ESDVALUEs>

Notice

<UNITID> Unit ID
<RECIPEID> Recipe ID
<LEVEL> Level of Recipe ID
<RECIPEIDVER> Recipe ID Version
<RECIPEIDTYPE>Type of Recipe
E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe
1,2,3,.. : Special Recipe (Defined with LG Display)
<RPARMNAME> Recipe Parameter Name
<RPARMVAL> Recipe Parameter Value
<ESDITEM> EQP Specific Data Item defined by LG. Display
<ESDVALUE> EQP Specific Data Value defined by LG. Display
<STEP> Recipe Step
0 = Step Not Used, Another Number is Recipe Step

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S15,F77	Recipe Parameter Change Send	ECS→EQP	reply

Description	This message commands Recipe Parameter change to the equipment
Structure	L, 8 1. U2[1] <UNITID1> 2. A[1] <RECIPEIDCMD1> 3. A[24] <RECIPEID1> 4. A[1] <LEVEL1> 5. A[12] <RECIPEIDVER1> 6. A[1] <RECIPEIDTYPE1> 7. L,p [p = Number of Step] 1. L,2 1. A[14] <STEP1> 2. L, n [n = Number of Parameter] 1. L,2 1. A[24] <RPARMNAME11> 2. A[24] <RPARMVAL11> : n.L,2 1. A[24] <RPARMNAME1n> 2. A[24] <RPARMVAL1n> : p. L,2 1. A[14] <STEPp> 2. L, q [q = Number of Parameter] 1. L,2 1. A[24] <RPARMNAMEp1 2. A[24] <RPARMVALp1> : q.L,2 1. A[24] <RPARMNAMEpq> 2. A[24] <RPARMVALpq> 8. L, r [r = Number of ESD] 1. L,2 1. A[24] <ESDITEM11> 2. A[40] <ESDVALUE11> : r. L,2 1. A[24] <ESDITEM1r> 2. A[40] <ESDVALUE1r>

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Notice	<UNITID> Unit ID <RECIPEIDCMD> RECIPEID Command B : Built C : change D : Delete R : Rename <RECIPEID> Recipe ID <LEVEL> Level of RECIPEID <RECIPEIDVER> RECIPEID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) < RPARMNAME> Recipe Parameter Name < RPARMVAL> Recipe Parameter Value <ESDITEM> EQP Specific Data Item defined by LG. Display <ESDVALUE> EQP Specific Data Value defined by LG. Display <STEP> Recipe Step 0 = Step Not Used, Another Number is Recipe Step
Exception	

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S15,F78	Recipe Change Command Acknowledge	ECS ←EQP	
Description	Recipe Parameter change command acknowledge		
Structure	L, 6 1. U2[1] <UNITID> 2. A[24] <RECIPEID> 3. A[1] <LEVEL> 4. A[12] <RECIPEIDVER> 5. A[1] <RECIPEIDTYPE> 6. B[1] <REACK>		
Notice	<UNITID> Unit ID <LEVEL> Level of RECIPEID <RECIPEIDVER> RECIPEID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <REACK> 0 : Accepted 1 : Permission not granted 2 : RECIPEID using 3 : Equipment not responding 4 : Version not found (Roll back) 5 : RECIPEID not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist		
Exception			

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S15,F79	Recipe Parameter Change Report	ECS ←EQP	Reply

Description	This message reports Recipe Parameter change from equipment
Structure	L, 8 1. U2[1] <UNITID1> 2. A[1] <RECIPEIDCMD1> 3. A[24] <RECIPEID1> 4. A[1] <LEVEL1> 5. A[12] <RECIPEIDVER1> 6. A[1] <RECIPEIDTYPE1> 7. L,p [p = Number of Step] 1. L,2 1. A[14] <STEP1> 2. L, n [n = Number of Parameter] 1. L,2 1. A[24] <RPARMNAME11> 2. A[24] <RPARMVAL11> : n.L,2 1. A[24] <RPARMNAME1n> 2. A[24] <RPARMVAL1n> : p. L,2 1. A[14] <STEP1p> 2. L, q [q = Number of Parameter] 1. L,2 1. A[24] <RPARMNAMEp1> 2. A[24] <RPARMVALp1> : q.L,2 1. A[24] <RPARMNAMEpq> 2. A[24] <RPARMVALpq> 8. L, r [r = Number of ESD] 1. L,2 1. A[24] <ESDITEM11> 2. A[40] <ESDVALUE11> : r. L,2 1. A[24] <ESDITEM1r> 2. A[40] <ESDVALUE1r>

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Notice	<UNITID> Unit ID <RECIPEIDCMD> RECIPEID Command B : Built C : change D : Delete R : Rename <RECIPEID> Recipe ID <LEVEL> Level of RECIPEID <RECIPEIDVER> RECIPEID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) < RPARMNAME> Recipe Parameter Name < RPARMVAL> Recipe Parameter Value <ESDITEM> EQP Specific Data Item defined by LG. Display <ESDVALUE> EQP Specific Data Value defined by LG. Display <STEP> Recipe Step 0 = Step Not Used, Another Number is Recipe Step
Exception	

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S15,F80	Recipe Change Report Acknowledge	ECS → EQP	
Description	Recipe Parameter change acknowledge		
Structure	L, 6 1. U2[1] <UNITID1> 2. A[24] <RECIPEID1> 3. A[1] <LEVEL1> 4. A[12] <RECIPEIDVER1> 5. A[1] <RECIPEIDTYPE1> 6. B[1] <PHACK1>		
Notice	<UNITID> Unit ID <LEVEL> Level of RECIPEID <RECIPEIDVER> RECIPEID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <PHACK> 0 : Accepted 1 : Permission not granted 2 : Length error 3 : RECIPEID not found 4 : Parameter range error 5 : Saved for additional use		
Exception			

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S15,F81	Equipment Operation Reference Data Send	ECS → EQP	Reply

Description	This message sends equipment operation reference data
Structure	L,6 1. U2[1] <UNITID> 2. A[8] <USEMODE> 3. A[16] <LOTID> 4. L,n [n = Number of Panel Data] 1. L, 2 1. A[288] <LOTPRITEMm1> 2. A[450] <LOTPRDATAm1> : n. L, 2 1. A[288] <LOTPRITEMmn> 2. A[450] <LOTPRDATAmn> 5. L, v [v = Number of LOT DV Item] 1. L, 2 1. A[14] <LOTDVNAME1> 2. A[14] <LOTDVVAL1> : v. L, 2 1. A[14] <LOTDVNAMEv> 2. A[14] <LOTDVVALv> 6. L,m [m = Number of Glass] 1. L,5 1. A[16] <GLSID1> 2. A[24] <RECIPEID1> 3. A[40] <EQPINFO1> 4. L, n [n = Number of Panel Data] 1. L, 2 1. A[288] <GLSPRITEM1> 2. A[450] <GLSPRDATA1> : n. L, 2 1. A[288] <GLSPPITEMn> 2. A[450] <GLSPRDATAn> 5. L, p [p = Number of Item] 1. L, 2 1. A[14] <GLSDVNAME11> 2. A[14] <GLSDVVAL11> : n. L, 2 1. A[14] <DVNAME1n> 2. A[14] <DVVAL1n> : m. L,5 1. A[16] <GLSIDm> 2. A[24] <RECIPEIDm> 3. A[40] <EQPINFOm> 4. L, n [n = Number of Panel Data] 1. L, 2 1. A[288] <GLSPRITEMm1> 2. A[450] <GLSPRDATAm1> : n. L, 2 1. A[288] <GLSPRITEMmn> 2. A[450] <GLSPRDATAmn> 5. L, p [p = Number of Item] 1. L, 2 1. A[14] <GLSDVNAMEm1> 2. A[14] <GLSDVVALm1> : p. L, 2 1. A[14] <GLSDVNAMEmp> 2. A[14] <GLSDVVALmp>

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Notice	<UNITID> Unit ID <USEMODE> Message Using Mode <RECIPEID> Recipe ID <EQPINFO> Equipment Information <LOTPRITEM> Lot Process Reference Item <LOTPRDATA> Lot Process Reference Data <LOTDVNAME> LOT Data Name <LOTDVVAL> LOT Data Value <GLSPRITEM> Glass Process Reference Item <GLSPRDATA> Glass Process Reference Data <GLSDVNAME> Data Name <GLSDVVAL> Data Value
Exception	

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S15,F82	Equipment Operation Reference Data Send Acknowledge	ECS ← EQP	
Description	Operation Reference Data Send Acknowledge		
Structure	L, m [m = Number of Unit] 1. L, 2 1. U2[1] <UNITID> 2. B[1] <DACK> : m. L, 2 1. U2[1] <UNITID> 2. B[1] <DACK>		
Notice	<UNITID> Unit ID <DACK> 0 : Accepted 1 : Accepted and changed parameter 2 : Invalid parameter 3 : Parameter range error 4 : Length error 5 : Saved for additional use 6 : Saved for additional use		
Exception			

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S15,F83	Equipment Operation Reference Data Apply Report	ECS → EQP	Reply
Description	When equipment applied operation reference data, equipment sends this message.		
Structure	L,4 1. U2[1] <UNITID> 2. A[24] <RECIPEID> 3. A[16] <LOTID> 4. B[1] <AACK>		
Notice	<UNITID> Unit ID <RCP> Recipe <LOTID> LOT ID <AACK> 0 : Updated 1 : Error occurred 2 : RCP not found 3 : The Recipe is Not Using Now (Request Recipe Unmatched) 4 : Disable Equipment Condition About R2R Request 5 : Overlapping R2R Request 6 : One Time Variation Range Over 7 : User Manual Cancel 8 : Parameter Range Error 9 : Invalid Parameter		
Exception			

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S15,F84	Operation Reference Data Send Acknowledge	ECS ← EQP	
Description	Equipment Operation Reference Data Apply Acknowledge		
Structure	Header Only		
Notice			
Exception			

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S15,F95	Equipment Constant Recipe Parameter Request	ECS→EQP	reply

Description	-This message is used when ECS requests Equipment Constant Recipe Parameter recipe parameter of the equipment
Structure	L,6 1. U2[1] <UNITID> 2. A[24] <RECIPEID> 3. A[1] <LEVEL> 4. A[1] <RECIPEIDTYPE> 5. A[12] <RECIPEIDVER> 6. L,m [m = the number of ESD] 1. L,2 1. A[24] <ESDITEM1> 2. A[40] <ESDVALUE1> : m. L,2 1.A[24] <ESDITEMm> 2.A[40] <ESDVALUEm>
Notice	<UNITID> Unit ID <RECIPEID> Recipe ID <LEVEL> Level of Recipe ID <RECIPEIDVER> Recipe ID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <ESDITEM> EQP Specific Data Item defined by LG. Display <ESDVALUE> EQP Specific Data Value defined by LG. Display
Exception	

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S15,F96	Equipment Constant Recipe Parameter Send	ECS ←EQP	

Description	This message reports the Equipment Constant Recipe Parameter of the equipment
Structure	L,2 1. L, 6 1. U2[1] <UNITID1> 2. A[24] <RECIPEID1> 3. A[1] <LEVEL1> 4. A[1] <RECIPEIDTYPE1> 5. A[12] <RECIPEIDVER1> 6. L,p [p = Number of Step] 1. L,2 1. A[14] <STEP1> 2. L, n [n = Number of Parameter] 1. L,2 1. A[24] <RPARMNAME11> 2. A[24] <RPARMVAL11> : n.L,2 1. A[24] <RPARMNAME1n> 2. A[24] <RPARMVAL1n> : p. L,2 1. A[14] <STEPp> 2. L, q [q = Number of Parameter] 1. L,2 1. A[24] <RPARMNAMEp1> 2. A[24] <RPARMVALp1> : q.L,2 1. A[24] <RPARMNAMEpq> 2. A[24] <RPARMVALpq> 2. L, s [s = Number of ESD] 1. L,2 1. A[24] <ESDITEM1> 2. A[40] <ESDVALUE1> : s. L,2 1. A[24] <ESDITEMs> 2. A[40] <ESDVALUEs>
Notice	<UNITID> Unit ID <RECIPEID> Recipe ID <LEVEL> Level of Recipe ID <RECIPEIDVER> Recipe ID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <ESDITEM> EQP Specific Data Item defined by LG. Display <ESDVALUE> EQP Specific Data Value defined by LG. Display <STEP> Recipe Step 0 = Step Not Used, Another Number is Recipe Step

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S15,F97	Equipment Constant Recipe Change Command	ECS→EQP	reply
Description	This message commands Equipment Constant Recipe Parameter change to the equipment		

Structure

L, 8
1. U2[1] <UNITID1>
2. A[1] <RECIPEIDCMD1>
3. A[24] <RECIPEID1>
4. A[1] <LEVEL1>
5. A[12] <RECIPEIDVER1>
6. A[1] <RECIPEIDTYPE1>
7. L,p [p = Number of Step]
    1. L,2
        1. A[14] <STEP1>
        2. L, n [n = Number of Parameter]
            1. L,2
                1. A[24] <RPARMNAME11>
                2. A[24] <RPARMVAL11>
            :
            n.L,2
                1. A[24] <RPARMNAME1n>
                2. A[24] <RPARMVAL1n>
    :
    p. L,2
        1. A[14] <STEPp>
        2. L, q [q = Number of Parameter]
            1. L,2
                1. A[24] <RPARMNAMEp1
                2. A[24] <RPARMVALp1>
            :
            q.L,2
                1. A[24] <RPARMNAMEpq>
                2. A[24] <RPARMVALpq>
8. L, r [r = Number of ESD]
    1. L,2
        1. A[24] <ESDITEM11>
        2. A[40] <ESDVALUE11>
    :
    r. L,2
        1. A[24] <ESDITEM1r>
        2. A[40] <ESDVALUE1r>

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Notice	<UNITID> Unit ID <RECIPEIDCMD> RECIPEID Command B : Built C : change D : Delete R : Rename <RECIPEID> Recipe ID <LEVEL> Level of RECIPEID <RECIPEIDVER> RECIPEID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) < RPARMNAME> Recipe Parameter Name < RPARMVAL> Recipe Parameter Value <ESDITEM> EQP Specific Data Item defined by LG. Display <ESDVALUE> EQP Specific Data Value defined by LG. Display <STEP> Recipe Step 0 = Step Not Used, Another Number is Recipe Step
Exception	

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S15,F98	Equipment Constant Recipe Change Command Acknowledge	ECS ←EQP	
Description	Equipment Constant Recipe Change Command Acknowledge		
Structure	L, 7 1. U2[1] <UNITID> 2. A[24] <RECIPEID> 3. A[1] <LEVEL> 4. A[12] <RECIPEIDVER> 5. A[1] <RECIPEIDTYPE> 6. B[1] <REACK> 7. A[180]<EQPMSG>		
Notice	<UNITID> Unit ID <LEVEL> Level of RECIPEID <RECIPEIDVER> RECIPEID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <REACK> 0 : Accepted 1 : Permission not granted 2 : RECIPEID using 3 : Equipment not responding 4 : Version not found (Roll back) 5 : RECIPEID not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist 9 : Invalid parameter 10 : Equipment in Off-line Mode		
Exception			

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S15,F99	Equipment Constant Recipe Change Report	ECS ←EQP	Reply

Description	This message reports Equipment Constant Recipe Parameter change from equipment
Structure	L, 8 1. U2[1] <UNITID1> 2. A[1] <RECIPEIDCMD1> 3. A[24] <RECIPEID1> 4. A[1] <LEVEL1> 5. A[12] <RECIPEIDVER1> 6. A[1] <RECIPEIDTYPE1> 7. L,p [p = Number of Step] 1. L,2 1. A[14] <STEP1> 2. L, n [n = Number of Parameter] 1. L,2 1. A[24] <RPARMNAME11> 2. A[24] <RPARMVAL11> : n.L,2 1. A[24] <RPARMNAME1n> 2. A[24] <RPARMVAL1n> : p. L,2 1. A[14] <STEP1p> 2. L, q [q = Number of Parameter] 1. L,2 1. A[24] <RPARMNAMEp1> 2. A[24] <RPARMVALp1> : q.L,2 1. A[24] <RPARMNAMEpq> 2. A[24] <RPARMVALpq> 8. L, r [r = Number of ESD] 1. L,2 1. A[24] <ESDITEM11> 2. A[40] <ESDVALUE11> : r. L,2 1. A[24] <ESDITEM1r> 2. A[40] <ESDVALUE1r>

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Notice	<UNITID> Unit ID <RECIPEIDCMD> RECIPEID Command B : Built C : change D : Delete R : Rename <RECIPEID> Recipe ID <LEVEL> Level of RECIPEID <RECIPEIDVER> RECIPEID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) < RPARMNAME> Recipe Parameter Name < RPARMVAL> Recipe Parameter Value <ESDITEM> EQP Specific Data Item defined by LG. Display <ESDVALUE> EQP Specific Data Value defined by LG. Display <STEP> Recipe Step 0 = Step Not Used, Another Number is Recipe Step
Exception	

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S15,F100	Equipment Constant Recipe Change Report Acknowledge	ECS → EQP	
Description	Equipment Constant Recipe Change Report Acknowledge		
Structure	L, 7 1. U2[1] <UNITID1> 2. A[24] <RECIPEID1> 3. A[1] <LEVEL1> 4. A[12] <RECIPEIDVER1> 5. A[1] <RECIPEIDTYPE1> 6. B[1] <PHACK1> 7. A[180]<HOSTMSG>		
Notice	<UNITID> Unit ID <LEVEL> Level of RECIPEID <RECIPEIDVER> RECIPEID Version <RECIPEIDTYPE>Type of Recipe E : Equipment Recipe / M : Machine Recipe / U : Unit Recipe 1,2,3,.. : Special Recipe (Defined with LG Display) <PHACK> 0 : Accepted 1 : Permission not granted 2 : Length error 3 : RECIPEID not found 4 : Parameter range error 5 : Saved for additional use		
Exception			

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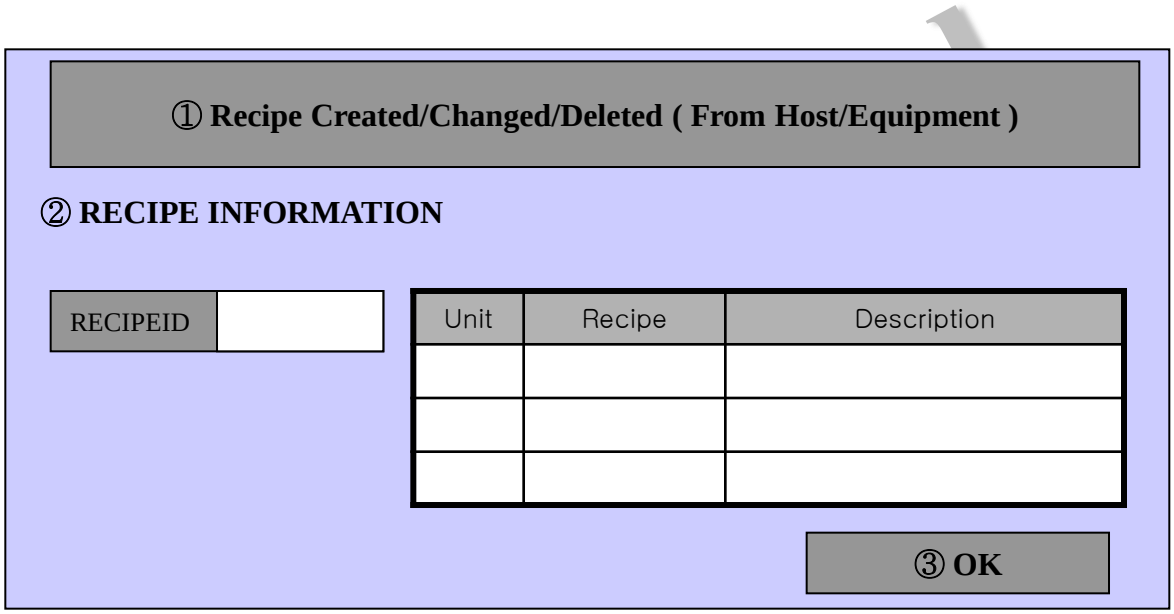
3. Attached page

1. The Confirming Screen Related to Recipe Command

이 장에서는 Recipe Command와 관련된 장비의 확인 화면에 대해 설명한다.
아래 예를 참고한다.

*** Recipe Creation/Change/Deletion confirmation window**

- 장비 recipe의 생성/변경/삭제를 표시한다. 이 window는 operation confirmation window와는 독립적으로 팝업된다. Recipe의 생성/변경/삭제 확인 window는 5초 후에 자동으로 닫힌다.



① Recipe Creation / Change / Deletion Description
- equipment message를 표시한다.

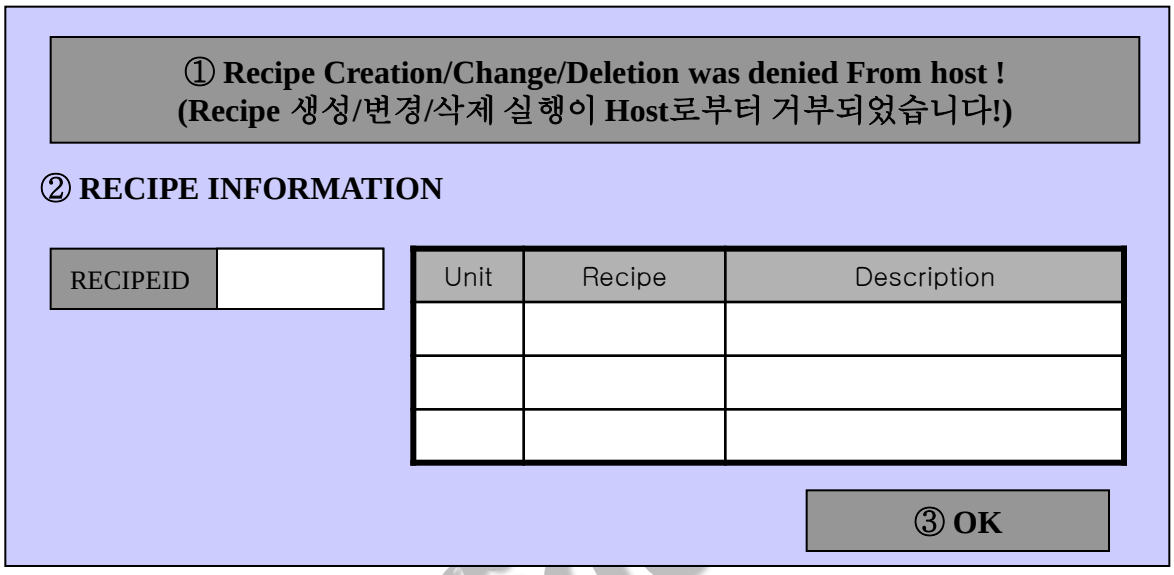
② Recipe Information
- equipment recipe의 생성/변경/삭제의 상세

③ OK Button
- 클릭하면 창 종료

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*** Recipe Creation/Change/Deletion confirmation window (denied from Host)**

- 장비 recipe의 생성/변경/삭제를 표시한다. 이 window는 operation confirmation window와는 독립적으로 팝업된다. Recipe의 생성/변경/삭제 확인 window는 5초 후에 자동으로 닫힌다.



① Recipe Creation / Change / Deletion Description
- 'Description'에는 denial reason / code가 표시된다.

② Recipe Information
- equipment recipe의 생성/변경/삭제의 상세

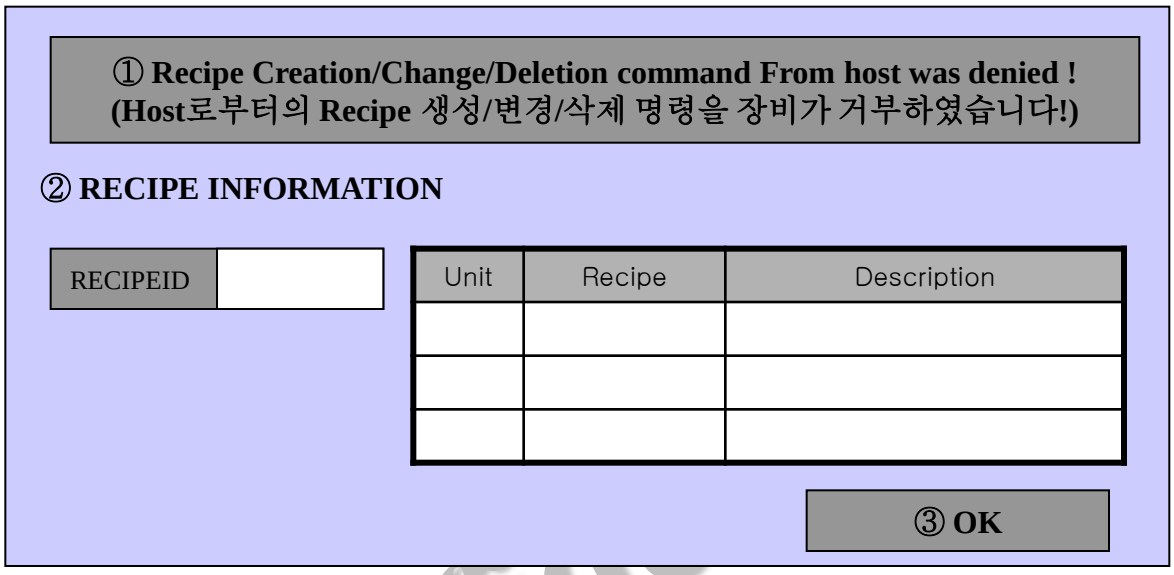
③ OK Button
- 클릭하면 창 종료

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*** Recipe Creation/Change/Deletion confirmation window**
(Equipment denied Recipe Command)

- 장비 recipe의 생성/변경/삭제를 표시한다. 이 window는 operation confirmation window와는 독립적으로 팝업된다. Recipe의 생성/변경/삭제 확인 window는 5초 후에 자동으로 닫힌다.



① Recipe Creation / Change / Deletion Description
- 'Description'에는 denial reason / code가 표시된다.

② Recipe Information
- equipment recipe의 생성/변경/삭제의 상세

③ OK Button
- 클릭하면 창 종료

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